

Financial Time Series Analysis

Lecture 1

- ▶ Class of 2022 MSCF admitted students may be watching the Time Series lecture on Thursday, March 25. You may wish to implement privacy protections.
- ▶ TA Session, Friday, March 26, 11:00am-12:00pm (EDT)

Outline

- ▶ Syllabus Material
- ▶ Introduction to Time Series
 - ▶ Time Series Topics to be Covered and Reference Material
 - ▶ Stylized Facts
 - ▶ Stationarity, White Noise, Martingale Difference Sequences

Course Topics Time Series Analysis

- ▶ Stylized facts
- ▶ Stationarity
- ▶ Standard Time Series Models
- ▶ Prediction
- ▶ ARCH/GARCH Models
- ▶ Multivariate Time Series Models
 - ▶ Vector autoregressive processes (VAR)
 - ▶ Dickey Fuller test for Stationarity
 - ▶ Cointegration
- ▶ State Space Models and Kalman Filtering

Time Series References: 1

- ▶ Box, G.E.P., Jenkins, G., Reinsel, G. and G.M. Ljung, *"Time Series Analysis: Forecasting and Control, 5th edition, 2015.*
- ▶ Brockwell, P.J. and R. Davis, *Introduction to Time Series and Forecasting, 3rd Edition, 2016.*
- ▶ Chan, N.H., *Time Series: Applications to Finance with R and S-Plus, 2010.*
- ▶ Hamilton, James, *Time Series Analysis, 1994.*
- ▶ Lai, T.L. and Xing, H., *Statistical Models and Methods for Financial Markets, 2008.*
- ▶ McKinney, W., *Python for Data Analysis: Data Wrangling with Pandas, Numpy and IPython, 2nd Edition, 2018.*

Time Series References: 2

- ▶ Ruppert, D. and D. Matteson, *Statistics and Data Analysis for Financial Engineering, 2nd Edition*, 2015.
- ▶ Sargent, T.J. and J. Stachurski, *Quantitative Economics with Python*, 2020. On-line <http://python.quantecon.org> .
- ▶ Tsay, R.S., *An Introduction to the Analysis of Financial Data with R*, 2013.
- ▶ Tsay, R.S., *Analysis of Financial Time Series, 3rd Edition*, 2019.
- ▶ Tsay, R.S. *Multivariate Time Series Analysis with R and Financial Applications*, 2014.
- ▶ Velu, R., Hardy, M. and D. Nehren, *Algorithmic Trading and Quantitative Strategies*, 2020.

Python Packages

- ▶ numpy, scipy, pandas, matplotlib
- ▶ statsmodels
- ▶ quantecon (Sargent and Stachurski text)
- ▶ pmdarima (claims to have functionality of R package `auto.arima`)
- ▶ arch (for ARCH/GARCH)
- ▶ johansen (for Johansen cointegration test)
- ▶ pyflux (time series analysis and prediction)
- ▶ state-space, pykalman
- ▶ See pypi.org, python package index

Time Series In Economics and Finance

- ▶ Time series analysis is a set of models and methodologies to handle non-independent data (univariate or multivariate) with a temporal dependence structure.
- ▶ Time series methods apply in many areas of finance:
 - ▶ Predicting future values of economic and financial time series like stock prices (prediction)
 - ▶ Identify sets of stocks that have a temporal dependence relationship, and use these in a convergence trading and strategy. (see next slides)
 - ▶ Model volatility clustering and use these models in risk management calculations. (GARCH - volatility clustering)

An Example: Convergence Trading: 1

- ▶ Consider two stocks that appear to be moving together (in standard units).
- ▶ If the price of stock 1 increases more than that of stock 2, one can short the relatively higher priced stock and long the lower price stock and wait for the two stock prices to revert to a common level.
- ▶ if this reversion occurs, the profit is the difference in prices at the time of the trade.

An Example: Convergence Trading: 2

- ▶ Such a trade is “zero-cost” and “market neutral”
- ▶ How do we determine that pairs (or baskets) of stocks are “moving together?”
- ▶ **Cointegration** invented by Engle and Granger is a concept to identify if a linear combination of non-stationary time series are stationary. There are statistical procedures to identify cointegrated series.

Gatev et al: Pairs Trading

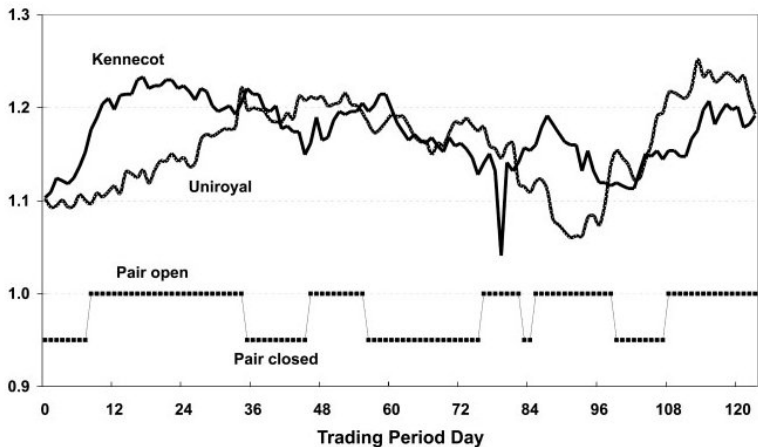


Figure 1
Daily normalized prices: Kennecott and Uniroyal (pair 5)
Trading period August 1963–January 1964.

Empirical Properties of Asset Returns: 1

- ▶ The availability of data and computing power is fundamental to understanding the price behavior of financial assets.
- ▶ Before 1960, there was no central database of security prices.
- ▶ In 1960, CRSP was founded at the University of Chicago, and a major project was the establishment of daily stock price data (open, high, low, close) for all NYSE stocks. This became available in 1962. Later expanded to other markets (AMEX, NASDAQ)

Empirical Properties of Asset Returns: 2

- ▶ In mathematical finance, the basic model of asset price processes is GBM, $S_{t+h} = S_t \exp((r - .5\sigma^2)h + \sigma W(h))$. This dates back to Bachelier in 1900.
- ▶ Calculating the log-returns over a time period of length j , we get $\log(S_{t+h}/S_t) = (r - .5\sigma^2)h + \sigma W(h)$, which has a $N((r - .5\sigma^2)h, \sigma^2 h)$ distribution.
- ▶ Thus we would expect a good model for log-returns would be a normal distribution.
- ▶ CRSP data allowed for a careful study (on a daily time scale) of the behavior of stock prices. Just like the microscope revolutionized biology and the telescope revolutionized astronomy, CRSP data revolutionized quantitative stock price modeling.

Empirical Properties of Asset Returns: 3

- ▶ In 1965, Fama published a major empirical study of the behavior of stock prices. He demonstrated that log-returns have much heavier tails than would be predicted from a normal distribution.
- ▶ Fama erroneously concluded that the log-returns should be modeled using **stable laws**.
- ▶ The stable laws (indexed by $0 < \alpha \leq 2$) (e.g. Cauchy, $\alpha = 1$) have no moments $\geq \alpha$.
- ▶ A 2001 paper by Rama Cont on “Empirical properties of asset returns and stylized facts” presents a number of observations that characterize the general behavior of financial asset prices.

Empirical Properties of Asset Returns: 4

- ▶ The Cont paper on stylized facts shows that the i.i.d. normal model is most often inappropriate for dealing with financial data. Some of the key departures from standard statistical models include
 - ▶ Lack of stability in long-run behavior for raw financial variables
 - ▶ Heavy tails in returns (but higher moments exist)
 - ▶ Temporal dependencies
 - ▶ Volatility clustering (GARCH models)

Stylized Facts: 1

Rama Cont, “Empirical properties of asset returns: stylized facts and statistical issues,” *Quantitative Finance*, **1**, 2001, 223-236. These are presented here, out of order.

1) Absence of autocorrelations:

“(linear) autocorrelations of asset returns are often insignificant, except for very small intraday time scales for which microstructure effects come into play.”

We will define the autocovariance and autocorrelation function later. However, a plot of the autocovariance function should show a spike at 0 for the variance of the process, possibly a significant, but smaller spike at lag 1 and no significant spikes beyond 1.

Stylized Facts: 2

2) Heavy Tails:

“the (unconditional) distribution of returns seems to display a power-law or Pareto-like tail, with a tail index which is finite, higher than two and less than five for most data sets studied. In particular this excludes stable laws with infinite variance and the normal distribution. However the precise form of the tails is difficult to determine.”

3) Gains/loss asymmetry:

“one observes large drawdowns in stock prices and stock index values but not equally large upward movements. (this property is not true for FX where there is symmetry).”

Stylized Facts: 3

4) Aggregational Gaussianity:

“as one increases the time scale over which returns are calculated, their distribution looks more like a normal distribution. The shape is not the same at different time scales.”

5) Intermittency:

“returns display, at any time scale, a high degree of variability. This is quantified by the presence of irregular bursts in time series of a wide variety of volatility estimators.”

Stylized Facts: 4

6) Volatility clustering:

“different measures of volatility display a positive autocorrelation over several days, which quantifies the fact that high-volatility events tend to cluster in time.”

7) Conditional heavy tails:

“even after correcting returns for volatility clustering (e.g. via GARCH-type models), the residual time series still exhibit heavy tails. However, the tails are less heavy than in the unconditional distribution of returns.”

8) Slow decay of autocorrelation in absolute returns:

“the autocorrelation function of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$. This is sometimes interpreted as a sign of long-range dependence.

Stylized Facts: 5

9) Leverage effect:

"most measures of volatility of an asset are negatively correlated with the returns of that asset."

P668 of Hamilton's text "A number of researchers have found evidence of asymmetry in stock price behavior - negative surprises seem to increase volatility more than positive surprises. Since a lower stock price reduces the value of equity relative to corporate debt, a sharp decline in stock prices increases corporate leverage and could thus increase the risk of holding stocks. For this reason, the apparent finding ... is called the **leverage effect**."

Stylized Facts: 6

10) Volume/volatility correlation:

"trading volume is correlated with all measures of volatility."

11) Asymmetry in time scales:

"coarse-grained measures of volatility predict fine-scale volatility better than the other way around."

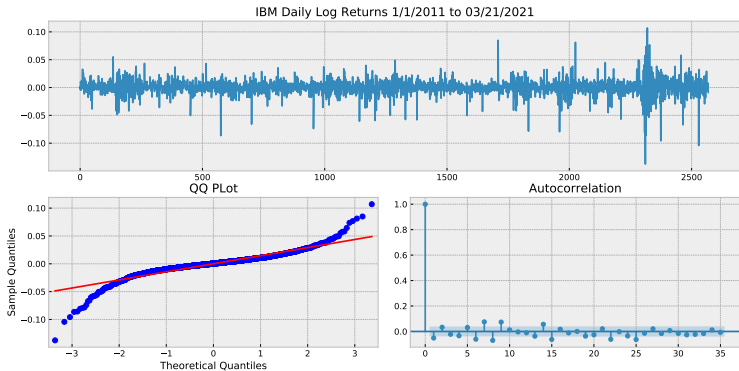
Log>Returns: 1

- ▶ The standard central limit theorem indicates that sums of i.i.d. random variables with finite variance, when properly normalized converge weakly to a normal distribution.
- ▶ There are many variations, and roughly the identical distribution part can be weakened as long as some of the summands don't predominate (i.e. an elephant among many ants).
- ▶ The independence assumption can be weakened as long as the dependence dies off quickly.

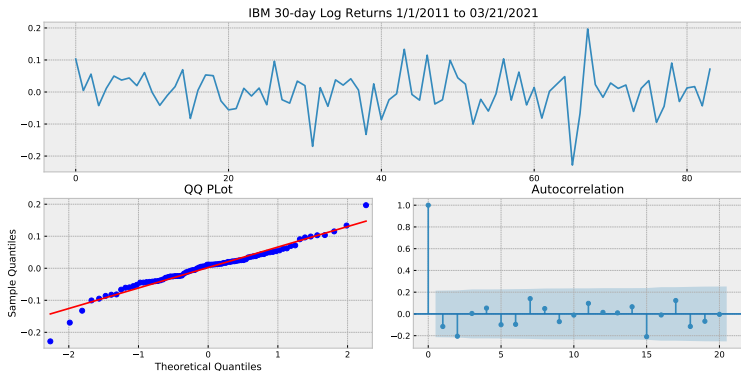
Log>Returns: 2

- ▶ Consider log-returns, $\log(S_{t+1}/S_t)$ and $\log(S_{t+N}/S_t)$
- ▶ Note $\log(S_{t+N}/S_t) = \log\left(\frac{S_{t+N}}{S_{t+N-1}} \cdots \frac{S_{t+1}}{S_t}\right) = \sum_{i=1}^N \log\left(\frac{S_{t+i}}{S_{t+i-1}}\right)$
- ▶ Thus log-returns should be normal over longer time horizons provided:
 - ▶ Variances are finite
 - ▶ The autocorrelation is small
 - ▶ No terms dominate the sum.

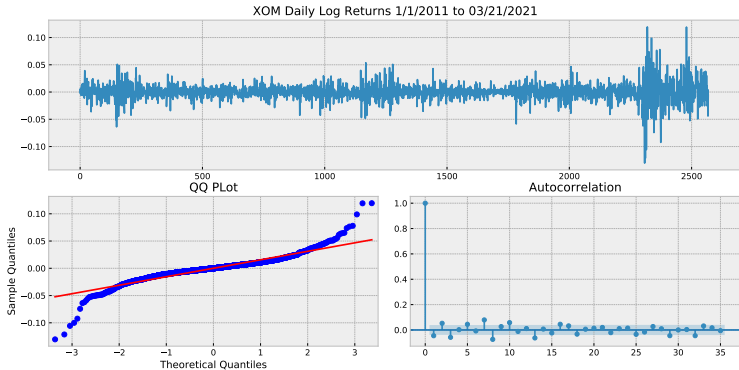
IBM Daily Log Returns 1/1/2011-3/20/2021



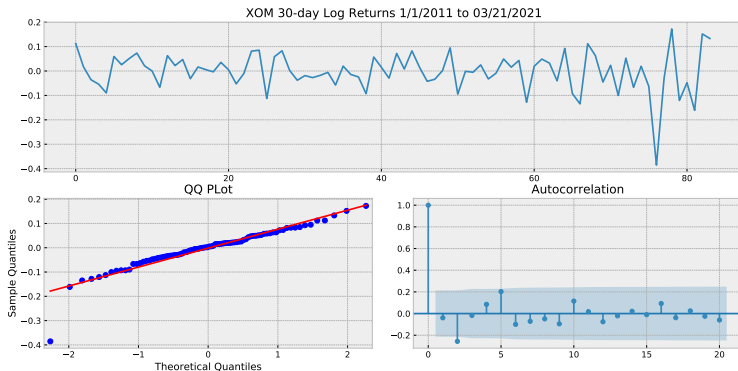
IBM 30-Day Log Returns 1/1/2011-3/20/2021



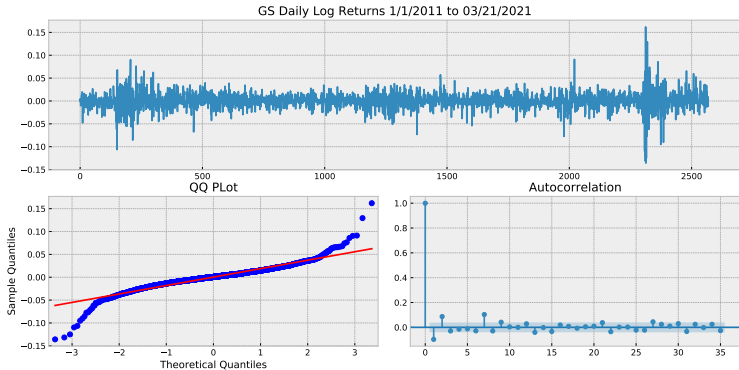
XOM Daily Log Returns 1/1/2011-3/20/2021



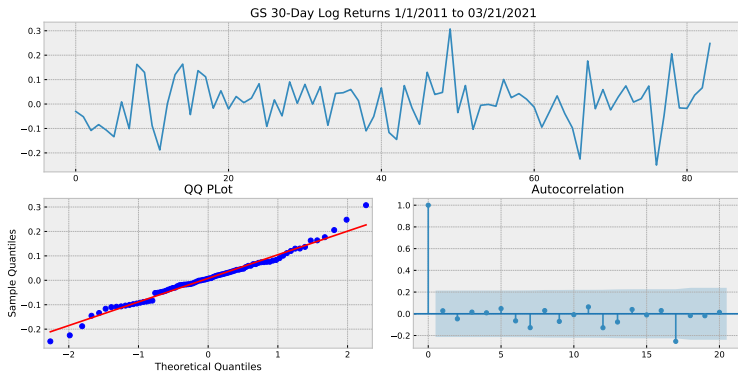
XOM 30-Day Log Returns 1/1/2011-3/20/2021



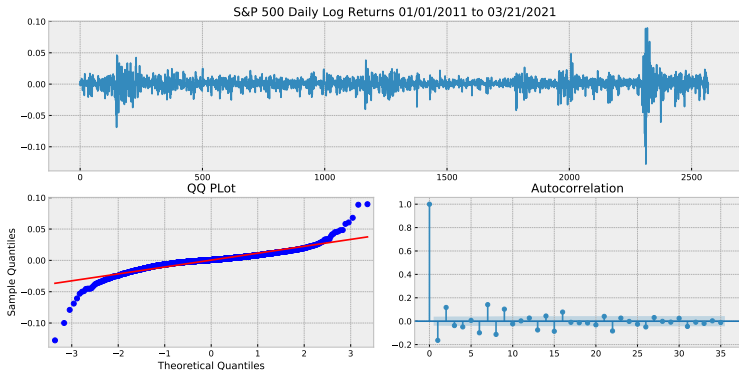
GS Daily Log Returns 1/1/2011-3/20/2021



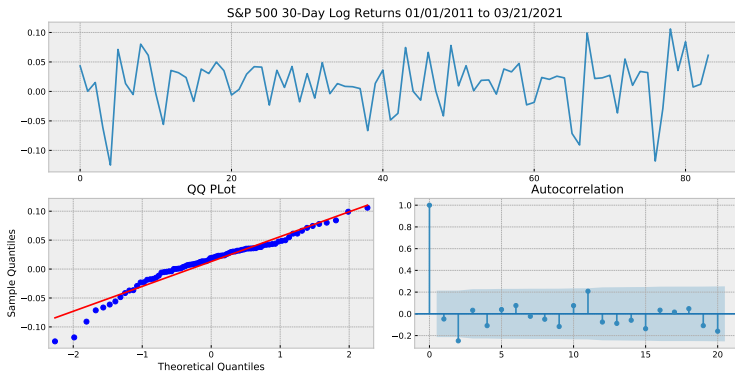
GS 30-Day Log Returns 1/1/2011-3/20/2021



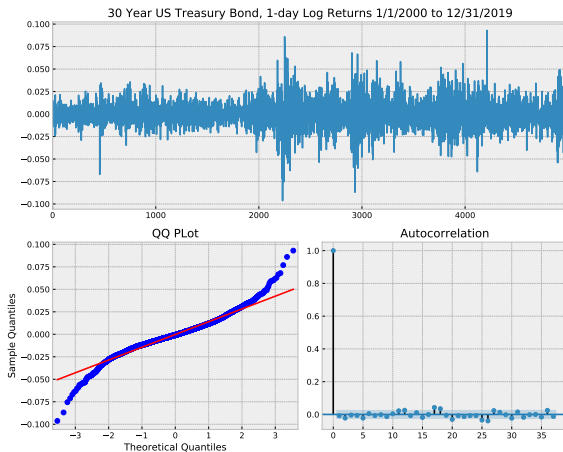
S&P 500 Daily Log Returns 1/1/2011-3/20/2021



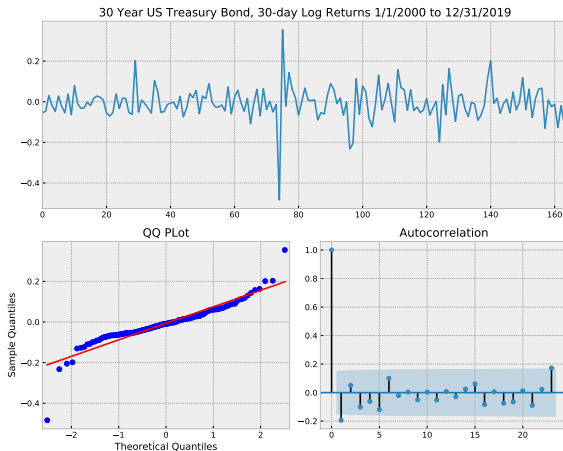
S&P 500 30-Day Log Returns 1/1/2011-3/20/2021



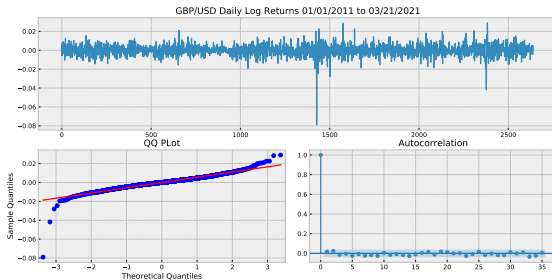
US 30-Year Treasury Bond Daily Returns 1/1/2000-12/31/2019



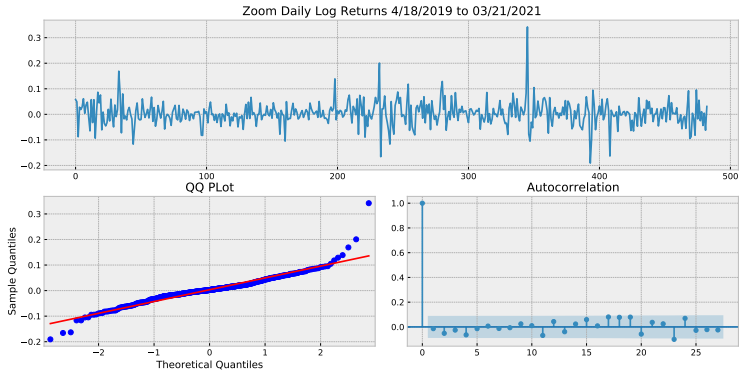
US 30-Year Treasury Bond 30-Day Returns 1/1/2000-12/31/2019



GBP/USD Daily Log>Returns 1/1/2011-3/21/2021



Zoom Video Daily Returns 4/16/2019-3/20/2021



Time Series Models/Graphics: 1

```
import os
import sys
import pandas as pd
import pandas_datareader as pdr
import numpy as np
import statsmodels as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.api as smt
import statsmodels.api as sm
import scipy.stats as scs
import matplotlib.pyplot as plt
import matplotlib as mp
import seaborn as sns
```

Time Series Models/Graphics: 2

```
def tsplot(y, lags=None, figsize = (10,8), style = 'bmh'):
    if not isinstance(y, pd.Series):
        y = pd.Series(y)
    with plt.style.context(style):
        fig = plt.figure(figsize=figsize)
        layout = (2,2)
        ts_ax = plt.subplot2grid(layout, (0,0), colspan = 2)
        acf_ax = plt.subplot2grid(layout, (1,1))
        qq_ax = plt.subplot2grid(layout, (1,0))
        y.plot(ax = ts_ax)
        ts_ax.set_title('Insert Title')
        smt.graphics.plot_acf(y, lags = lags, ax=acf_ax,
            alpha = 0.05)
        sm.qqplot(y, line = 's', ax=qq_ax)
        qq_ax.set_title('QQ PLOT')
        plt.tight_layout()

    return
```

Time Series Models/Graphics: 3

```
start = '2011-01-01'
end = '2021-03-21'
price = pdr.get_data_yahoo('gs',start,end)
adjclose = price_data['Adj Close']
shf = 1
logret = np.log(adjclose/adjclose.shift(shf)).dropna()
L = len(logret)
LL = int(L/shf)
logret_shf = np.zeros(LL)
for i in range(int(L/shf)):
    logret_shf[i] = logret[i*shf]
.
tsplot(logret_shf)
plt.show()
```

STATIONARITY

Stationarity: 1

- ▶ Consider a time series $\{X_t\}$ for $t = 0, 1, 2, \dots$ or $t = 0, \pm 1, \pm 2, \dots$. This time series is called
 - ▶ **Strictly Stationary** if for all $n \geq 1$, $t_1 < t_2 < \dots < t_n$, and $i = 1, 2, \dots$ the joint distribution of $(X_{t_1}, X_{t_2}, \dots, X_{t_n})$ is the same as $(X_{t_1+i}, X_{t_2+i}, \dots, X_{t_n+i})$.
 - ▶ **Weakly Stationary** if for all $t_1 < t_2$ and $i = 1, 2, \dots$, $\text{Cov}(X_{t_1}, X_{t_2}) = \text{Cov}(X_{t_1+i}, X_{t_2+i})$.
- ▶ We will always use the wider or less restrictive version of stationarity, weak stationarity also known as covariance stationarity or second order stationarity.

Stationarity: 2

- ▶ A covariance stationary process $\{X_t, t \in \mathcal{Z}\}$ satisfies:
 - ▶ $E(X_t) = \mu, t \in \mathcal{Z},$
 - ▶ $Cov(X_s, X_t) = \gamma(s, t) = \gamma(s + k, t + k), s, t, k \in \mathcal{Z}$
- ▶ Since $\gamma(s, t) = \gamma(0, t - s) = \gamma(t - s, 0)$, the function γ is actually a function of only one variable, $|t - s|$ the **lag**.
- ▶ We define the **autocovariance function**, $\gamma(h) = \gamma(h, 0)$
- ▶ Note that $\gamma(0) = Var(X_t)$ and the autocorrelation function, $\rho(h)$, is defined by $\rho(h) = \gamma(h)/\gamma(0)$.

Stationarity: 3

- ▶ The backbone of much of basic statistics is the i.i.d. process. In time series analysis, the corresponding backbone is the **White Noise Process**
- ▶ A **white noise process** is a weakly stationary process with constant mean 0, and autocovariance function $\gamma(h) = \sigma^2$ if $h = 0$ and 0 if $h \neq 0$.
- ▶ A white noise process is denoted by $WN(0, \sigma^2)$.
- ▶ The autocorrelation function for a white noise process, $\rho(h) = 1$ if $h = 0$ and 0 if $h \neq 0$

Financial Time Series Analysis

Lecture 2

- ▶ Welcome class of 2022 MSCF admitted students.
- ▶ TA Session, Friday, March 26, 11:00am-12:00pm (EDT)

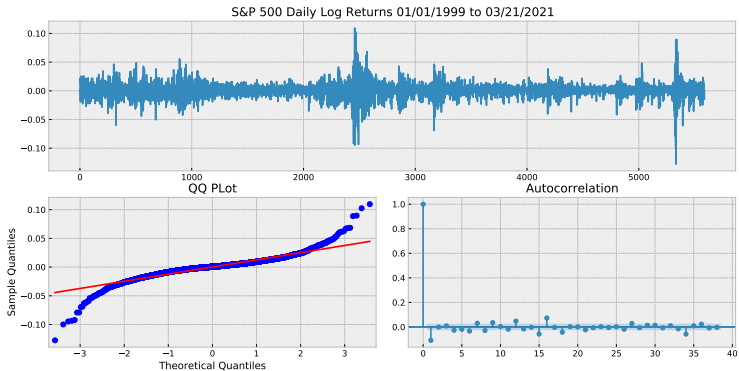
Outline

- ▶ Review
- ▶ Stationarity
 - ▶ Strong/Weak/White Noise
 - ▶ Martingale Difference Sequences (MDS)
 - ▶ Wold Decomposition
- ▶ Standard Time Series Models: $AR(p)$, $MA(q)$, $ARMA(p,q)$

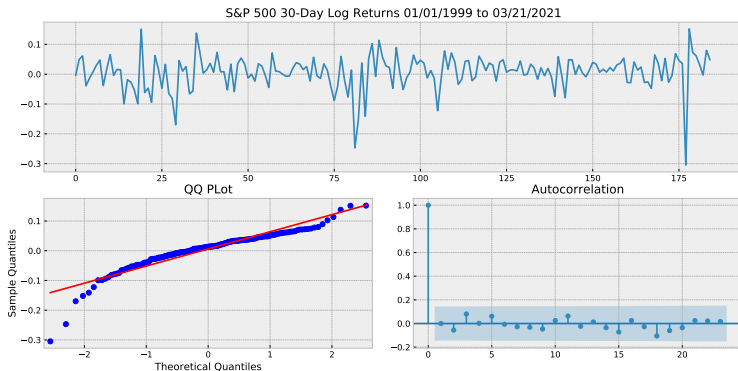
Brief Review

- ▶ Eleven stylized facts about asset prices
 - ▶ Daily log-returns have heavier tails than the normal distribution
 - ▶ As time interval increases log-returns become more normal
 - ▶ Little autocorrelation at daily time scale
 - ▶ Gains/loss asymmetry (larger drawdowns than upward moves)
 - ▶ Volatility clustering
 - ▶ Leverage effect, volatility is negatively correlated with returns.

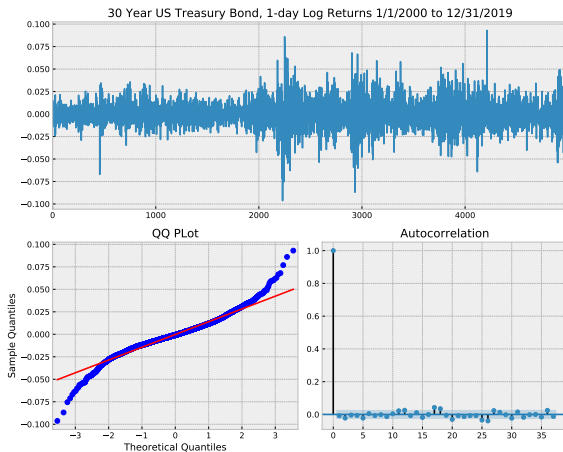
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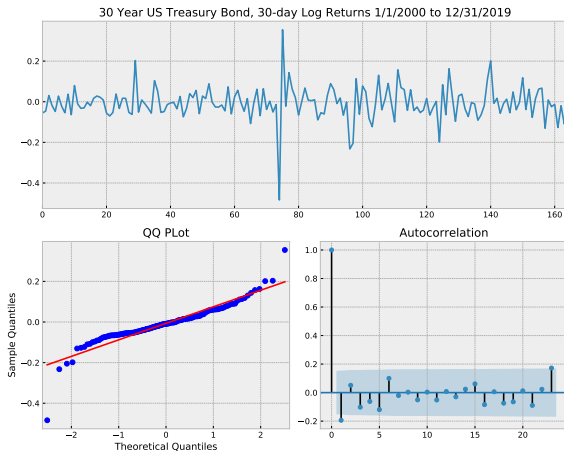
S&P 500 30-Day Log Returns 1/1/1999-3/20/2021



US 30-Year Treasury Bond Daily Returns 1/1/2000-12/31/2019



US 30-Year Treasury Bond 30-Day Returns 1/1/2000-12/31/2019



STATIONARITY

Stationarity: 1

- ▶ Consider a time series $\{X_t, -\infty < t < \infty\}$ or $\{X_t, 0 < t < \infty\}$. This time series is called
 - ▶ **Strictly Stationary** if for all $n \geq 1$, $t_1 < t_2 < \dots < t_n$, and $i = 1, 2, \dots$ the joint distribution of $(X_{t_1}, X_{t_2}, \dots, X_{t_n})$ is the same as $(X_{t_1+i}, X_{t_2+i}, \dots, X_{t_n+i})$.
 - ▶ **Weakly Stationary** if for all $t_1 < t_2$ and $i = 1, 2, \dots$, $\text{Cov}(X_{t_1}, X_{t_2}) = \text{Cov}(X_{t_1+i}, X_{t_2+i})$.
- ▶ Weak stationarity and jointly normal distributions imply strict stationarity.
- ▶ We will always use weak stationarity, which is also known as covariance stationarity, second-order stationarity, and wide-sense stationarity.

Stationarity: 2

- ▶ A weakly stationary process $\{X_t, t \in \mathcal{Z}\}$ satisfies:
 - ▶ $E(X_t) = \mu, t \in \mathcal{Z},$
 - ▶ $Cov(X_s, X_t) = \gamma(s, t) = \gamma(s + k, t + k), s, t, k \in \mathcal{Z}$
- ▶ Since $\gamma(s, t) = \gamma(0, t - s) = \gamma(t - s, 0)$, the function γ is actually a function of only one variable, $|t - s|$ is the **lag**.
- ▶ We define the **autocovariance function**, $\gamma(h) = \gamma(h, 0)$
- ▶ Note that $\gamma(0) = Var(X_t)$ and the autocorrelation function, $\rho(h)$, is defined by $\rho(h) = \gamma(h)/\gamma(0)$.

Stationarity: 3

- ▶ The backbone of much of basic statistics is the i.i.d. process. In time series analysis, the corresponding backbone is the **White Noise Process**
- ▶ A **white noise process** is a weakly stationary process with constant mean 0, and autocovariance function

$$\gamma(h) = \begin{cases} \sigma^2 & \text{if } h = 0, \\ 0 & \text{if } h \neq 0. \end{cases}$$

- ▶ A white noise process is denoted by $WN(0, \sigma^2)$.
- ▶ The autocorrelation function for a white noise process is,

$$\rho(h) = \begin{cases} 1 & \text{if } h = 0, \\ 0 & \text{if } h \neq 0. \end{cases}$$

Asymptotics of Stationary Sequences: 1

- ▶ While it is certainly possible to define stochastic sequences that are more general than i.i.d. sequences or independent sequences, we need to have sequences that have good asymptotic properties like laws of large numbers and central limit theorems.
- ▶ These theorems are important for having the consistency of statistical procedures, the asymptotic distributions of estimators of model parameters, the critical values for test statistics, or the width of prediction/forecast intervals being characterizable.
- ▶ We need the asymptotic behavior of our models to be characterizable.

Asymptotics of Stationary Sequences: 2

- ▶ Consider the example where X_0 is a Bernoulli(.5) random variable, and $X_n = X_0$ for all n . This results in two possible outcomes for the entire process: $(0, 0, 0, \dots)$ and $(1, 1, 1, \dots)$.
- ▶ This process is both strictly stationary and weakly stationary with $\gamma(h) = .25$, $\rho(h) = 1$, $\forall h \geq 0$.
- ▶ $S_n/n = (X_1 + \dots + X_n)/n = X_0$.
- ▶ This is an example in which the long run average is identical to X_0 almost surely, not the mean value of .5.
- ▶ Suppose X_t , $t \geq 0$ is weakly stationary with $E(X_t) = \mu$ and $\text{Var}(X_n) = \sigma^2$. If $\sum_{i=1}^{\infty} |\gamma(i)| < \infty$, then $\sum_{i=1}^n X_i/n \rightarrow \mu$ in probability.

Standard Facts on Conditional Expectation: 1

Consider a time series $\{X_t, -\infty < t < \infty\}$. We define $\mathcal{F}_t$ to be the σ -algebra generated by $(\cdots, X_{t-1}, X_t)$. Intuitively this represents “all the information in the random variables $(\cdots, X_{t-1}, X_t)$ ”. Important properties include

- ▶ $\mathcal{F}_s \subset \mathcal{F}_t$, for $s \leq t$
- ▶ $E(X|\mathcal{F}_s)$ is $\mathcal{F}_s$ measurable
- ▶ $E(Y) = E(E(Y|\mathcal{F}_t)) \forall t$ and Y with $E|Y| < \infty$.
- ▶ $E(X|\mathcal{F}_t) = X$ a.s. if X is $\mathcal{F}_t$ measurable
- ▶ $E(XY|\mathcal{F}_t) = YE(X|\mathcal{F}_t)$, a.s. if Y is $\mathcal{F}_t$ measurable.
- ▶ $E(E(X|\mathcal{F}_s)|\mathcal{F}_t) = E(E(X|\mathcal{F}_t)|\mathcal{F}_s) = E(X|\mathcal{F}_s)$ for $s < t$.

Martingale Difference Sequences: 1

- ▶ Recall that a **martingale sequence** is a pair $\{(X_t, \mathcal{F}_t), t \geq 0\}$ with $\mathcal{F}_s \subset \mathcal{F}_t, \forall s \leq t$ and $E|X_t| < \infty$, satisfying

$$E(X_{t+1}|\mathcal{F}_t) = X_t, a.s. \forall t \geq 0.$$

- ▶ In a similar fashion we define a **martingale difference sequence** (MDS) to be a pair $\{(Y_t, \mathcal{F}_t), t \geq 0\}$ with $\mathcal{F}_s \subset \mathcal{F}_t, \forall s \leq t$ and $E|Y_t| < \infty$, satisfying

$$E(Y_{t+1}|\mathcal{F}_t) = 0, a.s., \forall t \geq 0.$$

- ▶ By taking expectations, $E(Y_t) = 0$.
- ▶ Note that we can rewrite the martingale property as $E(X_{t+1} - X_t|\mathcal{F}_t) = 0, a.s. \forall t \geq 0$. If we let $Y_{t+1} = X_{t+1} - X_t$, then $\{Y_t, t \geq 0\}$ is an MDS.

Martingale Difference Sequences: 2

There are three different types of noise processes that we will be considering:

1. Independent sequences, $\{X_t, t \geq 0\}$ with common mean μ and variance σ^2 . Sometimes this is specialized to i.i.d. sequences and even further to i.i.d. with $N(\mu, \sigma^2)$ distribution.
 2. Martingale difference sequences (MDS):
 $E(X_{t+1}|\mathcal{F}_t) = 0, a.s.$
 3. Weakly stationary processes in which $E(X_t) = \mu$,
 $\text{Cov}(X_s, X_t) = \gamma(t - s)$.
- An independent sequence with $\mu = 0$ is clearly an MDS.

$$\{\text{Independent Sequences}\} \subset \{\text{MDS}\}.$$

Martingale Difference Sequences: 3

- ▶ MDS with common finite variance are white noise processes, therefore are weakly stationary, i.e.

$\{\text{Common Finite Variance MDS}\} \subset \{\text{White Noise Processes}\}.$

- ▶ If $\{X_t, t \geq 0\}$ is an MDS, and $s < t$

$$\begin{aligned}\text{Cov}(X_s, X_t) &= E(X_s X_t) - 0 \cdot 0 \\ &= E(E(X_s X_t | \mathcal{F}_{t-1})) \\ &= E(X_s E(X_t | \mathcal{F}_{t-1})) \\ &= E(X_s \cdot 0) \\ &= 0\end{aligned}$$

Martingale Difference Sequences: 4

- ▶ Why introduce this “intermediate” category of stochastic processes (between independent sequences and weakly stationary sequences)?
- ▶ There are dependencies in financial and economic data series, so the independence assumption is often not warranted.
- ▶ Much of statistical analysis involving confidence intervals and hypothesis testing relies on asymptotic normality.
- ▶ We need to have processes for which there is a central limit theorem.
- ▶ Given $\{(X_t, \mathcal{F}_t), t \geq 0\}$ is a MDS with common finite variance σ^2 , under some additional conditions

$$\sqrt{T}\bar{X} \Rightarrow N(0, \sigma^2).$$

Wold Decomposition: 1

- ▶ A very important theorem by Wold describes the structure of weakly stationary processes. This is referred to as the Wold decomposition.
- ▶ This decomposition begins with a general weakly stationary process $\{X_t, -\infty < t < +\infty\}$.
- ▶ We define $\mathcal{F}_t$ to be the σ -algebra generated by $\{X_s, -\infty < s \leq t\}$.
- ▶ Suppose $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\Phi, \Omega\}$.
- ▶ Then the $\{X_t\}$ process can be written as a **moving average process of infinite order** with respect to some white noise process $\{\epsilon_t, -\infty < t < \infty\}$. That is

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j},$$

where $\psi_0 = 1$ and $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

Wold Decomposition: 2

- ▶ The process is linear and causal and is represented as a moving average of infinite order with respect to a white noise process.
- ▶ Of course there are infinitely many parameters, so we must look for a simpler model that can be estimated with a finite data set, e.g. a moving average with q lags (MA(q)), an autoregressive process with p lags (AR(p)), or a combination of both, ARMA(p, q).
- ▶ Note, the condition $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \{\Phi, \Omega\}$ is restrictive and there are cases where this is false. In this case, the Wold decomposition has two components: the moving average of infinite order plus a “deterministic process.” We give an example of this next.

Wold Decomposition: 3

- ▶ Suppose we have two uncorrelated random variables, U and V with $E(U) = E(V) = 0$ and $Var(U) = Var(V) = \sigma^2$.
Let

$$X_t = U \cos(\theta t) + V \sin(\theta t).$$

- ▶ $E(X_t) = 0$ and $Var(X_t) = \sigma^2(\cos^2(\theta t) + \sin^2(\theta t)) = \sigma^2, \forall t$

$$\begin{aligned} Cov(X_{t_1}, X_{t_2}) &= Cov(U \cos(\theta t_1) + V \sin(\theta t_1), U \cos(\theta t_2) + V \sin(\theta t_2)) \\ &= \sigma^2(\cos(\theta t_1) \cos(\theta t_2) + \sin(\theta t_1) \sin(\theta t_2)) \\ &= \sigma^2 \cos(\theta(t_2 - t_1)). \end{aligned}$$

So $\gamma(h) = \sigma^2 \cos(\theta h)$ and $\rho(h) = \cos(\theta h)$

Wold Decomposition: 4

- ▶ Thus the $\{X_t\}$ process is weakly stationary.
- ▶ However, $\mathcal{F}_s = \sigma(U, V)$, $\forall s$ so $\cap_{j=1}^{\infty} \mathcal{F}_{t-j} = \sigma(U, V)$.
- ▶ This process is perfectly predictable. Given $\mathcal{F}_s$ for any s , the rest of the process is deterministic.
- ▶ This process is called “deterministic” even though it is created by two random variables U and V .
- ▶ Leaving aside the possibility of predictable components, how can we construct a white noise process, $\{\epsilon_t, -\infty < t < \infty\}$, and a set of constants, $\{\psi_j\}$, so that

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$$

Wold Decomposition: 5

Define $\{\epsilon_t, -\infty < t < \infty\}$ by

$$\epsilon_t = X_t - E(X_t | \mathcal{F}_{t-1}).$$

Define $\{\psi_t, 0 < t < \infty\}$ by

$$\psi_i = \gamma_X(i) / \sigma^2 = \rho_X(i).$$

It follows that $\{\epsilon_t, -\infty < t < \infty\}$ is a white noise process (see Course Notes, pp 16-19). Finally,

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$$

Wold Decomposition: 6

- ▶ Under the vanishing σ -algebra condition, we find that all weakly stationary processes can be represented as a **linear, causal** process.
- ▶ A moving average process of infinite order.
- ▶ Of course, all datasets are finite, and we cannot fit processes of infinite order. Thus we need to develop a family of simpler models with a small number of parameters.
- ▶ Those basic models include autoregressive models of order p , $AR(p)$, moving average models of order q , $MA(q)$, and autoregressive moving average models of order (p,q) , $ARMA(p,q)$.

AUTOREGRESSIVE PROCESSES

$$X_t = \phi_0 + \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t,$$

where $\{\epsilon_t\}$ is $WN(0, \sigma^2)$.

Time Series Ideas: AR(1) Processes: 1

- ▶ Consider an autoregressive process of order 1, namely:

$$X_t = \alpha + \beta X_{t-1} + \epsilon_t,$$

where $\{\epsilon_t\}$ is a stationary process with

$$E(\epsilon_t) = 0,$$

$$\text{Var}(\epsilon_t) = \sigma_\epsilon^2.$$

- ▶ One can find $E(X_t)$ and $\text{Var}(X_t)$ as solutions to simple difference equations, which can be iterated back to time 0:
- ▶ $E(X_t) = \alpha + \beta E(X_{t-1})$
- ▶ $\text{Var}(X_t) = \beta^2 \text{Var}(X_{t-1}) + \sigma_\epsilon^2$

Time Series Ideas: AR(1) Processes: 2

Assuming the process is stationary, hence with common mean μ and variance σ^2 ,

► $\mu = \alpha + \beta\mu$, or

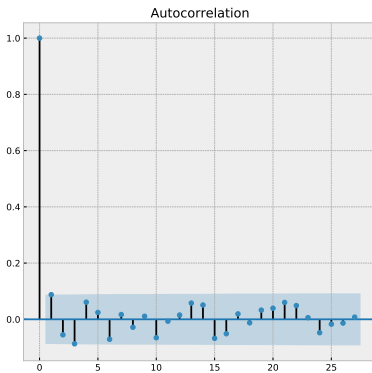
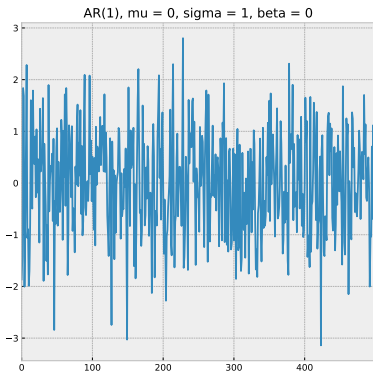
$$\mu = \alpha/(1 - \beta).$$

► $\sigma_X^2 = \beta^2\sigma_X^2 + \sigma_\epsilon^2$ or

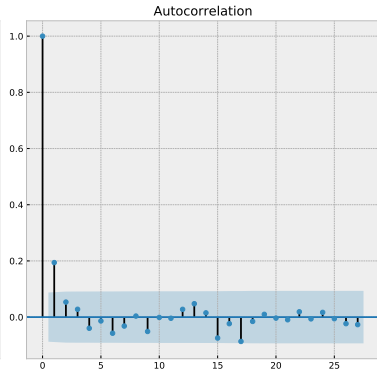
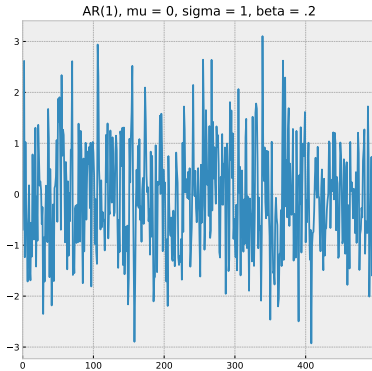
$$\sigma_X^2 = \sigma_\epsilon^2/(1 - \beta^2).$$

- Both require $|\beta| < 1$. Backward iteration of the AR(1) equation will show the necessity and sufficiency of this condition.

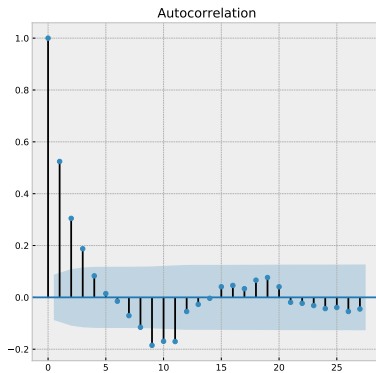
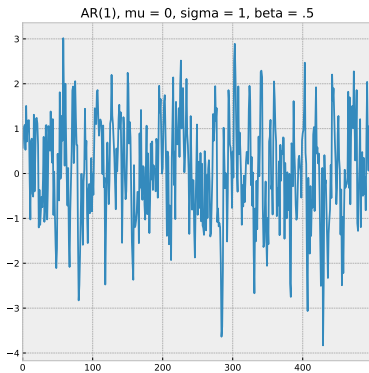
AR(1), $\beta = 0$



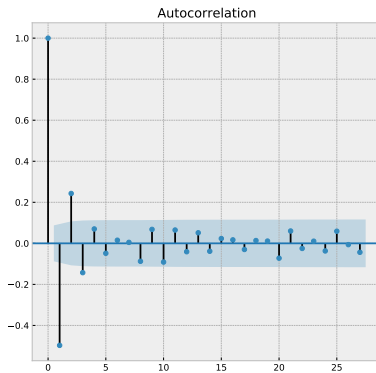
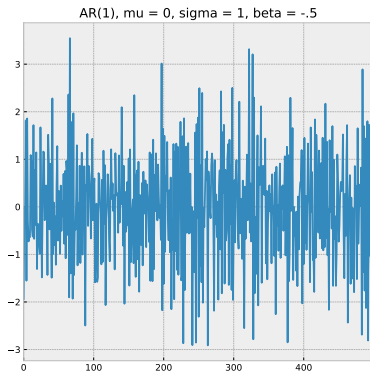
AR(1), $\beta = .2$



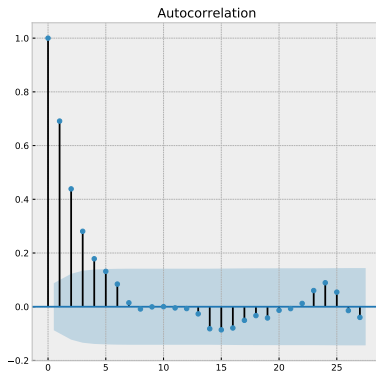
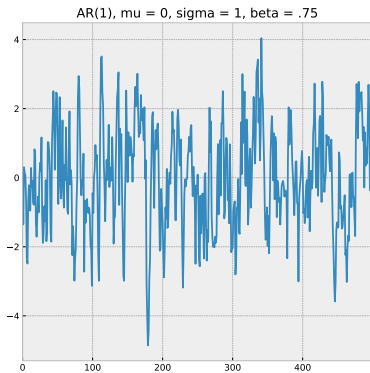
AR(1), $\beta = .5$



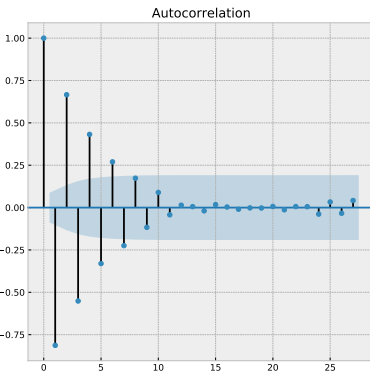
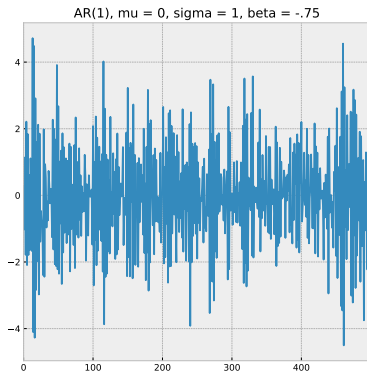
AR(1), $\beta = -0.5$



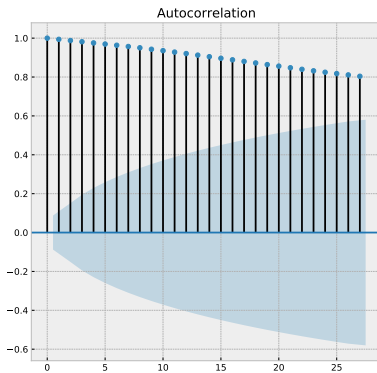
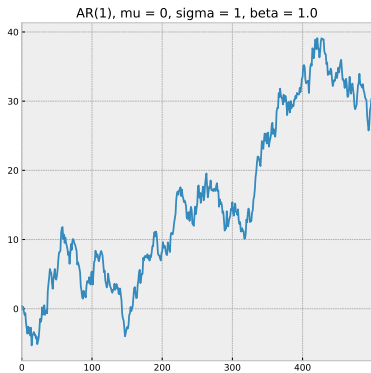
AR(1), $\beta = .75$



AR(1), $\beta = -.75$



AR(1), $\beta = 1$



Financial Time Series Analysis

Lecture 3

- ▶ TA Session Friday April 2, 2021, 11:00am - 12:00pm (EDT)

Outline

- ▶ Autocovariance/autocorrelation functions
 - ▶ Bartlett's formula
 - ▶ Ljung-Box test
- ▶ Basic time series models
 - ▶ Autoregressive Processes (AR)
 - ▶ Partial autocorrelation function (PACF)

Autocorrelation Function: 1

- ▶ The autocovariance and autocorrelation functions are central to time series analysis. They form the basis of **time domain analysis**.
- ▶ The autocovariance and autocorrelations are Fourier transforms of spectral density functions (over frequencies). These form the basis of **frequency domain analysis**.
- ▶ While frequency domain analysis is widely used, especially in engineering applications, it is very infrequently used in economics and finance. We pursue time domain analysis in this course, but the Course Notes will have a section on this approach.

Autocorrelation Function: 2

- ▶ We will want to estimate this function from data.
- ▶ The sample autocovariance function is computed in exactly the same way as a sample covariance function:

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (X_{t+h} - \bar{X})(X_t - \bar{X}).$$

- ▶ Some textbooks (and software packages) use $n - h$ as the divisor rather than n . $n - h$ gives an unbiased estimator.
- ▶ The denominator (n or $n - h$) is a parameter in `statsmodels.tsa.stattools.acovf (unbiased = True)` gives $n - k$, (False gives n), the default is False.
- ▶ Same for `statsmodels.tsa.stattools.acf`
- ▶

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Autocorrelation Function: 3

- ▶ We are concerned with the structure/shape of the autocorrelation/autocorrelation function. The shape can help to identify the underlying process.
- ▶ First, we ask if the autocorrelation is consistent with an independent process with mean 0 and variance σ^2 . That is, at any lag h , we want to test whether $\rho(h) = 0, h \neq 0$ using $\hat{\rho}(h)$.
- ▶ The problem of testing for the presence of autocorrelation at a particular lag, h , was first addressed by M. Bartlett in 1946, and he derived “Bartlett’s formula.” This formula gives the asymptotic behavior of $\hat{\rho}(h)$ for large sample sizes, and it is the source of the parallel lines denoting the 95% confidence intervals in ACF plots.
- ▶ There is a “rule of thumb” concerning the amount of data needed in order to estimate $\rho(h)$. Specifically, $n \geq \max(4h, 50)$.

Bartlett's Formula: 1

Following Brockwell and Davis:

- ▶ Bartlett assumed $X_t - \mu = \sum_{j=-\infty}^{\infty} \psi_j Z_{t-j}$ where $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, $\{Z_t\}$ iid, and $E(Z_t^4) < \infty$, then

$$\sqrt{n}(\hat{\rho}(h) - \rho(h)) \Rightarrow N_h(\mathbf{0}, \mathbf{W}),$$

where

$$\hat{\rho} = (\hat{\rho}(1), \dots, \hat{\rho}(h)),$$

$$\rho = (\rho(1), \dots, \rho(h))$$

and $\mathbf{W}$ is a covariance matrix whose elements are given by

Bartlett's Formula: 2

Bartlett's formula:

$$\begin{aligned}w_{ij} &= \sum_{k=-\infty}^{\infty} (\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j)) \\ &+ 2\rho(i)\rho(j)\rho^2(k) - 2\rho(i)\rho(k)\rho(k+j) \\ &- 2\rho(j)\rho(k)\rho(k+i))\end{aligned}$$

- ▶ For a given model, e.g. AR(1), $\gamma(h)$ can be determined, hence the correlation structure of the empirical correlation coefficients can be determined.
- ▶ For example, suppose, we want to test whether or not $\rho(h) = 0$ if $h \neq 0$, i.e. the process is white noise.

Bartlett's Formula: 3

- ▶ Consider the asymptotic distribution of $\sqrt{n}(\hat{\rho}(h) - 0)$. From Bartlett's formula, $w_{hh} = 1$ ($i = j = 1$). The first term is 1, while all the others are 0. Hence, if the process is white noise,

$$\sqrt{n}(\hat{\rho}(h) - 0) \Rightarrow N(0, 1).$$

- ▶ For a 95% confidence interval, the confidence limits for $\rho(h)$ are $\pm 1.96\sqrt{n}$.
- ▶ The w_{ij} can be calculated for some standard time series models, for example for an AR(1), MA(q) etc. We will display those formulas later.

Confidence Intervals for $\hat{\rho}_k$

- ▶ Suppose we have a AR(1) process with $\rho(h) = \phi_1^{|h|}$.
- ▶ One can apply Bartlett's formula to find the asymptotic distribution for $\hat{\rho}(h)$,

$$\sqrt{n}(\hat{\rho}(h) - \rho(h)) \Rightarrow N(0, w_{hh}),$$

where

$$w_{hh} = 1 + 2 \sum_{i=1}^{h-1} \rho(i)^2.$$

Ljung-Box Test: 1

- ▶ A shortcoming of the Bartlett approach is that we are testing many separate hypotheses simultaneously, checking to see whether the sample autocorrelation plot lies within the upper and lower limit. If so, we conclude that the process (residuals) are white noise.
- ▶ The difficulty with this is that we are conducting simultaneous statistical tests, each at a particular α level (usually .05) and making a simultaneous judgment about all the tests. Instead, we need a test that will keep the experimentwise error rate to the desired level (usually .95).

Ljung-Box Test: 2

- ▶ A simultaneous test of whether a sample autocorrelation function comes from an iid white noise process was first developed by Box and Pierce in 1970. They developed a test for the composite null hypothesis;

$$H_0 : \rho(1) = \rho(2) = \cdots = \rho(m) = 0$$

H_1 : at least one of $\rho(i)$, $1 \leq i \leq m$ is not 0.

- ▶ For a data set of size n , the test statistics was

$$Q^*(m) = n \sum_{i=1}^m \hat{\rho}^2(i).$$

- ▶ Under H_0 the test statistics had an asymptotic χ_m^2 distribution.

Ljung-Box Test: 3

- ▶ In 1978, Ljung and Box published an improved test that had greater power in finite samples. The test statistic was

$$Q(m) = n(n+2) \sum_{i=1}^m \frac{\hat{\rho}_i^2}{n-i},$$

and was tested using the same χ_m^2 distribution.

- ▶ This can be called using Python statsmodels
statsmodels.stats.diagnostic.acorr_ljungbox(x,lags)
with other parameters available.

Ljung-Box Test: 4

IID Normal $N(0,1)$ errors, $n = 1,000$

```
import statsmodels as smm
```

```
smm.stats.diagnostic.acorr_ljungbox(sim, lags = 10)
```

- ▶ (array([0.18728497, 3.07830832, 3.12318292,
4.65185238, 8.69101277, 10.48872061, 11.11405683,
12.85301371, 12.89994, 12.9098376]),
- ▶ array([0.66518598, 0.21456251, 0.37301997,
0.32491944, 0.12204167, 0.10552291, 0.13372072,
0.11700801, 0.16718639, 0.22875669]))

Ljung-Box Test: 5

AR(1), $\phi_1 = .2$, IID $N(0,1)$ errors, $n = 200$

```
import statsmodels as smm
```

```
smm.stats.diagnostic.acorr_ljungbox(x, lags = 10)
```

- ▶ (array([6.64749532, 6.94627588, 7.08514479, 8.27915957, 11.64544475, 11.66216157, 11.66359782, 15.4305607 , 19.57099227, 20.5614858])),
- ▶ (array([0.00992953, 0.03101954, 0.06923287, 0.08187116, 0.03998259, 0.06994412, 0.1121786 , 0.05129471, 0.02075263, 0.02436759]))

Ljung-Box Test: 6

AR(1), $\phi_1 = .2$, IID $N(0,1)$ errors, $n = 100$

```
import statsmodels as smm
```

```
smm.stats.diagnostic.acorr_ljungbox(x, lags = 10)
```

- ▶ array([2.52656454, 4.05288395, 4.52154401, 5.31449319, 7.28748766, 7.51645729, 7.81052663, 7.90728538, 8.36476559, 8.36484189]),
- ▶ array([0.1119437 , 0.13180365, 0.21037664, 0.25652324, 0.20012234, 0.27571037, 0.34959777, 0.44257887, 0.4978452 , 0.5932462])

Autoregressive Processes

Autoregressive Processes: 1

- ▶ An autoregressive process, $\{X_t, t \geq 0\}$, is a stochastic model in which the process is regressed on the lagged values of itself plus an additive noise term, so AR(p) model is given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t,$$

where $\{\epsilon_t, t \geq 0\}$ is a $WN(0, \sigma_\epsilon^2)$.

- ▶ There are a number of issues to address including
 - ▶ Whether the X_t process is weakly stationary,
 - ▶ Determining the moments of the X_t process, especially $\gamma(h)$
 - ▶ Identifying the order of an AR process from data
 - ▶ Parameter estimation
 - ▶ Predicting future values of an AR process.
- ▶ We start with AR(1)

Autoregressive Processes: 2

We begin with the AR(1) process:

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t,$$

where the ϵ_t process is $WN(0, \sigma^2)$.

- We iterate the above expression backward to time 0 (or $-\infty$) depending upon the index set)

$$\begin{aligned} X_t &= \phi_0 + \phi_1 X_{t-1} + \epsilon_t \\ &= \phi_0 + \phi_1(\phi_0 + \phi_1 X_{t-2} + \epsilon_{t-1}) + \epsilon_t \\ &= \phi_0(1 + \phi_1) + \phi_1^2 X_{t-2} + \sum_{i=0}^1 \phi_1^i \epsilon_{t-i} \\ &= \phi_0(1 + \phi_1 + \phi_1^2) + \phi_1^3 X_{t-3} + \sum_{i=0}^2 \phi_1^i \epsilon_{t-i} \\ &= \phi_0 \sum_{i=0}^{t-1} \phi_1^i + \phi_1^t X_0 + \sum_{i=0}^{t-1} \phi_1^i \epsilon_{t-i} \end{aligned}$$

Autoregressive Processes: 3

Assuming the index set is $\{t \geq 0\}$:

$$X_t = \begin{cases} \phi_0 \frac{1-\phi_1^t}{1-\phi_1} + \phi_1^t X_0 + \sum_{i=0}^{t-1} \phi_1^{i-1} \epsilon_{t-i} & \text{if } |\phi_1| < 1, \\ \phi_0 t + X_0 + \sum_{i=0}^t \epsilon_{t-i} & \text{if } |\phi_1| = 1, \\ \phi_0 \frac{\phi_1^t - 1}{\phi_1 - 1} + \phi_1^t X_0 + \sum_{i=0}^{t-1} \phi_1^i \epsilon_{t-i} & \text{if } |\phi_1| > 1, \end{cases}$$

- ▶ It is clear with the above expressions, if $|\phi_1| > 1$, X_t will explode.
- ▶ If $\phi_1 = 1$ then the process is a random walk and the variance will not be constant (see later), then the process will not be stationary.

Autoregressive Processes: 4

- ▶ It is clear that $|\phi_1| < 1$ is a necessary condition for the X_t to be weakly stationary, whether the index set is $\{t \geq 0\}$ or $\{-\infty < t < \infty\}$.
- ▶ For weak stationarity we need the mean and variance to be constant as well as an appropriate covariance function. Defining $E(X_t) = m_t$ and recalling $E(\epsilon_k) = 0$.

$$\begin{aligned}m_t &= \phi_0 + \phi_1 m_{t-1} + 0 \\&= \phi_0(1 + \phi_1) + \phi_1^2 m_{t-2} \\&\dots \\&= \phi_0 \frac{1 - \phi_1^t}{1 - \phi_1} + \phi_1^t m_0\end{aligned}$$

Autoregressive Processes: 5

- ▶ Interestingly, if we take $m_0 = \phi_0/(1 - \phi_1)$, then $m_t = m_0 = \phi_0/(1 - \phi_1)$, $\forall t$.
- ▶ Thus the constant mean property is fulfilled if we initialize $E(X_0) = \phi_0/(1 - \phi_1)$. If we start with a different initial value and $|\phi_1| < 1$, then the process is asymptotically stationary, provided the variance and covariance are appropriate.
- ▶ We can use the same iteration procedure for the variance, Letting $\text{Var}(X_t) = \sigma_t^2$,

Autoregressive Processes: 6

$$\begin{aligned}\sigma_t^2 &= \text{Var}(\phi_0 + \phi_1 X_{t-1} + \epsilon_t) \\ &= \phi_1^2 \sigma_{t-1}^2 + \sigma_\epsilon^2 \\ &= \phi_1^4 \sigma_{t-2}^2 + \sigma_\epsilon^2 (1 + \phi_1^2) \\ &\dots \\ &= \phi_1^{2t} \sigma_0^2 + \sigma_\epsilon^2 \sum_{i=0}^{t-1} \phi_1^{2i}\end{aligned}$$

Thus

$$\sigma_t^2 = \begin{cases} \sigma_\epsilon^2 \frac{1-\phi_1^{2t}}{1-\phi_1^2} + \phi_1^{2t} \sigma_0^2 & \text{if } |\phi_1| < 1, \\ t\sigma_\epsilon^2 + \sigma_0^2 & \text{if } |\phi_1| = 1, \\ \sigma_\epsilon^2 \frac{\phi_1^{2t}-1}{\phi_1^2-1} + \phi_1^{2t} \sigma_0^2 & \text{if } |\phi_1| > 1, \end{cases}$$

Autoregressive Processes: 7

We now assume that $|\phi_1| < 1$, a necessary condition for weak stationarity of an AR(1) process.

- ▶ Interestingly, if we let $\sigma_0^2 = \sigma_\epsilon^2 / (1 - \phi_1^2)$, then

$$\text{Var}(X_t) = \sigma_t^2 = \sigma_\epsilon^2 / (1 - \phi_1^2), \forall t \geq 0.$$

- ▶ We next calculate the autocovariance function for an AR(1) process, assuming $|\phi_1| < 1$, $E(X_0) = \phi_0 / (1 - \phi_1)$ and $\text{Var}(X_0) = \sigma^2 / (1 - \phi_1^2)$

Autoregressive Processes: 8

$$\begin{aligned}\text{Cov}(X_t, X_{t+h}) &= \text{Cov}\left(\sum_{i=0}^t \phi_1^i \epsilon_{t-i}, \sum_{i=0}^{t+h} \phi_1^i \epsilon_{t+h-i}\right) \\&= \text{Cov}\left(\sum_{i=0}^t \phi_1^i \epsilon_{t-i}, \sum_{i=0}^{h-1} \phi_1^i \epsilon_{t+h-i} + \sum_{i=h}^{t+h} \phi_1^i \epsilon_{t+h-i}\right) \\&= \text{Cov}\left(\sum_{i=0}^t \phi_1^i \epsilon_{t-i}, \sum_{i=0}^{h-1} \phi_1^i \epsilon_{t+h-i} + \phi_1^h \sum_{i=0}^t \phi_1^i \epsilon_{t-i}\right) \\&= 0 + \phi_1^h \text{Var}(X_t) \\&= \phi_1^h \frac{\sigma_\epsilon^2}{1 - \phi_1^2}\end{aligned}$$

Autoregressive Processes: 9

We summarize properties of an AR(1) process $\{X_t, t \geq 0\}$

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t,$$

where $\{\epsilon_t, t \geq 0\}$ is $WN(0, \sigma_\epsilon^2)$, $E(X_0) = \phi_0/(1 - \phi_1)$, and $Var(X_0) = \sigma_\epsilon^2/(1 - \phi_1^2)$, $|\phi_1| < 1$

- ▶ The X_t process is weakly stationary if and only if $|\phi_1| < 1$.
- ▶ $E(X_t) = \phi_0/(1 - \phi_1)$,
- ▶ $Var(X_t) = \sigma_\epsilon^2/(1 - \phi_1^2)$, $|\phi_1| < 1$
- ▶ $\gamma(h) = \phi_1^{|h|} \frac{\sigma_\epsilon^2}{1 - \phi_1^2}$, $|\phi_1| < 1$
- ▶ $\rho(h) = \phi_1^{|h|}$, $|\phi_1| < 1$

Autoregressive Processes: 10

- ▶ Note, if either $E(X_0)$ or $Var(X_0)$ differ from the stationary values, but $|\phi_1| < 1$, then the process is not weakly stationary (because the mean and variance are not constant); however, it is **asymptotically stationary**.
- ▶ Sometimes, it is convenient to remove the mean. We can do that by defining $\mu = \phi_0/(1 - \phi_1)$ and then defining $Y_t = X_t - \mu$. The resulting sequence $\{Y_t, t \geq 0\}$ now has mean 0.

Causal Processes

- ▶ As an aside, we have a representation of a weakly stationary process X_t as

$$X_t = \sum_{i=0}^t \phi_1^i \epsilon_{t-i}, \quad t \geq 0, \text{ or}$$

$$X_t = \sum_{i=0}^{\infty} \phi_1^i \epsilon_{t-i}, \quad -\infty < t < \infty$$

- ▶ More generally, if

$$X_t = \sum_{i=0}^{\infty} \psi_i \epsilon_{t-i},$$

satisfying $\sum_{i=0}^{\infty} |\psi_i| < \infty$, then the process is called **causal**.

- ▶ Causality is the same concept as non-anticipate that arises in stochastic calculus, “not looking into the future.”

AR(2) Processes: 1

- ▶ We now consider AR(2) processes:

$$X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t,$$

for a linear regression model whose explanatory (independent) variables are the first two lagged values of the time series.

- ▶ Rather than repeating the iterations as we did before and then determining the mean, variance and autocovariance, we assume that the moment structure is constant and work with the definition of X_t .
- ▶ Taking expectations, and assuming that $E(X_t) = \mu$, $\forall t$, we find $\mu = \phi_0 + \mu(\phi_1 + \phi_2) + 0$, or $\mu = \phi_0 / (1 - \phi_1 - \phi_2)$, assuming $\phi_1 + \phi_2 \neq 1$.

AR(2) Processes: 2

- ▶ By substituting out ϕ_0 and defining $Y_t = X_t - \mu$, we find

$$Y_{t+h} = \phi_1 Y_{t+h-1} + \phi_2 Y_{t+h-2} + \epsilon_{t+h}$$

- ▶ Multiplying each term by (Y_t) and taking expectations, we find

$$\gamma(h) = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}, h > 0$$

because the covariance of ϵ_{t+h} and Y_t is 0 and the means are 0.

- ▶ Setting $h = 1$, recognizing that $\gamma(h) = \gamma(-h)$, and dividing by $\gamma(0)$ to obtain the autocorrelation,

$$\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2),$$

and for $h = 1$, $\rho(1) = \phi_1 \rho(0) + \phi_2 \rho(1)$

AR(2) Processes: 3

- ▶ Since $\rho(0) = 1$, we can find the AR(2) autocorrelation function

$$\rho(h) = \begin{cases} 1 & \text{if } h = 0, \\ \phi_1/(1 - \phi_2) & \text{if } h = 1, \\ \phi_1\rho(h-1) + \phi_2\rho(h-2) & \text{if } h \geq 2. \end{cases}$$

- ▶ If we define the **backshift operator B**, $BX_t = X_{t-1}$, $B^2X_t = X_{t-2}$, $\dots$. Using this, the autocorrelation function satisfies the a second order difference equation:

$$(1 - \phi_1 B - \phi_2 B^2)\rho(h) = 0.$$

- ▶ We will see that the roots of the associated quadratic equation

$$1 - \phi_1 x - \phi_2 x^2 = 0,$$

as being central to conditions on ϕ_1, ϕ_2 to have weak stationarity.

AR(2) Processes: 4

- ▶ Consider an AR(2) process, $\{X_t, -\infty < t < \infty\}$:

$$X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t,$$

where $\{\epsilon_t\}$ is $WN(0, \sigma^2)$.

- ▶ We want to determine conditions under which the X_t process is weakly stationary.
- ▶ We convert this to a vector process for $(X_t, X_{t-1})^T$ using the defining equation give above for the X_t component and $X_t = X_{t-1}$ for the X_{t-1} component.
- ▶ This gives the following representation:

$$\mathbf{X}_t = \boldsymbol{\mu} + \mathbf{M}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t,$$

where $\boldsymbol{\mu} = (\phi_0, 0)^T$, $\boldsymbol{\epsilon}_t = (\epsilon_t, 0)^T$.

AR(2) Processes: 5

Finally,

$$\mathbf{M} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix}.$$

One can iterate backward as before:

$$\begin{aligned} \mathbf{X}_t &= \boldsymbol{\mu} + \mathbf{M}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t \\ &= \boldsymbol{\mu} + \mathbf{M}(\boldsymbol{\mu} + \mathbf{M}\mathbf{X}_{t-2} + \boldsymbol{\epsilon}_{t-1}) + \boldsymbol{\epsilon}_t \\ &= \dots \\ &= \sum_{i=0}^{t-1} \mathbf{M}^i \boldsymbol{\mu} + \mathbf{M}^t \mathbf{X}_0 + \sum_{i=0}^{t-1} \mathbf{M}^i \boldsymbol{\epsilon}_{t-i} \end{aligned}$$

AR(2) Processes: 6

- ▶ To ensure stationarity, we need the matrix $\mathbf{M}^t \rightarrow \mathbf{0}$ as $t \rightarrow \infty$.
- ▶ Consequently, we need the two eigenvalues of this matrix to converge to 0.
- ▶ The eigenvalues are the roots of the determinantal equation:

$$(\phi_1 - \lambda)(0 - \lambda) - (\phi_2)(-1) = 0.$$

- ▶ Writing this as a quadratic equation in λ

$$\lambda^2 - \phi_1\lambda - \phi_2 = 0.$$

- ▶ The roots are easily found as are the conditions for $|\lambda| < 1$.

AR(2) Processes: 7

- ▶ One can introduce a change of variable, $z = \frac{1}{\lambda}$ to give



$$\phi_2 z^2 + \phi_1 z - 1 = 0.$$

- ▶ The left side of this equation is called the **autoregressive polynomial**.
- ▶ Stationarity of the AR(2) process is equivalent to the eigenvalues of **M** having modulus strictly less than 1. This is equivalent to the roots of the autoregressive polynomial having modulus strictly greater than 1.
- ▶ This concept of the magnitude of the eigenvalues being less than 1 and the roots of the autoregressive polynomial having magnitude greater than 1 extends to AR(p) processes.

AR(2) Processes: 8

- ▶ For an AR(2) process, the autoregressive polynomial is a quadratic

$$1 - \phi_1 z - \phi_2 z^2 = 0.$$

- ▶ The roots are

$$\frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}.$$

- ▶ These roots are called the **characteristic roots** of the AR(2) model.
- ▶ For weak stationarity, we need the modulus of these roots to be larger than 1.

Financial Time Series Analysis

Lecture 4

- ▶ TA Session Friday April 2, 2021, 11:00am - 12:00pm (EDT)
- ▶ Homework assignment due dates are being moved to Thursday

Outline

- ▶ Weak Stationarity of $AR(p)$
- ▶ Partial autocorrelation function (PACF)
- ▶ Basic time series models (AR, MA, ARMA)
- ▶ Simulating and fitting AR/MA/ARMA using statsmodels

Weak Stationarity of AR(p): 1

Consider an AR(p) model

$$X_t = \phi_0 + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t,$$

where the ϵ_t process is $WN(0, \sigma^2)$.

► If $E(X_t) = \mu$, $\forall t$, then

$$\mu = \phi_0 / (1 - \sum_{i=1}^p \phi_i), \quad \sum_{i=1}^p \phi_i < 1.$$

► Now consider $Y_t = X_t - \mu$.

Weak Stationarity of AR(p): 2

$$\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \epsilon_t \\ Y_{t-1} &= Y_{t-1} \\ &\cdot \quad \cdot \quad \cdot \\ &\cdot \quad \cdot \quad \cdot \\ Y_{t-p+1} &= Y_{t-p+1} \end{aligned}$$

If we define $\mathbf{Y}_t = (Y_t, \dots, Y_{t-p+1})^T$, $\epsilon_t = (\epsilon_t, 0, \dots, 0)^T$ and

Weak Stationarity of AR(p): 3

$$\mathbf{M} = \begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix},$$

then the resulting AR(p) model now becomes in matrix form

$$\begin{aligned} \mathbf{Y}_t &= \mathbf{M}\mathbf{Y}_{t-1} + \epsilon_t \\ &= \mathbf{M}(\mathbf{M}\mathbf{Y}_{t-1} + \epsilon_{t-1}) + \epsilon_t \\ &\dots \\ &= \mathbf{M}^t\mathbf{Y}_0 + \sum_{i=0}^{t-1} \mathbf{M}^i\epsilon_{t-i}, \end{aligned}$$

Weak Stationarity of AR(p): 4

- ▶ For weak stationary, we need $\mathbf{M}^t \rightarrow 0$ as $t \rightarrow \infty$. This requires that the eigenvalues of $\mathbf{M}$ have modulus strictly less than 1.
- ▶ The eigenvalues can be characterized as the roots of a p-dimensional polynomial

$$\lambda^p - \sum_{i=1}^{p-1} \lambda^{p-i} \phi_i = 0,$$

and the p roots of this equation must have modulus less than 1.

Weak Stationarity of AR(p): 5

- ▶ Alternatively one can define $z = 1/\lambda$ and rewrite the above equation as

$$1 - \sum_{i=1}^p z^i \phi_i = 0.$$

- ▶ This is the autoregressive polynomial, and for weak convergence, its roots have to have modulus greater than 1.

Statsmodels Example: 1

From statsmodels.tsa.arima_process.ArmaProcess

```
import statsmodels.api as sm
```

```
import numpy as np
```

```
arparams = np.array([.75,-.25])
```

```
ar = np.r_[1,-arparams]
```

```
arma_process = sm.tsa.ArmaProcess(ar,ma = None)
```

```
arma_process.isstationary
```

```
»> yes
```

```
arma_process.arroots
```

```
»> array([1.5 -1.32287566j, 1.5 + 1.3287566j])
```

```
y = arma_process.generate_sample(500) model =
```

```
sm.tsa.ARMA(y,(2,0)).fit(trend = 'nc',disp = 0)
```

```
model.params
```

```
»> array([.69085419, -.2715547])
```

Statsmodels Example: 2

```
import statsmodels.api as sm
import numpy as np

from statsmodels.tsa.arima_process import ArmaProcess
from statsmodels.tsa.arima_model import ARMA

import matplotlib.pyplot as plt

# Simulate the AR series
arparams = np.array([.30, .20, .10]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
ar_process = ArmaProcess(ar)
```

Statsmodels Example: 3

```
ar_process.isstationary
```

```
True
```

```
ar_process.arroots
```

```
array([1.6694 - 2.1640j, 1.6694 - 2.1640j, 1.33874])
```

```
#Generate data and Fit the model
```

```
y = ar_process.generate_sample(250)
```

```
model = ARMA(y, (3, 0)).fit(trend='nc', disp=0)
```

```
model.params # Show the parameters
```

```
(0.310, 0.208, 0.150)
```

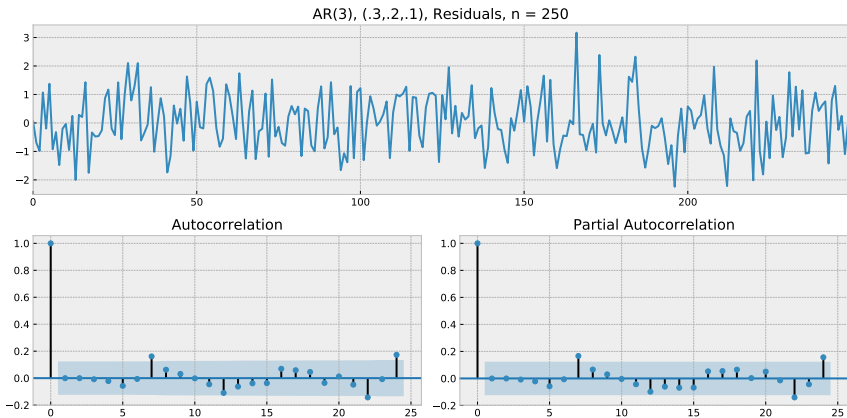
```
plt.plot(model.resid) # Residual Plot
```

```
plt.show()
```

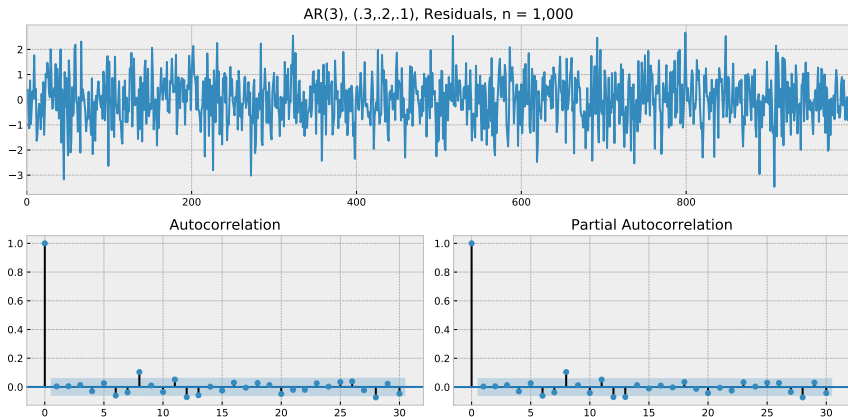
```
plt.acorr(model.resid) #residuals acf plot
```

```
plt.show()
```


Residuals from AR(3), $n = 250$



Residuals from AR(3), $n = 1,000$



Partial Correlation and PACF

Partial Correlation Coefficients: 1

- ▶ In multivariate statistical analysis, one tries to characterize the dependency among a set of variables. For example, consider (X, Y, Z) .
- ▶ The familiar pairwise correlation coefficient, e.g. $\rho(X, Y)$, gives the linearly relationship of X and Y without any consideration of Z .
- ▶ For example, one could start with the joint distribution, $f_{X,Y,Z}(x, y, z)$, integrate out Z leaving $f_{X,Y}(x, y)$ and then compute $\rho(X, Y)$.
- ▶ However, we might want to first eliminate any effects that Z may have on X and Y , and then compute the correlation. This is known as the **partial correlation coefficient**.

Partial Correlation Coefficients: 2

- ▶ The partial correlation of $\rho(X, Y|Z)$ can be found by regressing X on Z and Y on Z separately, then computing the correlation coefficient of the residuals from each of those regressions.
- ▶ Note that Z could be set of random variables, i.e. $\mathbf{Z} = (Z_1, \dots, Z_m)$. One would again regress X and Y on $\mathbf{Z}$ and then find the sample correlation of the two residuals from the regressions.

Partial Autocorrelation Function (PACF): 1

- ▶ The concept of partial correlation can be useful in determining the order of a time series model, especially in the AR process case.
- ▶ Suppose that the data came from a true autoregressive process with order p . We fit an autoregressive model but want to estimate the order.
- ▶ If the true model has order p , then

$$X_t = \mu + \beta_1 X_{t-1} + \cdots + \beta_p X_{t-p} + \epsilon_t.$$

- ▶ The model indicates that the correlation of X_{t-k} and X_t will be 0 conditional on $X_{t-1}, \dots, X_{t-p}$ for $k > p$ whereas these will be correlated if $k < p$.
- ▶ How can we use this to devise a method to determine the order of an autoregressive process?

Partial Autocorrelation Function (PACF): 2

- ▶ Suppose the true order of the AR process is p for some $p \geq 1$.
- ▶ The k th partial autocorrelation, $\phi_{k,k}$, is estimated by $\hat{\phi}_{k,k}$ in the regression

$$X_t = \phi_{0,k} + \phi_{1,k}X_{t-1} + \cdots + \phi_{k,k}X_{t-k} + \epsilon_{k,t}.$$

- ▶ Note that we contemplate running a series of regressions of the type above as we increase $k = 2, 3, \dots$. The $\epsilon_{k,t}$ have the same structure as the errors in the actual autoregressive process but are different, as are the regression coefficients.
- ▶ If X_t is an AR(p) process, then $\phi_{k,k} = 0$ for $k > p$. Testing that $\phi_{k,k} = 0$ can be used to determine the value of p .

Partial Autocorrelation Function (PACF): 3

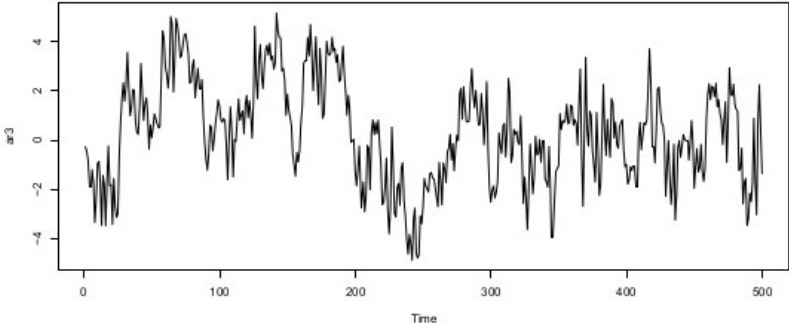
- ▶ Thus, given time series data from an AR(p) process, $\{X_t, 0 \leq t \leq T\}$ One can consider a series of regressions of the form

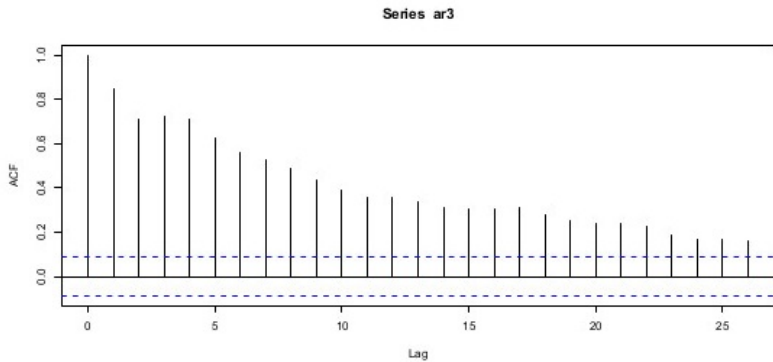
$$X_t = \phi_{0,k} + \phi_{1,k}X_{t-1} + \cdots + \phi_{k,k}X_{t-k} + \epsilon_{k,t},$$

each time testing whether $\phi_{k,k} = 0$.

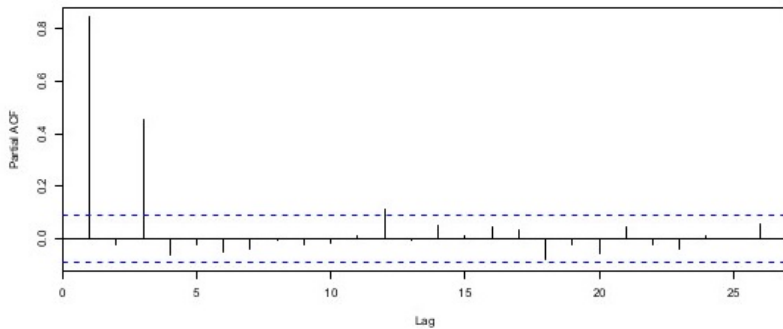
- ▶ The smallest k for which the test allows us to conclude $\phi_{k,k} = 0$ will then give us our estimated order for the process.
- ▶ More generally, one can use the AIC (Akaike Information Criterion) or the BIC (Bayesian Information criterion) based on the log likelihood to determine the order of the autoregression.

AR(3) series

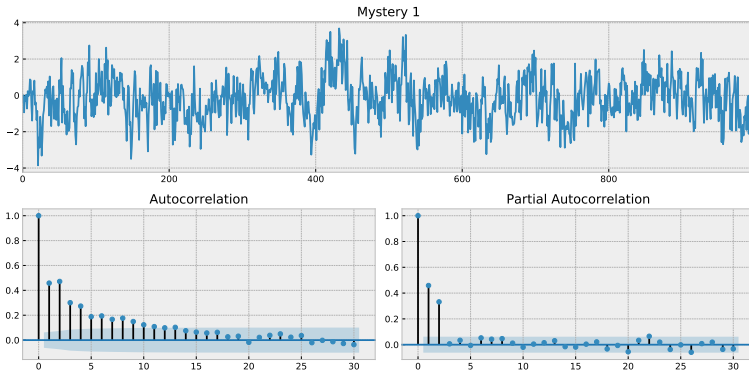




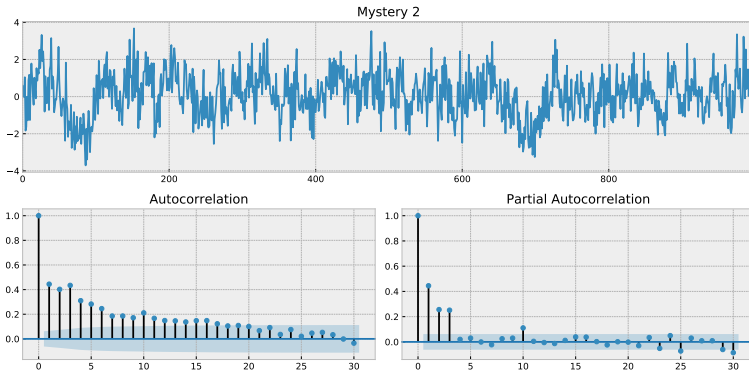
Series ar3



Mystery 1



Mystery 2



Parameter Estimation for AR(p): 1

Starting with an AR(p) process

$$X_t = \sum_{i=1}^p \beta_i X_{t-i} + \epsilon_t,$$

and writing it in vector form

$$\mathbf{X}_t = \boldsymbol{\beta} \mathbf{X}_{t-1} + \epsilon_t,$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$ and $\mathbf{X}_{t-1} = (X_{t-1}, \dots, X_{t-p})^T$

- ▶ The equation has a strong resemblance to the familiar regression equation.

Parameter Estimation for AR(p): 2

- ▶ The least squares estimator of β is given by

$$\hat{\beta} = \left(\sum_{t=p+1}^n \mathbf{x}_{t-1} \mathbf{x}_{t-1}^T \right)^{-1} \left(\sum_{t=p+1}^n \mathbf{x}_{t-1} x_t \right).$$

- ▶ In the simple case with $p = 1$,

$$\hat{\beta} = \sum_{t=p+1}^n x_t x_{t-1} / \sum_{t=p+1}^n x_{t-1}^2.$$

- ▶ Most of the standard regression methodology carries over to this autoregressive case, e.g. the asymptotic normality of $\hat{\beta}$ and the asymptotic variance is a function of σ_ϵ and $\gamma(h)$.

Moving Average Processes

Moving Average Processes: 1

- ▶ A moving average process (MA(q)) has the following representation:

$$X_t = \mu + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t,$$

where $\{\epsilon_t, -\infty < t < \infty\}$ is a white noise process.

- ▶ For all $(\theta_1, \dots, \theta_q)$, the $\{X_t, -\infty < t < \infty\}$ process is weakly stationary.
- ▶ Taking expectations across the defining equation, we find $E(X_t) = \mu + \sum_{i=1}^q \theta_i E(\epsilon_{t-i}) + E(\epsilon_t) = \mu + 0 = \mu$.
- ▶ Similarly, computing the variance of each side, $Var(X_t) = \sum_{i=1}^q \theta_i^2 \sigma_\epsilon^2 + \sigma_\epsilon^2 = \sigma_\epsilon^2 (1 + \sum_{i=1}^q \theta_i^2), \forall t$.

Moving Average Processes: 2

- ▶ Thus the MA(q) process has constant mean and variance. Now we must evaluate the autocovariance function. Ignoring the constant μ :

$$\text{Cov}(X_{t+k}, X_t) = \text{Cov}\left(\sum_{i=1}^q \theta_i \epsilon_{t+k-i} + \epsilon_{t+k}, \sum_{j=1}^q \epsilon_{t-j} + \epsilon_t\right)$$

- ▶ X_{t+k} involves $(\epsilon_{t+k}, \dots, \epsilon_{t+k-q})$ while
- ▶ X_t involves $(\epsilon_t, \dots, \epsilon_{t-q})$. If $k > q$
- ▶ If $0 < k \leq q$, then $\gamma(k) = \sigma_\epsilon^2 \sum_{i=0}^{q-k} \theta_i \theta_{i+k}$ which is independent of t , hence an MA(q) is weakly stationary.

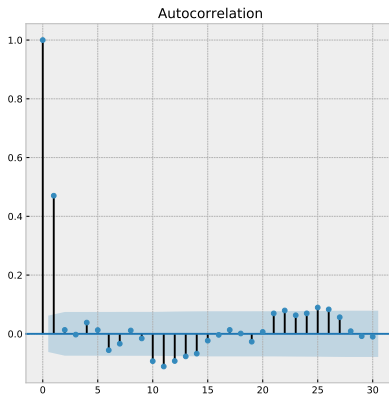
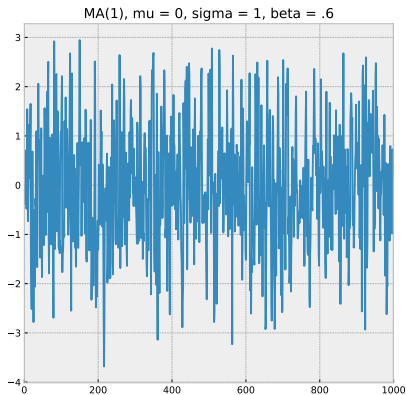
Moving Average Processes: 3

We summarize the properties of an MA(q) process:

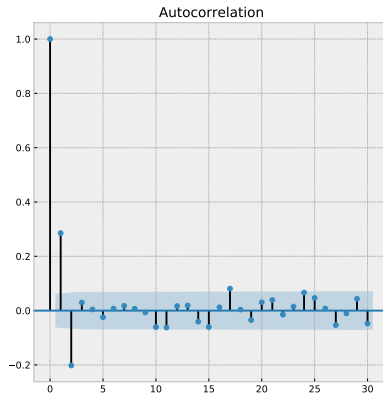
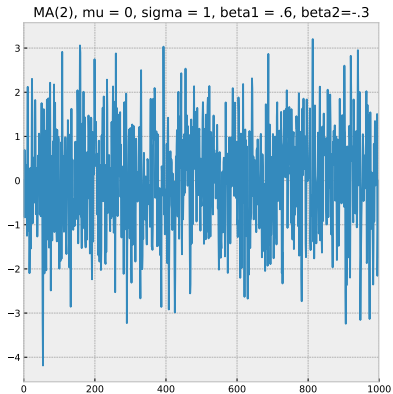
- ▶ Weakly stationary for all θ_i , $1 \leq i \leq q$,
- ▶ $E(X_t) = 0$,
- ▶ $\text{Var}(X_t) = \sigma_\epsilon^2(1 + \sum_{i=1}^q \theta_i^2)$, $\forall t$.
- ▶

$$\gamma(h) = \begin{cases} \sigma_\epsilon^2 \sum_{i=0}^{q-|h|} \theta_i \theta_{i+|h|} & \text{if } q \leq |h|, \\ 0 & \text{if } q > |h|. \end{cases}$$

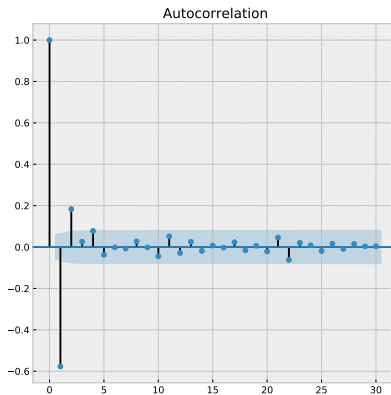
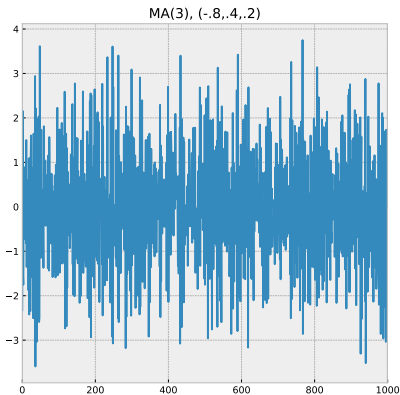
MA(1), (.6)



MA(2), (.6, -.3)



MA(3), (-.8, .4, .2)



MA(q) Analysis

```
# Simulate an MA series
maparams = np.array([.50, -.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
ma_process = ArmaProcess(ma = ma)

ma_process.isstationary
True

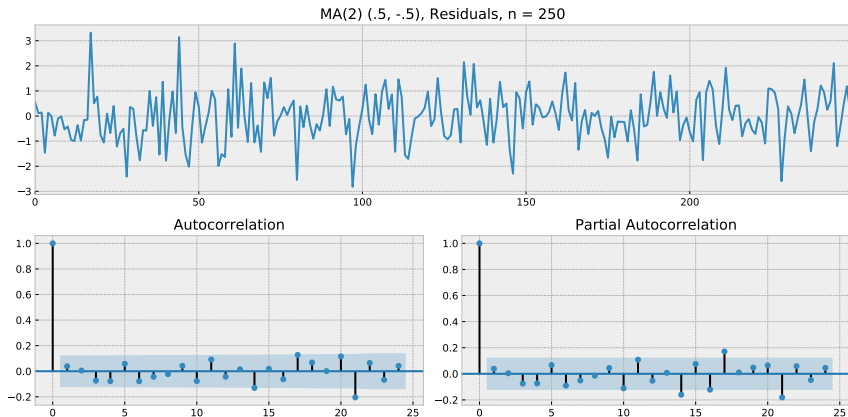
ma_process.isinvertible
True

y = ma_process.generate_sample(250) # Fit the model
model = ARMA(y, (0, 2)).fit(trend='nc', disp=0)

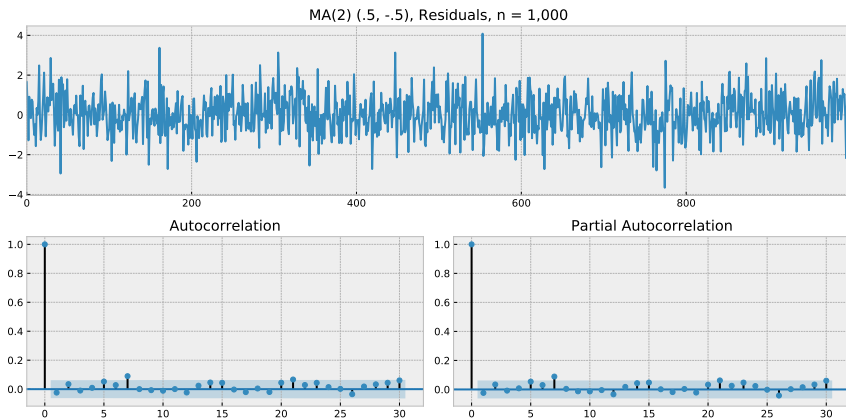
model.params # Show the parameters
array([0.402, -.599]) # n = 250
array([0.504, -.496]) # n = 1,000

plt.plot(model.resid) # Residual Plot
plt.show()
```

Residuals from MA(2), $n = 250$



Residuals from MA(2), n= 1,000



Financial Time Series Analysis

Lecture 5

- ▶ TA Session Friday April 9, 2021, 11:00am - 12:00pm (EDT)
- ▶ Homework assignment due dates are being moved to Thursday

Outline

- ▶ Homework #2, problem 3
- ▶ Formal notation for AR, MA, ARMA, ARIMA
- ▶ Properties of ARMA(p,q)
- ▶ Dickey Fuller test

Invertibility of MA Processes: 1

- ▶ We have determined conditions on the coefficients of an AR(p) process for weak stationarity, namely the roots of the autoregressive polynomial, $1 - \sum_{i=1}^p \phi_i x^i = 0$ have modulus greater than 1.
- ▶ There is a similar condition for the coefficients of the MA(q) process. We learned that all MA(q) processes are weakly stationary.
- ▶ Consider two MA(1) processes:

$$Y_t^{(1)} = \epsilon_t - \theta_1 \epsilon_{t-1},$$

$$Y_t^{(2)} = \epsilon_t - \frac{1}{\theta_1} \epsilon_{t-1}$$

- ▶ $E(Y_t^{(i)}) = 0$ for $i = 1, 2$. While the variances are different,

Invertibility of MA Processes: 2

both processes have identical acfs, $i = 1, 2$:

$$\rho_i(h) = \begin{cases} 1 & \text{if } h = 0, \\ -\theta_1/(1 + \theta_1^2) & \text{if } |h| = 1, \\ 0 & \text{if } |h| > 1. \end{cases}$$

From the point of view of the mean, variance and autocovariance, these two processes are identical. Which parametrization is preferable? We can express the residual, ϵ_t , by iterating backward:

$$\begin{aligned} \epsilon_t &= Y_t^{(1)} + \theta_1 \epsilon_{t-1} \\ &= Y_t^{(1)} + \theta_1 (Y_{t-1}^{(1)} + \theta_1 Y_{t-2}^{(1)}) \\ &= \dots \\ &= Y_t^{(1)} + \sum_{i=1}^{\infty} \theta_1^i Y_{t-i}^{(1)}. \end{aligned}$$

Invertibility of MA Processes: 3

Similarly

$$\epsilon_t = Y_t^{(2)} + \sum_{i=1}^{\infty} \frac{1}{\theta_1^i} Y_{t-i}^{(2)}$$

- ▶ Now if $|\theta_1| < 1$ then the coefficients involving $Y_t^{(1)}$ are a convergent series; however, the coefficients involving $Y_t^{(2)}$ are a divergent series.
- ▶ The MA(q) process is said to be **invertible** if the residuals can be represented by an AR process with convergent coefficients.
- ▶ This is equivalent to the roots of $\theta(B) = 0$ having all roots with modulus greater than 1, i.e.

$$1 - \theta_1 z - \theta_2 z^2 - \dots - \theta_q z^q = 0.$$

Formal Notation for AR Processes

There is a formal representation for AR(p) processes, $\{X_t\}$.
Defining $Y_t = X_t - \mu$ the process satisfies

$$Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \epsilon_t.$$

This can be represented as

$$\phi(B)Y_t = \epsilon_t,$$

where

$$\phi(B) = (1 - \phi_1 B - \cdots - \phi_p B^p).$$

```
arparams = np.array([.30, .20, .10]) #Ar coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
ar_process = ArmaProcess(ar)
```

Formal Notation for MA Processes

Let X_t be a MA(q) with mean μ and let $Y_t = X_t - \mu$, to eliminate the mean. This can be written as

$$Y_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}.$$

This can be concisely symbolized as

$$Y_t = \theta(B)\epsilon_t,$$

where

$$\theta(B) = (1 + \theta_1 B + \cdots + \theta_q B^q).$$

```
maparams = np.array([.50,-.50]) # MA coefficients  
ma = np.r_[1, maparams] # add zero-lag
```


Formal Notation for ARMA(p,q)

For an ARMA(p,q) model given by

$$Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p} = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q},$$

we can express this using the backshift notation as

$$\phi(B)Y_t = \theta(B)\epsilon_t.$$

```
arparams = np.array([.60, -.40]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
maparams = np.array([.50, -.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
arma_process = ArmaProcess(ar, ma)
```

Formal Notation for ARIMA(p,d,q)

A process X_t is an ARMA(p,q) process if

- ▶ X_t is weakly stationary,
- ▶ $\phi(B)X_t = \theta(B)\epsilon_t, \forall t$ where ϵ_t is $WN(0, \sigma^2)$
- ▶ A process X_t is an ARMA(p,q) process with mean μ if Y_t is an ARMA(p,q) process where $Y_t = X_t - \mu$.
- ▶ Suppose W_t is an ARMA(p,q) process, so $\phi(B)W_t = \theta(B)\epsilon_t, \forall t$ where ϵ_t is $WN(0, \sigma^2)$.
- ▶ Now suppose $W_t = (1 - B)^d Y_t$, then Y_t is an ARIMA(p,d,q) process. This can be written in the notation

$$\phi(B)(1 - B)^d Y_t = \theta(B)\epsilon_t.$$

ARMA Models

ARMA Models: 1

- ▶ Last lecture we introduced and described the properties of two standard time series models::

$$AR(p) : X_t = \phi_0 + \phi_1 X_{t-1} + \cdots \phi_p X_{t-p} + \epsilon_t.$$

$$MA(q) : X_t = \mu + \theta_1 \epsilon_{t-1} + \cdots \theta_q \epsilon_{t-q} + \epsilon_t.$$

- ▶ One problem is that fitting data, especially financial data, with either a pure AR(p) model or pure MA(q) model, may require a relatively large number of parameters..
- ▶ One potential solution is to combine both into a single model, ARMA(p,q) defined by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \cdots \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}.$$

ARMA Models: 2

- ▶ For stationarity, the mean and variance must be constant. Taking expectations on both sides of the equation, we find

$$E(X_t) = \mu = \phi_0 + (\phi_1 + \cdots + \phi_p)\mu$$

or

$$E(X_t) = \mu = \frac{\phi_0}{1 - (\phi_1 + \cdots + \phi_p)}.$$

- ▶ To simplify, we set $\mu = 0$, which can be achieved by simply substituting $Y_t = X_t - \mu$.
- ▶ While this model has $p+q$ parameters as opposed to just p or q , it is often the case that ARMA(1,1), a two-parameter model, can fit the data quite well.

ARMA(1,1): 1

The ARMA(1,1) model with $E(X_t) = \mu$ is given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t,$$

with ϵ_t is a $WN(0, \sigma_\epsilon^2)$ process.

- ▶ Assuming stationarity and taking expectations of both sides, we find $E(X_t) = \mu = \phi_0 + \phi_1 E(X_{t-1})$ or $\mu = \phi_0 / (1 - \phi_1)$
- ▶ To compute the variance and autocovariance, we first note

$$\begin{aligned} \text{Cov}(X_t, \epsilon_t) &= \text{Cov}(\phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t, \epsilon_t) \\ &= 0 + 0 + 0 + \sigma_\epsilon^2 \\ &= \sigma_\epsilon^2 \end{aligned}$$

because the first three terms are uncorrelated with ϵ_t .

ARMA(1,1): 2

Assuming weak stationarity, we have constant variances, σ_X^2 , and:

$$\begin{aligned}\sigma_X^2 &= \text{Var}(\phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t) \\ &= \phi_1^2 \sigma_X^2 + \theta_1^2 \sigma_\epsilon^2 + \sigma_\epsilon^2 + 2\text{Cov}(\phi_1 X_{t-1}, \theta_1 \epsilon_{t-1}) \\ &= \phi_1^2 \sigma_X^2 + (1 + \theta^2 + 2\phi_1 \theta_1) \sigma_\epsilon^2\end{aligned}$$

Solving for σ_X^2 , we find

$$\sigma_X^2 = \frac{(1 + \theta^2 + 2\phi_1 \theta_1) \sigma_\epsilon^2}{1 - \phi_1^2}.$$

ARMA(1,1): 3

- Having found the mean and variance of the ARMA(1,1), we move on to the autocovariance and autocorrelation functions. $\gamma(h)$ is found by multiplying the defining equation for the ARMA(1,1) process by X_{t-h} and taking expectations on both sides of the equation. The autocorrelation is then found by dividing by σ_X^2 . It is easiest to set $\mu = \theta_0 = 0$ as the constant term will not change $\gamma(h)$. First, $\gamma(1) = E(X_t X_{t-1})$, and

$$\begin{aligned} E(X_t X_{t-1}) &= \phi_1 E(X_{t-1}^2) + \theta_1 E(X_{t-1} \epsilon_{t-1}) + E(X_{t-1} \epsilon_t) \\ &= \phi_1 \sigma_X^2 + \theta_1 \sigma_\epsilon^2 + 0 \\ &= \frac{(\phi_1(1 + \theta_1^2 + 2\phi_1\theta_1) + \theta_1(1 - \phi_1^2))\sigma_\epsilon^2}{1 - \phi_1^2} \\ &= \frac{(\phi_1 + \theta_1)(1 + \phi_1\theta_1)\sigma_\epsilon^2}{1 - \phi_1^2} \end{aligned}$$

ARMA(1,1): 4

We divide by σ_X^2 to find $\rho(1)$

- ▶ For $h = 1$, the result is

$$\rho(1) = \frac{(1 + \phi_1\theta_1)(\phi_1 + \theta_1)}{1 + \theta_1^2 + 2\phi_1\theta_1}.$$

- ▶ For $h \geq 1$, the calculation is simple. Recall that for an MA(q) process, the autocorrelation function is 0 when $h > q$. In this case, the MA parameter is 1, so the autocorrelation function will have the same geometric decay that we observed in the AR(1) process.

ARMA(1,1): 5

Specifically, for $h \geq 2$, multiplying by x_{t-h} and taking expectations:

$$\gamma(h) = E(\phi_1 X_{t-1} X_{t-h} + \theta_1 X_{t-h} \epsilon_{t-1} + X_{t-h} \epsilon_t) = \phi_1 \gamma(h-1)$$

and

$$\rho(h) = \phi_1 \rho(h-1), \quad h \geq 2.$$

► Having found $\rho(1)$, we conclude that for $h \geq 2$,

$$\rho(h) = \phi_1^{h-1} \rho(1).$$

ARMA(p,q) Analysis: 1

```
# Simulate the ARMA series
arparams = np.array([.60, -.40]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
maparams = np.array([.50, -.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
arma_process = ArmaProcess(ar, ma)

arma_process.isstationary
True

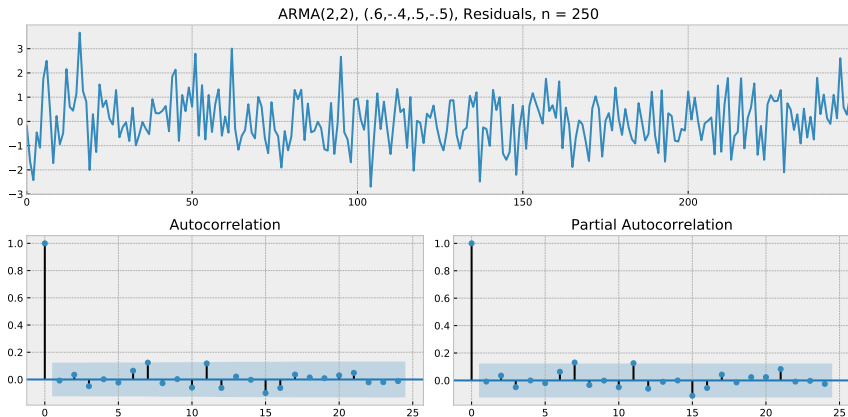
arma_process.isinvertible
True
```

ARMA(p,q) Analysis: 2

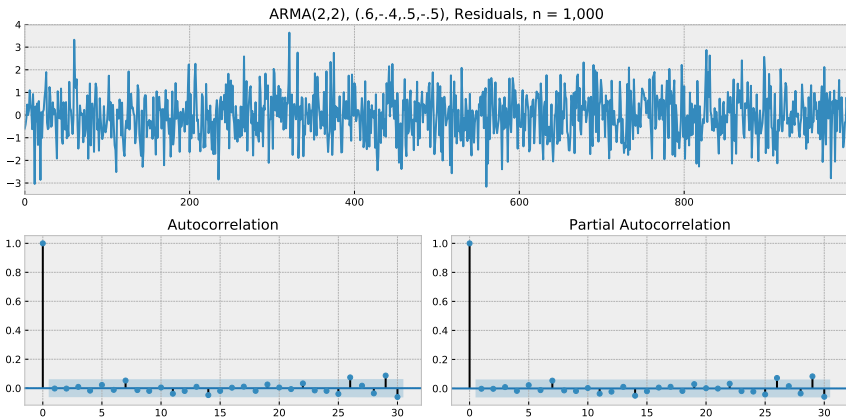
```
# Fit the model
y = arma_process.generate_sample(250)
model = ARMA(y, (2, 2)).fit(trend='nc', disp=0)
array([ 0.657,-.305, .377, -.622]) #n = 250
array([.657,-.402,465,-.535]) # n = 1,000

print(model.params)
tsplot(model.resid)
plt.show()
```

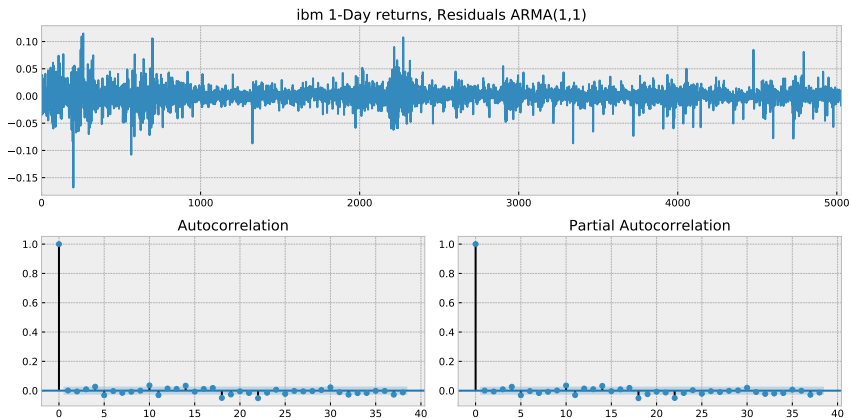
Residuals from ARMA(2,2), $n = 250$



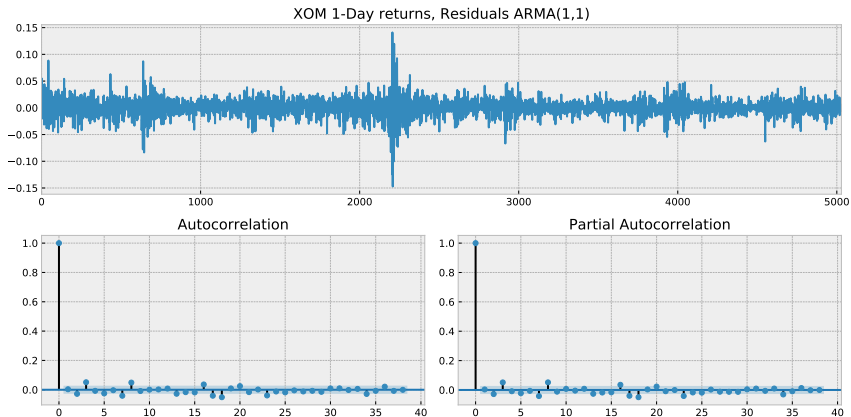
Residuals from ARMA(2,2), $n = 1,000$



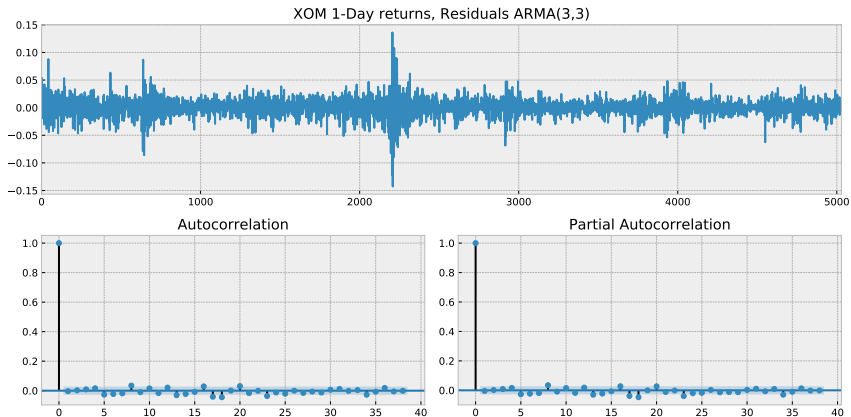
IBM One-Day Log Returns, ARMA(1,1) Residuals



XOM One-Day Log Returns, ARMA(1,1) Residuals



XOM One-Day Log Returns, ARMA(3,3) Residuals



ARIMA and Differencing: 1

- ▶ There is yet a broader class of basic time series models, the ARIMA(p,d,q) models. Here the “i” in ARIMA stands for “integrated,” and the middle parameter, d, stands for “difference.”
- ▶ Recall problem #1 on Homework #2, that differencing a weakly stationary time series yields a new time series that is also weakly stationary.
- ▶ It is also possible that differencing a non-weakly stationary times series might transform it to become weakly stationary.

Arima and Differencing: 2

- ▶ A simple example of this is a random walk model. Suppose we begin with $\{\epsilon_t, t > 0\}$, a $WN(0, \sigma^2)$, and we create an AR(1) sequence with $\phi_1 = 1$

$$X_t = X_{t-1} + \epsilon_t.$$

- ▶ This is not a weakly stationary sequence as the characteristic polynomial has a root with modulus 1 (a unit root).
- ▶ Now consider the first difference sequence, $Y_t = X_t - X_{t-1}$. Note the Y_t is just the ϵ_t sequence, which is weakly stationary by assumption.

Arima and Differencing: 3

- ▶ This simple example shows that taking differences of non-stationary time series is one approach to transforming a non-weakly stationary series into a weakly stationary one.
- ▶ Of course this works in only one direction, doing the opposite of differencing (integrating) can turn an otherwise weakly time series into a non-stationary one.
- ▶ Consequently, the d parameter indicates how many differences need to be applied in order to transform the time series so it is weakly stationary.

Differencing to obtain Stationarity: 1

- ▶ We introduce the “Backshift” or “lag” operator, B , and the differencing operator, ∇ , defined by:

$$B^k(X_t) = B^k X_t = X_{t-k}, \quad k = 1, 2, \dots,$$

$$\nabla X_t = X_t - X_{t-1} = X_t - B X_t = (1 - B) X_t.$$

- ▶ The last formula can be iterated to give

$$\nabla^k X_t = (1 - B)^k X_t.$$

- ▶ Note

$$\nabla^2 X_t = (1 - 2X_{t-1} + X_{t-2}).$$

Differencing to obtain Stationarity: 2

- ▶ Definition: A univariate time series $\{X_t\}$ is **integrated of order k** (denoted by $\mathcal{I}(k)$) if $\nabla^{k-1}X_t$ is non-stationary (hence $\nabla^{k-i}X_t$ is non-stationary for $i = 1, \dots, k-1$); however $\nabla^k X_t$ is stationary.
- ▶ An m -dimensional time series is **integrated of order k** (denoted $\mathcal{I}(k)$), if at least one of its coordinates is $\mathcal{I}(k)$ and all the others are $\mathcal{I}(j)$ for some $j \leq k$.

Testing for Stationarity

Dickey-Fuller Test

- ▶ Begin with the AR(1) process, $X_t = \phi_1 X_{t-1} + \epsilon_t$, with ϵ_t assumed to be a stationary process. The $\{X_t\}$ process is stationary if and only if $|\phi_1| < 1$.
- ▶ Test: $H_0 : |\phi_1| = 1$ versus $H_1 : |\phi_1| < 1$.
- ▶ There are two approaches:
 1. Regress X_t on X_{t-1} and set up a “t-statistic,”
 2. Subtract X_{t-1} from both sides, so $\nabla X_t = (\phi_1 - 1)X_{t-1} + \epsilon_t$, regress ∇X_t on X_{t-1} , and set up a “t-statistic”.

Dickey-Fuller Test: 2

- ▶ Both approaches give identical estimates of β and identical “t-statistics”. To test for the unit root, we need the distribution of the test statistics under $H_0 : \phi_1 = 1$. In this case, the time series behaves like a random walk, or for long series, when properly scaled like a Wiener process. This gives rise to the Dickey-Fuller test.

Dickey-Fuller Test: 3

$$\hat{\phi}_1 = \frac{\sum_{t=1}^T X_{t-1} X_t}{\sum_{t=1}^T X_{t-1}^2}$$

$$\hat{\phi}_1 - 1 = \frac{\sum_{t=1}^T X_{t-1} (X_t - X_{t-1})}{\sum_{t=1}^T X_{t-1}^2}$$

$$s_T^2 = \frac{1}{T-1} \sum_{t=1}^T (X_t - \hat{\phi}_1 X_{t-1})^2$$

$$t_T = \frac{\hat{\phi}_1 - 1}{s_T / \sqrt{\sum_{t=1}^T X_{t-1}^2}}$$

Under H_0 , $\phi_1 = 1$ and the time series is a random walk. For large samples, both numerator and denominator can be approximated using central limit theorem ideas.

Dickey-Fuller Test: 4

$$\phi_1 = 1, T \rightarrow \infty$$

$$Y_t = \frac{X_t}{\sigma\sqrt{T}} \Rightarrow W\left(\frac{t}{T}\right) = W(s), \quad s = \frac{t}{T}, \quad 0 \leq s \leq 1.$$

$$T(\hat{\phi}_1 - 1) = \frac{\sum_{t=1}^T Y_{t-1}(Y_t - Y_{t-1})}{\sum_{t=1}^T Y_{t-1}^2 \frac{1}{T}} \Rightarrow \frac{\int_0^1 W(s) dW(s)}{\int_0^1 W^2(s) ds}.$$

$$t_T = \frac{\hat{\phi}_1 - 1}{s_T / \sqrt{\sum_{t=1}^T X_{t-1}^2}}$$

$$t_T \Rightarrow \frac{\int_0^1 W(s) dW(s)}{\sqrt{\int_0^1 W^2(s) ds}} = \frac{\frac{1}{2}(W^2(1) - 1)}{\sqrt{\int_0^1 W^2(s) ds}}.$$

Dickey Fuller Test: 5

- ▶ The limiting probability distribution is different from the t-distribution. For example $P(t_T < 0) = P(W^2(1) < 1) = P(-1 < Z < 1) = .68$, so the distribution is not symmetric, unlike the t-distribution.
- ▶ This distribution is tabled, and used in statistical packages. Remember that it is derived under certain specific assumptions. The analysis is correct for $T \rightarrow \infty$ as long as the ϵ_t has finite second moments. However, in smaller samples and heavy tailed distributions, the distribution will not be correct.
- ▶ Can use Monte Carlo methods to simulate the distribution of t_T under any specific model.

Dickey Fuller Test: 6

```
import numpy as np
import statsmodel.tsa.api as smt

n = 1000
phi0 = 0
phi1 = 1
sigma = 1
z = sigma*np.random.randn(n)
x[0] = sigma*z[0]
for i in range(n-1):
     $x[i + 1] = \phi_0 + \phi_1 * x[i] + z[i + 1]$ 
smt.stattools.adfuller(x)
(-0.96, 0.77, 0, 999, ('1%': -3.44, '5%': -2.86, '%10%': -2.57))
smt.stattools.adfuller(z)
(-23.75, 0.0, 1, 998, ('1%': -3.44, '5%': -2.86, '%10%': -2.57))
```

Financial Time Series Analysis

Lecture 6

- ▶ TA Session Friday April 9, 2021, 11:00am - 12:00pm (EDT)

Outline

- ▶ Parameter Estimation
 - ▶ Least Squares
 - ▶ Yule Walker Equations
 - ▶ Likelihood Methods
- ▶ Statsmodels fitting ARMA Models
- ▶ Forecasting

Parameter Estimation: 1

- ▶ We begin with the estimation of autoregressive models. If we have assumptions about the distribution of ϵ_t , then we can use standard maximum likelihood methods. On the other hand, if the error process is only assumed to be $WN(0, \sigma^2)$, then we need to fall back on moment methods.
- ▶ We want to assume we are working with a weakly stationary model
- ▶ The standard autoregressive model for Y_t with $E(Y_t) = 0$ can be written in vector form as

$$Y_t = \mathbf{Y}_{t-1}^T \phi + \epsilon_t,$$

where $\mathbf{Y}_{t-1} = (Y_{t-1}, \dots, Y_{t-p})^T$ and $\phi = (\phi_1, \dots, \phi_p)^T$.

Parameter Estimation: 2

- ▶ Putting this together with data $(Y_1, \dots, Y_n)$, we can find the least squares solution for ϕ :

$$\hat{\phi} = \left(\sum_{t=p+1}^n \mathbf{Y}_{t-1} \mathbf{Y}_{t-1}^T \right)^{-1} \left(\sum_{t=p+1}^n \mathbf{Y}_{t-1} Y_t \right).$$

- ▶ If the ϵ_t sequence is iid $N(0, \sigma^2)$, then the estimator is also the MLE.
- ▶ In the simple case that $p = 1$, the formula simplifies to

$$\hat{\phi} = \sum_{t=1}^{n-1} Y_t Y_{t+1} / \sum_{t=1}^{n-1} Y_t^2.$$

Parameter Estimation: 3

- ▶ The least squares estimators do not make any assumptions about the error distribution, yet given that the errors are white noise, we can work out the asymptotic distribution of those estimators. In particular they are asymptotically normal:

$$\sqrt{n}(\hat{\phi} - \phi) \Rightarrow N_p(\mathbf{0}, \sigma_\epsilon^2 \mathbf{\Gamma}_p^{-1}),$$

where

$$\mathbf{\Gamma}_p = E(\mathbf{Y}^T \mathbf{Y}),$$

with $\mathbf{Y} = (Y_1, \dots, Y_p)$.

- ▶ The (ij) element of this matrix is $E(Y_i Y_j) = \gamma(i - j)$.

Parameter Estimation: 4

- ▶ For $p=1$, the AR(1) model, we have

$$\Gamma_1 = (\gamma(0)) = \sigma_\epsilon^2 / (1 - \phi_1^2).$$

- ▶ This gives

$$\sqrt{n}(\hat{\phi}_1 - \phi_1) \Rightarrow N(0, \sigma_\epsilon^2 / \sigma_\epsilon^2 / (1 - \phi_1^2)) = N(0, (1 - \phi_1^2)).$$

- ▶ Approximate confidence intervals can be generated and hypothesis tests conducted about the parameter ϕ_1 or the parameters ϕ using methods based on the normal distribution.
- ▶ We next turn to the familiar Yule-Walker (YW) equations approach.

Yule Walker Equations: 1

- ▶ From Homework #2 students are familiar with the YW equations.
- ▶ Multiply the standard autoregressive equation, $\phi(B)Y_t = \epsilon_t$ by Y_{t-k} and take expectations. This leads to a set of linear equations for the autocovariance values, and if one normalizes by dividing by γ_0 , we get linear equations for the autocorrelations. Specifically, we get for $1 \leq k \leq p$:

$$\gamma(k) = \phi_1\gamma(k-1) + \cdots + \phi_p\gamma(k-p),$$

$$\rho(k) = \phi_1\rho(k-1) + \cdots + \phi_p\rho(k-p).$$

Yule Walker Equations: 2

- ▶ If we let $\boldsymbol{\rho} = (\rho(1), \dots, \rho(p))^T$ and $\mathbf{R}$ be a $p \times p$ matrix whose ij element is $\rho(i - j)$, then

$$\boldsymbol{\rho} = \mathbf{R}\boldsymbol{\phi}.$$

- ▶ We want to use these equations, plugging in estimates of $\boldsymbol{\rho}$ to generate an estimate of $\boldsymbol{\phi}$.
- ▶ Replacing $\rho(k)$ by $\hat{\rho}(k) = r_k$, $1 \leq k \leq p$ (of course $\rho(0) = \hat{\rho}(0) = 1$) we have the YW estimates for $\boldsymbol{\phi}$:

$$\hat{\boldsymbol{\phi}} = \mathbf{R}^{-1}\mathbf{r}.$$

Yule Walker Equations: 3

- ▶ If appropriate, one would prefer to use MLE methods for parameter estimation. However, as we will see, the MLEs often must be solved numerically by a search algorithm. This sort of algorithm requires a starting value, and the estimates from the YW equation approach can often provide a good starting value.
- ▶ We will soon turn to forecasting/prediction. For this problem, we have data, from which we can fit a model and estimate the model parameters, then we will use that model to forecast future values.

Yule Walker Equations: 4

- ▶ For example, if one fit an AR(P) model to data $Y_1, \dots, Y_t$ and then wished to predict a future value, say Y_{t+k} , the minimum mean square error predictor would be given by $E(Y_{t+k}|\mathcal{F}_t)$. This will often be given by extrapolating the model from time t to time $t+k$ using the estimated coefficients. This will be demonstrated later.
- ▶ If one is going to engage in forecasting on a daily (or even shorter) basis, updating the data each time, it would be computationally much more efficient to have a recursive or on-line method to take the old results and update them with the new data.

Yule Walker Equations: 5

- ▶ As a simple example of this, consider a stream of numbers, $X_1, X_2, \dots$. Suppose we want to compute the mean as each new number arrives, so we have the running mean, $\bar{X}_n = \sum_{i=1}^n X_i/n$. Each time, rather than going back to the beginning and adding all the numbers up (having to store them all), one can note that

$$\bar{X}_n = \bar{X}_{n-1} + \frac{X_n - \bar{X}_{n-1}}{n}.$$

Thus, one only needs to know the current value of $n - 1$ and $\bar{X}_{n-1}$ and the new number, x_n in order to compute $\bar{X}_n$.

- ▶ There is a similar on-line algorithm to update the sample variance knowing only its current value, the sample size and the new value.

Yule Walker Equations: 6

- ▶ There is a recursive algorithm of this sort for the solution to the YW equations given by the Durbin Levinson algorithm. We do not present it here, but it is of practical use, especially if one has a current $AR(p)$ model estimated using YW, adds new data and updates the estimated model.
- ▶ As we saw in estimating the parameters from a $MA(2)$ model, the problem is far more difficult and must be done numerically. This, of course, carries over to $ARMA$ models which have a MA component. In such cases, the estimation is done using likelihood methods, and standard packages based this on iid white noise errors.

Likelihood Methods: 1

- ▶ We begin with the simplest model, AR(1) with a single parameter with iid Normal $WN(0, \sigma_\epsilon^2)$ with three unknown parameters, $\mu, \phi_1, \sigma_\epsilon^2$.
- ▶ Suppose the AR(1) data is $(Y_1, Y_2, \dots, Y_n)$.
- ▶ We must make an assumption concerning the distribution of Y_1 , the first data point that initiates the process. We might assume that it has the stationary distribution, $N(\mu, \sigma_\epsilon^2 / (1 - \phi_1^2))$, or that it is unknown.
- ▶ Conditional on the starting value, Y_1 , the sequence is a Markov process as the conditional distribution of $Y_{i+1} | \mathcal{F}_i$ has a $N(\mu + \phi_1(Y_i - \mu), \sigma_\epsilon^2)$ distribution. Consequently, we can easily write out the likelihood function for the data:

Likelihood Methods: 3

The MLE for σ_ϵ^2 is found by the usual methods and is given by

$$\hat{\sigma}_\epsilon^2 = \frac{1}{n-1} \sum_{i=2}^n (Y_i - (\hat{\mu} + \hat{\phi}_1(Y_{i-1} - \hat{\mu})))^2,$$

the residuals from comparing the data with the fitted values for the data using the model.

- ▶ The extension of likelihood methods for AR(1) to AR(p) processes is in some sense straightforward, but it will again rely on numerical solutions.
- ▶ For AR(p) models and a data set $(Y_1, \dots, Y_n)$, the model is no longer a Markov process.

Likelihood Methods: 4

- ▶ The next value in the time series, Y_{n+1} will depend upon $Y_n, Y_{n-1}, \dots, Y_{n+1-p}$, but not on any data further in the past. Also there will be an initial part of the data set, $Y_1, \dots, Y_p$ for which we must determine their joint distribution. One obvious choice is the p -dimensional stationary distribution of the $AR(p)$ process.
- ▶ This again calls for numerical methods to find the maximum likelihood, and it is common to use conditional least squares. Nevertheless, the MLE of σ_ϵ^2 is still given by the sum of squared residuals divided by the appropriate denominator.
- ▶ Having given this description of the MLE approaches used for the simple $AR(1)$ and $AR(p)$ models, it should be clear that using likelihood methods for $MA(1)$ and $ARMA(p,q)$ models will be more complicated and will certainly require numerical methods.

Statsmodels: 1

- ▶ Comment on
`statsmodels.tsa.arima_process.arma_generate_sample`
There is a parameter in this function, `"distrvs"` which generates the random variables that drive the process. The default is `"np.random.randn"` which gives iid $N(0,1)$ errors. This can be changed to other distributions including changing the variance if desired.
- ▶ for `statsmodels.tsa.arima_model.ARIMA.fit` there is a parameter called `"method"` which allow for two choices, `"mle"` which maximizes the full likelihood including the initial distributions which will be taken from the stationary distribution, or `"css"` the conditional sum of squares where the initial factors in the likelihood are omitted.

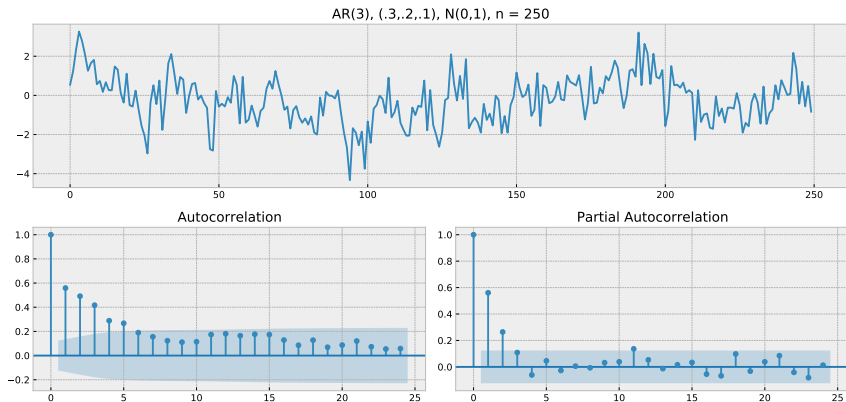
Statsmodels: 2

```
from statsmodels.tsa.arima_process import ArmaProcess
from statsmodels.tsa.arima_model import ARMA
from statsmodels.tsa.stattools import arma_order_select_ic
arparams = np.array([.30, .20, .10])
ar = np.r_[1, -arparams]
ar_process = ArmaProcess(ar)
ar_process.isstationary
yes
y = ar_process.generate_sample(250)
tsplot(y)
plt.show()
res = arma_order_select_ic(y,ic=['aic','bic'],trend = 'nc')
res.aic_min_order
[2,0]
res.bic_min_order
[2,0]
```

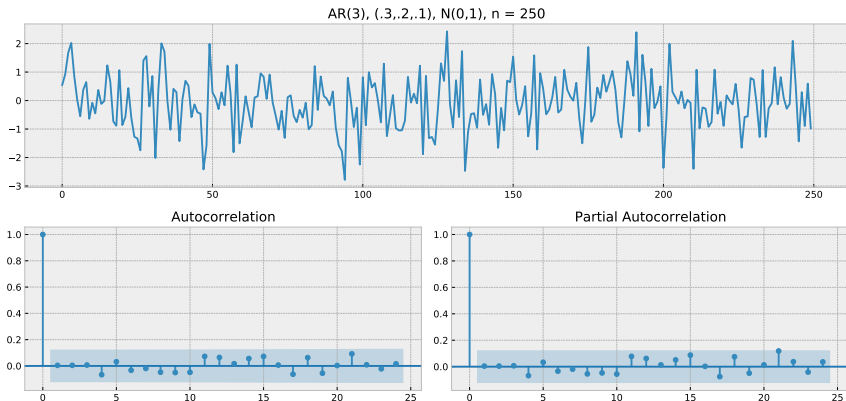
Statsmodels: 3

```
model1 = ARMA(y, (3, 0)).fit(trend='nc', disp=0)
print(model1.params)
tsplot(model1.resid)
plt.show()
model2 = ARMA(y, (2,0)).fit(trend = 'nc', disp = 0)
print(model2.params)
tsplot(model2.resid)
plt.show()
model3 = ARMA(y, (1,1)).fit(trend = 'nc', disp = 0)
print(model3.params)
tsplot(model3.resid)
plt.show()
```

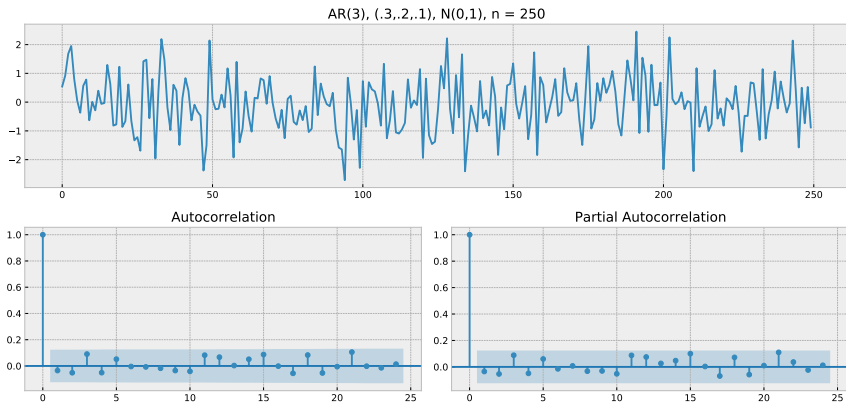
AR(3), (.3,.2,.1), N(0,1), n = 250



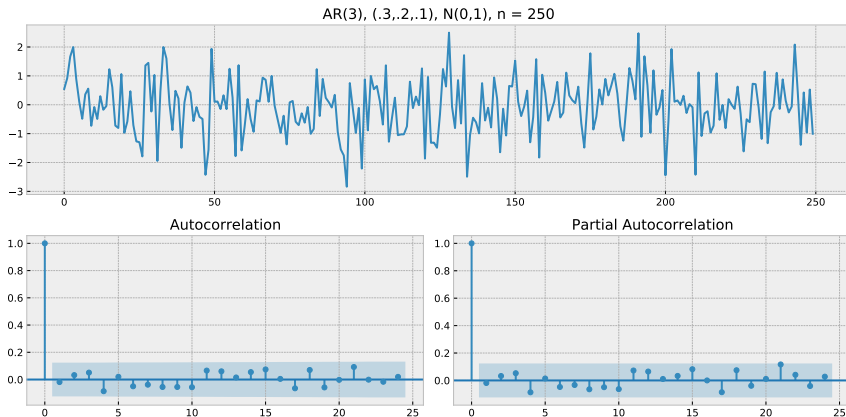
AR(3), (.3,.2,.1), Residuals (3,0),(.38,.22,.12)



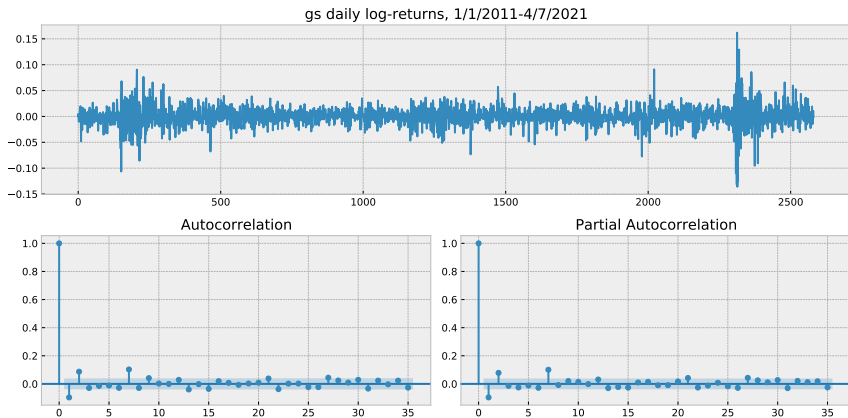
AR(3), (.3,.2,.1), Residuals (2,0), (.42,.27)



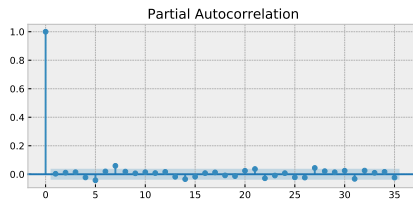
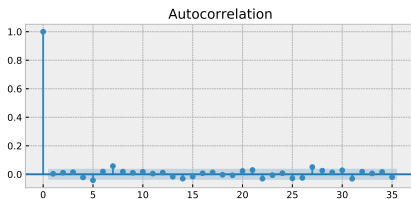
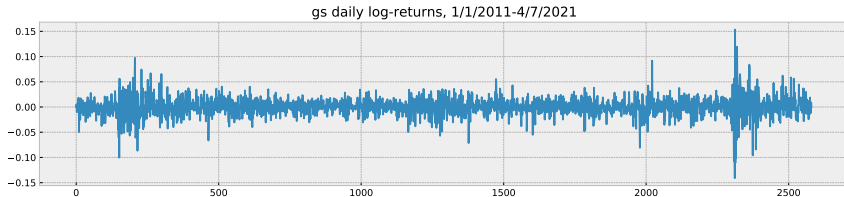
AR(3), (.3,.2,.1), Residuals (1,1),(.86,-.45)



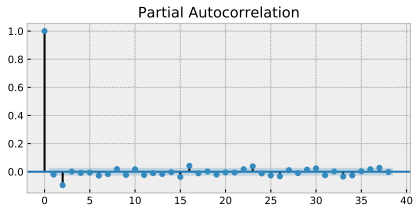
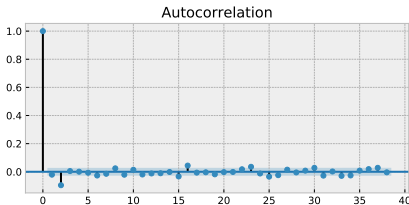
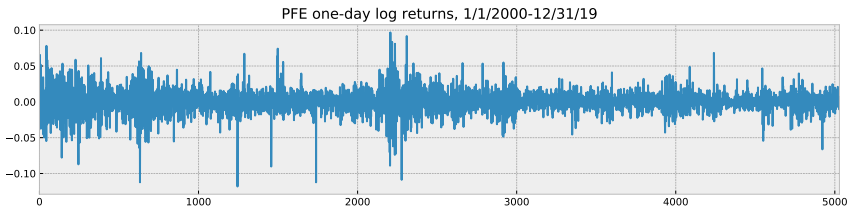
GS daily log-returns 1/1/2011-4/7/2021



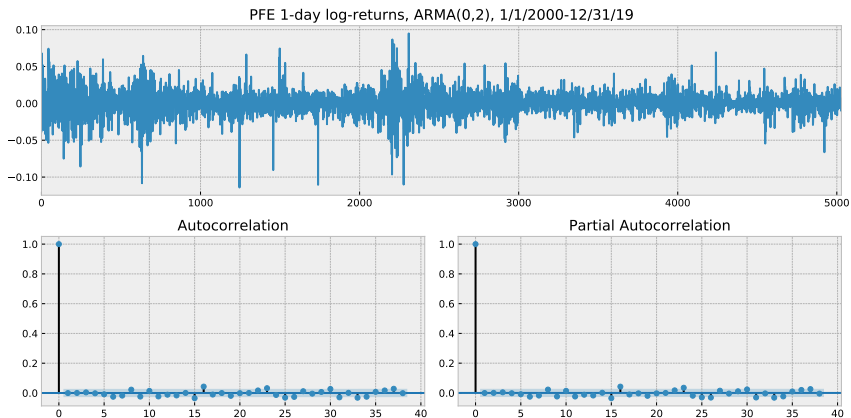
GS daily log-returns residuals from (2,2)



Pfizer (pfe) 1-day log returns, 1/1/2000-12/31/2019



PFE 1-day residuals fit with ARMA(0,2), 1/1/2000-12/31/2019



Forecasting: 1

- ▶ We begin with a basic result about prediction. In the context of time series, we have data, $\mathbf{X} = (X_1, X_2, \dots, X_n)$, and we wish to predict a future value of the series, for example X_{n+k} .
- ▶ The prediction must be made using only the available information. We symbolize that knowledge by $\mathcal{F} = \mathcal{F}(\mathbf{X})$.
- ▶ The goal is to find the best predictor that is a function of $\mathbf{X}$, that is the function must be measurable with respect to $\mathcal{F}$.
- ▶ Our criterion for “best” is the predictor that minimizes the mean square error, $E((Y - \text{predictor})^2)$.
- ▶ The answer is $E(Y|\mathcal{F})$.

Forecasting: 2

- ▶ Suppose we consider any predictor of Y based on $g(\mathbf{X})$ and we consider its mean square error, $E((Y - g(\mathbf{X}))^2)$.

$$\begin{aligned} E((Y - g(\mathbf{X}))^2) &= E((Y - E(Y|\mathcal{F}) + E(Y|\mathcal{F}) - g(\mathbf{X}))^2) \\ &= E((Y - E(Y|\mathcal{F}))^2) + E(E(Y|\mathcal{F}) - g(\mathbf{X}))^2 \\ &= +2E((Y - E(Y|\mathcal{F}))(E(Y|\mathcal{F}) - g(\mathbf{X}))) \\ &= E((Y - E(Y|\mathcal{F}))^2) + E(E(Y|\mathcal{F}) - g(\mathbf{X}))^2 \\ &\geq E((Y - E(Y|\mathcal{F}))^2), \end{aligned}$$

where the cross term is 0 because $E(Y - E(Y|\mathcal{F})) = 0$, and $E((Y - E(Y|\mathcal{F}))(E(Y|\mathcal{F}) - g(\mathbf{X}))) = 0$ by conditioning the unconditional expectation on $\mathcal{F}$.

Forecasting: 2.5

- ▶ Letting $f = E(Y|\mathcal{F})$, an $\mathcal{F}$ -measurable function, for the cross term we want to show $E((Y - f)(f - g(\mathbf{X}))) = 0$.

$$\begin{aligned} E((Y - f)(f - g(\mathbf{X}))) &= E(E((Y - f)(f - g(\mathbf{X})))|\mathcal{F}) \\ &= E((f - g(\mathbf{X}))E(Y - f|\mathcal{F})) \\ &= E((f - g(\mathbf{X}))(f - f)) \\ &= 0 \end{aligned}$$

$$E((Y - g(\mathbf{X}))^2) \geq E((Y - E(Y|\mathcal{F}))^2),$$

for all functions g , that is to say all estimators that are a function of the data $\mathbf{X}$.

- ▶ Thus $E(Y|\mathcal{F})$ is the minimum mean square error estimator of Y .

Forecasting: 3

- ▶ Given we know the form of the optimal forecasting/prediction function, in full generality it may not be easy to determine.
- ▶ We begin with the simplest time series model, AR(1):

$$Y_t = \mu + \phi_1(Y_{t-1} - \mu) + \epsilon_t.$$

- ▶ Suppose we have data $(Y_1, \dots, Y_t)$ and we wish to predict Y_{t+1} , a one-step-ahead forecast.
- ▶ We usually assume that the noise process, ϵ_t , is $WN(0, \sigma_\epsilon^2)$. If we specialize to the case where it is an independent sequence or an MDS, the $E(\epsilon_{t+1}|\mathcal{F}_t) = 0$. So our best prediction of the noise term, ϵ_{t+1} , given the present information is 0.

Forecasting: 4

- ▶ The AR(1) model gives $Y_{t+1} = \mu + \phi_1(Y_t - \mu) + \epsilon_{t+1}$, so the best linear forecast will be

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\phi}_1(Y_t - \hat{\mu}) + 0,$$

- ▶ Extending to forecasts for times further in the future is straightforward; however, we note that the forecast must be based only on the current dataset, $(Y_1, \dots, Y_t)$, only.

Forecasting: 5

- ▶ To forecast Y_{t+2} , the same idea can be used, except we need to forecast both Y_{t+1} and Y_{t+2} :

$$\begin{aligned}\hat{Y}_{t+2} &= \hat{\mu} + \hat{\phi}_1(\hat{Y}_{t+1} - \mu) \\ &= \hat{\mu} + \hat{\phi}_1(\hat{\phi}_1(Y_t - \hat{\mu})) \\ &= \hat{\mu} + \hat{\phi}_1^2(Y_t - \hat{\mu}).\end{aligned}$$

In general, the k-step-ahead predictor for an AR(1) sequence is

$$\hat{Y}_{t+k} = \hat{\mu} + \hat{\phi}_1^k(Y_t - \hat{\mu}).$$

Forecasting: 6

- ▶ While we don't write out the derivation here, all the above ideas extend to AR(p) processes, except the model now contains p parameters whose estimates are used in the forecast.
- ▶ A new aspect arises when forecasting a MA(q) time series, therefore also in forecasting an ARMA(p,q) or ARIMA(p,d,q) time series model.
- ▶ To see the issue, consider a MA(1) model:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}.$$

Forecasting: 7

- ▶ If we wish to forecast one-step-ahead, Y_{t+1} , then we note that $Y_{t+1} = \mu + \epsilon_{t+1} + \theta_1 \epsilon_t$ and the forecast, $\hat{Y}_{t+1}$ must be based only on $\mathcal{F}_t$.
- ▶ Clearly, our best estimate of ϵ_{t+1} is 0, and we will obtain estimates of μ and θ_1 and plug them into the equation. But what do we do with ϵ_t ?
- ▶ We replace ϵ_t by the residual, $\hat{\epsilon}_t$. Thus the forecast of Y_{t+1} is

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_t.$$

Forecasting: 8

- ▶ For a MA(1) model, the autocorrelation is 0 for lags greater than 1. To forecast Y_{t+2} we note that the model gives

$$Y_{t+2} = \mu + \theta_1 \epsilon_{t+1} + \epsilon_{t+2}.$$

- ▶ Our best estimates of the two error terms, ϵ_{t+1} and ϵ_{t+2} is 0. This gives

$$\hat{Y}_{t+k} = \hat{\mu}, \quad k \geq 2.$$

Forecasting: 9

In summary:

- ▶ We have a weakly stationary time series, $\{X_t\}$, we have observations from that time series, $(X_1, X_2, \dots, X_n)$, and we wish to predict a future value of the series, say X_{n+k} , a k -step-ahead predictor, $\hat{X}_{n+k}$.
- ▶ The criterion we are trying to optimize is minimum mean square error, $E((X_{n+k} - \hat{X}_{n+k})^2)$.
- ▶ We know that the optimal estimator is $E(X_{t+k} | \mathcal{F}_n)$ to minimize mean square error.
- ▶ In practice, we use the optimum linear forecast, i.e.

$$\hat{X}_{n+k} = a_0 + a_1 X_1 + \dots + a_n X_n.$$

- ▶ As n changes, the coefficients may change, so we actually need to represent this forecast equation as

$$\hat{X}_{n+k} = a_{0n} + a_{1n} X_1 + \dots + a_{nn} X_n.$$

Financial Time Series Analysis

Lecture 7

- ▶ TA Session Friday April 16, 2021, 11:00am - 12:00pm (EDT)
- ▶ TA Session Monday April 19, 2021, 7:30pm - 8:30pm (EDT)

Outline

- ▶ Forecasting
 - ▶ Brief Review: AR(1)
 - ▶ MA and ARMA forecasts
 - ▶ Linear Forecasting
- ▶ Transforming from non-stationarity to stationarity
- ▶ New models (e.g. SARIMA) and packages (e.g. pmdarima)

Review: 1

- ▶ We begin with a basic result about forecasting. In the context of time series, we have data, $\mathbf{X} = (X_1, X_2, \dots, X_n)$, and we wish to forecast a future value of the series, for example X_{n+k} .
- ▶ The forecast must be made using only the available information. We symbolize that knowledge by $\mathcal{F} = \mathcal{F}(\mathbf{X})$.
- ▶ The goal is to find the best predictor that is a function of $\mathbf{X}$, that is the function must be measurable with respect to $\mathcal{F}$.
- ▶ Our criterion for “best” is the predictor that minimizes the mean square error, $E((Y - \text{predictor})^2)$.
- ▶ The answer is $E(Y|\mathcal{F})$, as $Y - E(Y|\mathcal{F})$ is uncorrelated with all $\mathcal{F}$ measurable functions.

Review: 2

- ▶ We begin with the simplest time series model, AR(1):

$$Y_t = \mu + \phi_1(Y_{t-1} - \mu) + \epsilon_t.$$

- ▶ Suppose we have data $(Y_1, \dots, Y_t)$ and we wish to forecast Y_{t+1} , the next value in the time series
- ▶ We usually assume that the noise process, ϵ_t , is $WN(0, \sigma_\epsilon^2)$. If we specialize to the case where it is an independent sequence or an MDS, the $E(\epsilon_{t+1}|\mathcal{F}_t) = 0$. So our best prediction of the noise term, ϵ_{t+1} , given the present information is 0.

Review: 3

- ▶ The AR(1) model gives $Y_{t+1} = \mu + \phi_1(Y_t - \mu) + \epsilon_{t+1}$, so the best linear forecast will be

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\phi}_1(Y_t - \hat{\mu}) + 0,$$

Note, we assume $|\phi_1| < 1$.

- ▶ To forecast Y_{t+2} , the same idea can be used, except we need to forecast both Y_{t+1} and Y_{t+2} :

$$\begin{aligned}\hat{Y}_{t+2} &= \hat{\mu} + \hat{\phi}_1(\hat{Y}_{t+1} - \mu) \\ &= \hat{\mu} + \hat{\phi}_1(\hat{\phi}_1(Y_t - \hat{\mu})) \\ &= \hat{\mu} + \hat{\phi}_1^2(Y_t - \hat{\mu}).\end{aligned}$$

In general, the k-step-ahead predictor for an AR(1) sequence is

$$\hat{Y}_{t+k} = \hat{\mu} + \hat{\phi}_1^k(Y_t - \hat{\mu}).$$

Forecasting: 1

- ▶ While we don't write out the derivation here, all the above ideas extend to AR(p) processes, except the model now contains p parameters whose estimates are used in the forecast.
- ▶ A new aspect arises when forecasting a MA(q) time series, therefore also in forecasting an ARMA(p,q) or ARIMA(p,d,q) time series model.
- ▶ To see the issue, consider a MA(1) model:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}.$$

Forecasting: 2

- ▶ If we wish to forecast one-step-ahead, Y_{t+1} , then we note that $Y_{t+1} = \mu + \epsilon_{t+1} + \theta_1 \epsilon_t$ and the forecast, $\hat{Y}_{t+1}$ must be based only on $\mathcal{F}_t$.
- ▶ Clearly, our best estimate of ϵ_{t+1} is 0, and we will obtain estimates of μ and θ_1 and plug them into the equation. But what do we do with ϵ_t ?
- ▶ We replace ϵ_t by the residual, $\hat{\epsilon}_t$,

$$\hat{\epsilon}_t = Y_t - \mu - \theta_1 \hat{\epsilon}_{t-1}.$$

Thus the forecast of Y_{t+1} is

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_t.$$

Forecasting: 3

- ▶ For a MA(1) model, the autocorrelation is 0 for lags greater than 1. To forecast Y_{t+2} we note that the model gives

$$Y_{t+2} = \mu + \theta_1 \epsilon_{t+1} + \epsilon_{t+2}.$$

- ▶ Our best estimates of the two error terms, ϵ_{t+1} and ϵ_{t+2} is 0. This gives

$$\hat{Y}_{t+k} = \hat{\mu}, \quad k \geq 2.$$

- ▶ These ideas easily extend to MA(q):

$$Y_{t+1} = \mu + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t,$$

and

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_{t-1} + \cdots + \hat{\theta}_q \hat{\epsilon}_{t-q} + \hat{\epsilon}_t,$$

- ▶ This carries over to ARMA(p,q) processes.

Forecasting: 4

In summary:

- ▶ We have a weakly stationary time series, $\{X_t\}$, we have observations from that time series, $(X_1, X_2, \dots, X_n)$, and we wish to predict a future value of the series, say X_{n+k} , a k -step-ahead predictor, $\hat{X}_{n+k}$.
- ▶ The criterion we are trying to optimize is minimum mean square error, $E((X_{n+k} - \hat{X}_{n+k})^2)$.
- ▶ We know that the optimal estimator is $E(X_{t+k} | \mathcal{F}_n)$ to minimize mean square error. This is straightforward when we have a structured model like AR, MA or ARMA. In general the conditional expectation may not be computable, and in that case we fall back onto linear estimation.

Forecasting: 5

- ▶ In practice, we use the optimum linear forecast, i.e.

$$\hat{X}_{n+k} = a_0 + a_1X_1 + \cdots + a_nX_n.$$

- ▶ As n changes, the coefficients may change, so we actually need to represent this forecast equation as

$$\hat{X}_{n+k} = a_{0n} + a_{1n}X_1 + \cdots + a_{nn}X_n.$$

- ▶ So there are two issues to address: 1) for a fixed n , find the optimal coefficients $\mathbf{a}_n = (a_{0n}, \cdots, a_{nn})$ and 2) if we have $\mathbf{a}_n$ and the next value in the series, X_{n+1} , how can we update $\mathbf{a}_n$ to $\mathbf{a}_{n+1}$?

Forecasting: 6

Finding the best linear predictor is very similar to finding the optimal control variable in Monte Carlo simulation. We wish to minimize

$$S(\mathbf{a}) = E((X_{n+k} - (a_0 + a_1X_n + \cdots + a_nX_1))^2),$$

over $\mathbf{a} = (a_0, a_1, \cdots, a_n)$.

- ▶ We differentiate $S(\mathbf{a})$ sequentially with respect to the components of $\mathbf{a}$, obtain a system of linear equations to solve and check that the solution is a minimum.
- ▶ In this case one can simplify the calculation by interchanging the order of expectation and differentiation, i.e.

$$\frac{\partial}{\partial \mathbf{a}} E(S(\mathbf{a})) = E\left(\frac{\partial}{\partial \mathbf{a}} ((X_{n+k} - (a_0 + a_1X_1 + \cdots + a_nX_n))^2)\right),$$

Forecasting: 7

For a_0 we have

$$\begin{aligned} 0 = \frac{\partial}{\partial a_0} E(S(\mathbf{a})) &= E\left(\frac{\partial}{\partial a_0} (X_{n+k} - a_0 - \sum_{j=1}^n a_j X_{n+1-j})^2\right) \\ &= -2E(X_{n+k} - a_0 - a_1 X_n - \cdots - a_n X_1) \\ &= -2(\mu - a_0 - \mu(a_1 + \cdots + a_n)) \\ &= -2(a_0 + \mu(1 - (a_1 + \cdots + a_n))). \end{aligned}$$

This is easily solved

$$\hat{a}_0 = \mu(1 - \sum_{j=1}^n a_j).$$

Forecasting: 8

Differentiation with respect to a_i , $1 \leq i \leq n$ gives

$$0 = \frac{\partial}{\partial a_i} = -2E\left((X_{n+k} - a_0 - \sum_{j=1}^n a_j X_{n+1-j})X_{n+1-i}\right).$$

Equivalently,

$$E(X_{n+k}X_{n+1-i} - a_0X_{n+1-i} - a_1X_nX_{n+1-i} - \cdots - a_nX_1X_{n+1-i}) = 0,$$

or

$$\gamma(k+i-1) + \mu^2 - a_0\mu - a_1(\gamma(1-i) + \mu^2) - \cdots - a_n(\gamma(n-i) + \mu^2) = 0.$$

Forecasting: 9

This can be rewritten as

$$\gamma(k+i-1) = a_0\mu - \mu^2 + \sum_{j=1}^n a_j\mu^2 + \sum_{j=1}^n a_j\gamma(j-i),$$

and note $a_0\mu - \mu^2 + \sum_{j=1}^n a_j\mu^2 = 0$. This results in a set of linear equations relating the a_i coefficients in terms of the $\gamma(h)$ coefficients:

$$\mathbf{\Gamma}_n \mathbf{a}_n = \gamma_n(n-1),$$

where $\mathbf{\Gamma}_n = (\gamma(i-j))_{i,j=1,\dots,n}$, $\gamma_n(n-1) = (\gamma(n-1), \dots, \gamma(0))$. These are the **prediction equations**. Here $\mathbf{\Gamma}_n$ is the same matrix that arose in the general Yule-Walker equations, and we can find the coefficients of the best predictor as

$$\mathbf{a}_n = \mathbf{\Gamma}_n^{-1} \gamma_n(k).$$

Forecasting: 10

The best linear predictor is

$$\hat{X}_{n+k} = \mu + \hat{a}_1(X_n - \mu) + \cdots + \hat{a}_n(X_1 - \mu),$$

and its mean square prediction error is

$$E((X_{n+k} - \hat{X}_{n+k})^2) = \gamma(0) - \mathbf{a}_n^T \boldsymbol{\gamma}_n(k).$$

Thus linear prediction for a weakly stationary time series model is straightforward:

- ▶ Solve the prediction equations using the autocovariance function of the process
- ▶ Write down the expression for the best linear predictor
- ▶ Evaluate its mean squared error.

Forecasting: 11

Predicting a white noise process, $WN(\mu, \sigma_\epsilon^2)$

- ▶ Here $\gamma(h) = \sigma_\epsilon^2$ if $h = 0$ and 0 if $h \neq 0$. Thus $\Gamma_n = \sigma_\epsilon^2 \mathbf{I}_n$.
- ▶ Thus the prediction equations become $\mathbf{a} = \gamma_n = \mathbf{0}$.
- ▶ Thus the best linear predictor of a white noise process with mean μ at a future point in time, ϵ_{n+k} , is given by $\hat{\mu} + 0 = \hat{\mu}$. The mean square error is $E((\epsilon_{n+k} - \mu)^2) = \sigma_\epsilon^2$.
- ▶ There is no information in the data that can improve the minimum mean square error prediction

Forecasting: 12

We next predict an AR(1) process, beginning with predicting X_{n+1} from $X_1, \dots, X_n$, the one-step-ahead predictor.

- ▶ Recall for an AR(1) process, $\gamma(h) = \frac{\sigma_\epsilon^2 \phi_1^{|h|}}{1 - \phi_1^2}$
- ▶ We need to find $\mathbf{a}_n$ to satisfy the prediction equations: $\mathbf{\Gamma}_n \mathbf{a}_n = \gamma_n(1)$. Both $\mathbf{\Gamma}_n$ and $\gamma_n(1)$ have a common factor $\sigma_\epsilon^2 / (1 - \phi_1^2)$ that we omit in writing the equations.
- ▶ The first row of $\mathbf{\Gamma}$ is $(1, \phi_1, \dots, \phi_1^{n-1})$. Multiplying this by ϕ_1 gives exactly $\gamma_n(1)$. Consequently, if we set

$$\hat{\mathbf{a}}_n = (\phi_1, 0, \dots, 0),$$

then we have a solution to the prediction equations.

Forecasting: 13

- ▶ Consequently, the best linear one-step-ahead predictor of X_{n+1} given $\mathcal{F}_n$ is $\hat{X}_{n+1} = \hat{\mu} + \phi_1(X_n - \hat{\mu})$.
- ▶ The minimum mean square error is

$$\begin{aligned} E((X_{n+1} - \hat{X}_{n+1})^2) &= \gamma(0) - \mathbf{\hat{a}}\gamma_n(1) \\ &= \frac{\sigma_\epsilon^2}{1 - \phi_1^2}(1 - \phi_1^2) \\ &= \sigma_\epsilon^2 \end{aligned}$$

- ▶ This is a reduction from $\sigma_\epsilon^2/(1 - \phi_1^2)$.

Forecasting: 14

- ▶ Now we can predict further in time, e.g. $\hat{Y}_{n+k}$ for $k > 1$. The analysis is essentially identical, as all that changes is $\gamma_n(k) = \phi_1^k \gamma_n(1)$.
- ▶ The prediction equations are easily solved, we simply multiply $\mathbf{a}_n$ found before by ϕ_1^k , $(\phi_1^k, 0, \dots, 0)$.
- ▶ The minimum mean square error for this k-step-ahead predictor is

$$\sigma_\epsilon^2 \frac{1 - \phi_1^{2k}}{1 - \phi_1^2}$$

- ▶ Note how the error increases as k increases, with the limit as $k \rightarrow \infty$ being the unconditional variance $\sigma_\epsilon^2 / (1 - \phi_1^2)$, an intuitive result.

Forecasting: 15

- ▶ Now the best linear predictor we derived for a weakly stationary time series (coming from solving the prediction equations) used the parameters of the time series as though we knew them.
- ▶ Obviously, in general we will not know those parameters (unless we generated the time series by simulation).
- ▶ We saw this problem in Simulation where we needed to use, say, a standard deviation of the data in defining the optimal control variable. In that case we used the data to estimate the standard deviation, resulting in a standard error (estimated standard deviation of an estimate).

Forecasting: 16

- ▶ This is the same situation. We have an optimal estimator (forecast) involving unknown parameters, so we substitute those for the unknown true values.
- ▶ However, the minimum mean squared error we computed does not reflect the added variability coming from the uncertainty in the estimated values, including σ_e .
- ▶ In addition to the additional uncertainty that arises from the estimation of the model parameter estimate, we should recognize that creating a “point” forecast is only part of the task.

Forecasting: 17

- ▶ We need to accompany the point forecast with a measure of uncertainty. Indeed, we really need to develop a **predictive distribution**.
- ▶ The predictive distribution is the probability distribution of our optimal forecast. We generally use the mean of that distribution if we must give a specific value, but the predictive distribution allows for the creation of confidence intervals for a prediction (“prediction intervals”).
- ▶ The generation of a predictive distribution can be done in theory for special favorable assumptions, or it can be done by Monte Carlo methods for more general assumptions.

Forecasting: 18

- ▶ Generally, with enough data, the model parameter estimates will be asymptotically normally distributed (e.g. MLE or moment estimators are so by the central limit theorem).
- ▶ If n is sufficiently large, the error from the parameter estimates likely dominates the error from the model and from the noise process in the model.
- ▶ In most cases, statistical software like statsmodels is used and will automatically compute forecasts and their limits.
- ▶ But for more careful work (for example, a concern for heavy tails), Monte Carlo simulation should be applied.

Forecasting: 19

- ▶ For example, given an AR(1) process, parameters ϕ_0, ϕ_1 and σ_ϵ^2 , a model for $\{\epsilon_t, -\infty < t < \infty\}$, and data $(X_1, \dots, X_n)$.
- ▶ Given this information, one can simulate many paths for the process using the AR(1) model, the parameter estimates and the assumptions about the noise process distribution.
- ▶ This would result in a set of values for X_{n+k} , and the distribution of these values gives the predictive distribution of the process. From that distribution, one can find a point prediction (usually the mean) and develop confidence limits using that distribution.

Forecasting Example: 1

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
start = '2011-01-01'
end = '2021-04-7'
stk = pdr.get_data_yahoo('ibm',start,end)
stk_adjclose = stk['Adj Close']
#More commands to get logret
#Order select ARMA Model (2,2)
model = sm.tsa.SARIMAX(logret, order = (2,0,2), trend = 'c')
res = model.fit()
model1= res.get_forecast(steps = 10)
print(model1_fcast.summary_frame())
```

Forecasting Example: 2

	coef	std err	[0.025	0.975]
intercept	0.0001	0.000	-0.001	0.001
ar.L1	-0.3246	0.057	-0.436	-0.214
ar.L2	0.1946	0.090	0.019	0.371
ma.L1	0.2762	0.057	0.165	0.388
ma.L2	-0.1794	0.096	-0.367	0.009
sigma2	0.0002	2.63e-06	0.000	0.000

Forecasting Example: 3

y	mean	mean_se	mean_ci_lower	mean_ci_upper
0	-0.000800	0.014499	-0.029216	0.027617
1	0.000654	0.014516	-0.027796	0.029104
2	-0.000266	0.014523	-0.028729	0.028198
3	0.000316	0.014525	-0.028153	0.028785
4	-0.000052	0.014526	-0.028523	0.028419
5	0.000181	0.014527	-0.028292	0.028653
6	0.000033	0.014527	-0.028439	0.028506
7	0.000126	0.014527	-0.028346	0.028599
8	0.000068	0.014527	-0.028405	0.028540
9	0.000105	0.014527	-0.028368	0.028578

Non-Stationary to Stationary

- ▶ Most series in financial economics and finance are not stationary. Since our methods are for stationary processes, we want to transform the original data set into one that is much closer to stationarity.
- ▶ There are several standard methods in common use to accomplish this including:
 - ▶ Note if the dataset exhibits unequal variance (other than volatility clustering), for example larger variance for higher values. [Consider a Box-Cox transformation.](#)
 - ▶ If the time series has a periodically repeating pattern, then remove a [seasonal component](#)
 - ▶ If the time series has a smooth upward or downward behavior, [fit and remove a trend component](#)
 - ▶ After removing trends and seasonal components, if the resulting series is not weakly stationary, consider [differencing](#)

Box-Cox Transformation

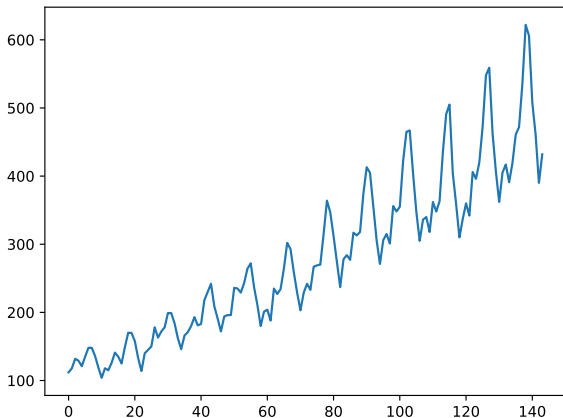
- ▶ Much of statistical analysis is done in the context of a symmetric error distribution. When there is a lack of symmetry, a Box-Cox or power transformation such as a square root or logarithmic transformation might help.
- ▶ The Box-Cox family is indexed by a parameter, λ mapping the data X into the transformed data X^λ , where

$$X^\lambda = \begin{cases} (X^\lambda - 1)/\lambda & \lambda \neq 0, \\ \log(X) & \lambda = 0. \end{cases}$$

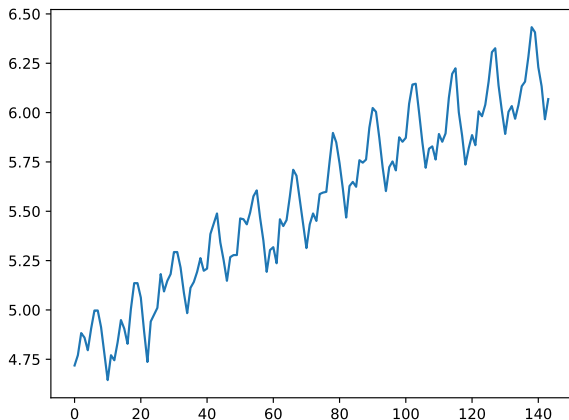
- ▶ The λ divisor is included to make the entire family continuous as $\lambda \rightarrow 0$, since

$$\lim_{\lambda \rightarrow 0} (X^\lambda - 1)/\lambda = \log(X).$$

International Airline Passengers, 1949-1960



Log(Int. Airline Passengers: 1949-1960)



Trend and Seasonal Components: 1

- ▶ After transforming to make the variances more equal, we must also consider trends and seasonality (periodicity).
- ▶ Let us first consider trends and let $X_t = m_t + Y_t$, where m_t is a trend and Y_t is weakly stationary.
- ▶ Differencing is a common way to handle the m_t term, but first we consider explicit modeling of m_t .
- ▶ One first decides on a parametric model for the trend, e.g.
 - ▶ Linear trend: $m_t = \beta_0 + \beta_1 t$,
 - ▶ Quadratic trend: $m_t = \beta_0 + \beta_1 t + \beta_2 t^2$
- ▶ One then estimates the coefficients using least squares.

Trend and Seasonal Components: 2

- ▶ An alternative approach is to use a **moving average smoother**. Choose a non-negative integer q , the half-bandwidth of the smoother). Then for $t = q + 1, \dots, n - q - 1$

$$\hat{m}_t = \frac{1}{2q + 1} \sum_{j=-q}^q X_{t+j}.$$

- ▶ Why does this work?

Trend and Seasonal Components: 3

$$\begin{aligned}\hat{m}_t &= \frac{1}{2q+1} \sum_{j=-q}^q X_{t+j} \\ &= \frac{1}{2q+1} \sum_{j=-q}^q (m_t + Y_{t+j}) \\ &= \frac{1}{2q+1} \sum_{j=-q}^q m_{t+j} + \frac{1}{2q+1} \sum_{j=-q}^q Y_{t+j} \\ &\approx m_t + \text{small error}\end{aligned}$$

Trend and Seasonal Components: 4

- ▶ The moving average smoother does preserve a linear trend. If $m_t = a_0 + a_1 t$:

$$\begin{aligned}\frac{1}{2q+1} \sum_{j=-q}^q m_{t+j} &= \frac{1}{2q+1} \sum_{j=-q}^q (a_0 + a_1(t+j)) \\ &= \frac{1}{2q+1} \sum_{j=-q}^q (a_0 + a_1 t) + \frac{1}{2q+1} \sum_{j=-q}^q a_1 j \\ &= a_0 + a_1 t + 0 \\ &= m_t\end{aligned}$$

- ▶ An alternative approach to the explicit modeling of a trend and seasonal (periodic) component is to difference the time series.

Trend and Seasonal Components: 5

- ▶ We next consider the case in which there is a seasonal (periodic) component with period d present.
- ▶ First, we define indicator functions,

$$I_j(t) == \begin{cases} 1 & \text{if } t \bmod d = j \\ 0 & \text{otherwise} \end{cases}$$

for $1 \leq j \leq d$.

- ▶ Second, we consider a model for the trend, for example a linear trend, $m_t = \beta_0 + \beta_1 t$. Thus our model becomes

$$\hat{X}_t = \beta_0 + \beta_1 t + \sum_{j=2}^d \beta_j I_j(t),$$

and we estimate the $d + 1$ parameters $\{\beta_i, i = 0, \dots, d\}$ using least squares by minimizing $\sum (X_t - \hat{X}_t)^2$.

Trend and Seasonal Components: 6

- Note that one of the indicators is omitted as the sum of all indicators must be 0. Define $\{\omega_j, 1 \leq j \leq d\}$ by $\omega_1 = 0, \omega_2 = \hat{\beta}_2, \dots, \omega_d = \hat{\beta}_d$. We need to adjust the seasonal components so that they sum to 0, consequently, we define $\bar{\omega} = \frac{1}{d} \sum_{j=2}^d \hat{\beta}_j$. The seasonal components are then given by $\hat{s}_j = \omega_j - \bar{\omega}, 1 \leq j \leq d$, with $\hat{s}_t = \hat{s}_{t-d}$ for $t > d$. The trend estimate is

$$\hat{m}_t = \hat{\beta}_0 + \hat{\beta}_1 t + \bar{\omega}.$$

The same ideas can be applied for more complicated trend models.

Differencing: 1

- ▶ We define the differencing operator ∇ by $\nabla X_t = X_t - X_{t-1} = (1 - B)X_t$. Note that one can create polynomials in ∇ and B for example:

$$\nabla^2 X_t = (1 - B)^2 X_t = (1 - 2B + B^2)X_t = X_t - 2X_{t-1} + X_{t-2}.$$

- ▶ Differencing can remove polynomial trends. Consider the linear and quadratic cases. Suppose $m_t = \beta_0 + \beta_1 t$, that is $X_t = \beta_1 + \beta_1 t + Y_t$, where Y_t is stationary.

$$\begin{aligned}\nabla X_t &= X_t - X_{t-1} \\ &= \beta_0 + \beta_1 t + Y_t - \beta_0 - \beta_1(t-1) - Y_{t-1} \\ &= \beta_1 + Y_t - Y_{t-1} \\ &= \beta_1 + \text{a stationary process}\end{aligned}$$

Differencing: 2

Similarly, suppose instead there is a quadratic term, i.e.

$X_t = \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t$. Applying differencing twice, $(1 - B)^2 = 1 - 2B + B^2$, to the time series, we find

$$\begin{aligned}(1 - B)^2 X_t &= X_t - 2X_{t-1} + X_{t-2} \\&= \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t \\&\quad - 2(\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2 + Y_{t-1}) \\&\quad + \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 + Y_{t-2} \\&= 2\beta_2 + (Y_t - 2Y_{t-1} + Y_{t-2}) \\&= 2\beta_2 + \text{a stationary process}\end{aligned}$$

Differencing: 3

- ▶ One can also use differencing to remove both a seasonal and a linear trend. Suppose that $X_t = \beta_0 + \beta_1 t + s_t + \epsilon_t$, and the seasonal term has period d , i.e. $s_t = s_{t-d}$.
- ▶ In this case the differencing should be done over d time periods rather than simply one time period using ∇ . Instead we define a “d-lagged difference operator”

$$\nabla_d = X_t - X_{t-d} = (1 - B^d) \neq \nabla^d.$$

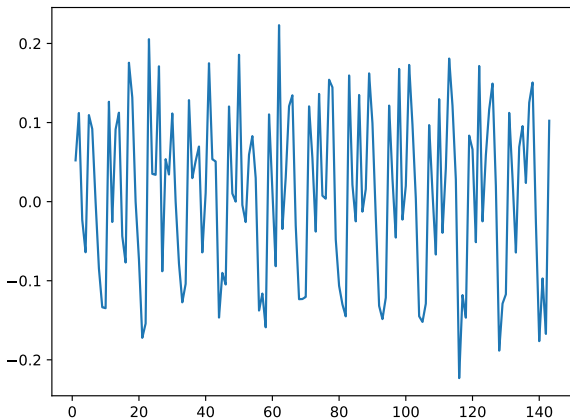
Differencing: 4

Applying this operator to the time series, we find

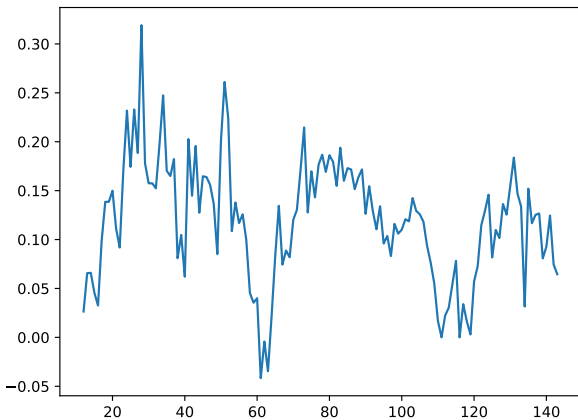
$$\begin{aligned}\nabla_d X_t &= \nabla_d m_t + \nabla_d S_t + \nabla_d \epsilon_t \\ &= m_t - m_{t-d} + S_t - S_{t-d} + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1(t - (t - d)) + 0 + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1 d + \text{a stationary process.}\end{aligned}$$

Thus this differencing operator has transformed the original time series into a weakly stationary process.

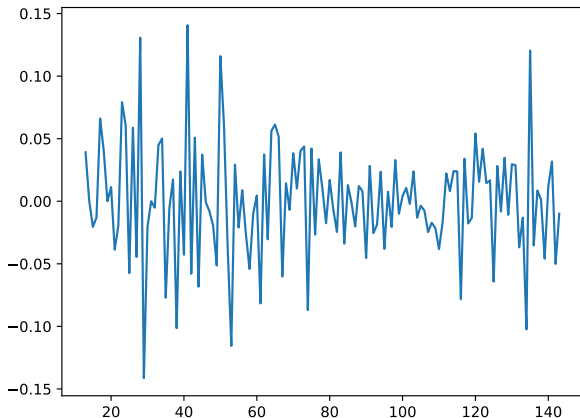
Difference(Log(Airline Passengers))



Seasonal Adj. (12) Log(Airline)



Diff(Seasonal12Log(Airline))



Financial Time Series Analysis

Lecture 8

- ▶ Homework #3 due Friday, April 16, 1:30pm
- ▶ TA Session Friday April 16, 2021, 11:00am - 12:00pm (EDT)
- ▶ TA Session Monday April 19, 2021, 7:30pm - 8:30pm (EDT)

Outline

- ▶ Transforming from non-stationarity to stationarity
 - ▶ Seasonality
 - ▶ Differencing
- ▶ Forecasting Example
- ▶ ARCH/GARCH

Non-Stationary to Stationary

- ▶ Most series in financial economics and finance are not stationary. Since our methods are for stationary processes, we want to transform the original data set into one that is much closer to stationarity.
- ▶ There are several standard methods in common use to accomplish this including:
 - ▶ Note if the dataset exhibits unequal variance (other than volatility clustering), for example larger variance for higher values. [Consider a Box-Cox transformation.](#)
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Trend and Seasonal Components: 5

- ▶ We next consider the case in which there is a seasonal (periodic) component with period d present.
- ▶ First, we define indicator functions,

$$I_j(t) = \begin{cases} 1 & \text{if } t \bmod d = j \\ 0 & \text{otherwise} \end{cases}$$

for $1 \leq j \leq d$.

- ▶ Second, we consider a model for the trend, for example a linear trend, $m_t = \beta_0 + \beta_1 t$. Thus our model becomes

$$X_t = \beta_0 + \beta_1 t + \sum_{j=2}^d \beta_j I_j(t),$$

and we estimate the $d + 1$ parameters $\{\beta_i, i = 1, \dots, d\}$ using least squares by minimizing $\sum (X_t - \hat{X}_t)^2$.

Trend and Seasonal Components: 6

- Note that one of the indicators is omitted as the sum of all indicators must be 0. Define $\{\omega_j, 1 \leq j \leq d\}$ by $\omega_1 = 0, \omega_2 = \hat{\beta}_2, \dots, \omega_d = \hat{\beta}_d$. We need to adjust the seasonal components so that they sum to 0, consequently, we define $\bar{\omega} = \frac{1}{d} \sum_{j=2}^d \hat{\beta}_j$. The seasonal components are then given by $\hat{s}_j = \omega_j - \bar{\omega}, 1 \leq j \leq d$, with $\hat{s}_t = \hat{s}_{t-d}$ for $t > d$. The trend estimate is

$$\hat{m}_t = \hat{\beta}_0 + \hat{\beta}_1 t + \bar{\omega}.$$

The same ideas can be applied for more complicated trend models.

Differencing: 1

- ▶ We define the differencing operator ∇ by $\nabla X_t = X_t - X_{t-1} = (1 - B)X_t$. Note that one can create polynomials in ∇ and B for example:

$$\nabla^2 X_t = (1 - B)^2 X_t = (1 - 2B + B^2)X_t = X_t - 2X_{t-1} + X_{t-2}.$$

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Differencing: 2

Similarly, suppose instead there is a quadratic term, i.e.

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$$\begin{aligned}(1 - B)^2 X_t &= X_t - 2X_{t-1} + X_{t-2} \\&= \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t \\&\quad - 2(\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2 + Y_{t-1}) \\&\quad + \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 + Y_{t-2} \\&= 2\beta_2 + (Y_t - 2Y_{t-1} + Y_{t-2}) \\&= 2\beta_2 + \text{a stationary process}\end{aligned}$$

Differencing: 3

- ▶ One can also use differencing to remove both a seasonal and a linear trend. Suppose that $X_t = \beta_0 + \beta_1 t + s_t + \epsilon_t$, and the seasonal term has period d , i.e. $s_t = s_{t-d}$.
- ▶ In this case the differencing should be done over d time periods rather than simply one time period using ∇ . Instead we define a “d-lagged difference operator”

$$\nabla_d = X_t - X_{t-d} = (1 - B^d) \neq \nabla^d.$$

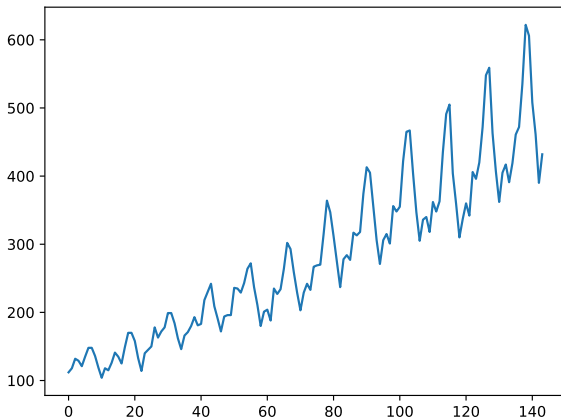
Differencing: 4

Applying this operator to the time series, we find

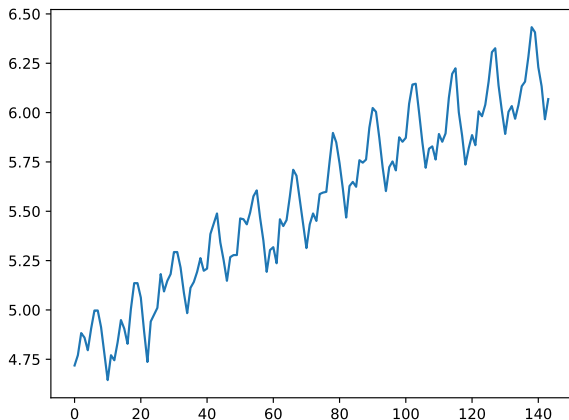
$$\begin{aligned}\nabla_d X_t &= \nabla_d m_t + \nabla_d S_t + \nabla_d \epsilon_t \\ &= m_t - m_{t-d} + S_t - S_{t-d} + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1(t - (t - d)) + 0 + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1 d + \text{a stationary process.}\end{aligned}$$

Thus this differencing operator has transformed the original time series into a weakly stationary process.

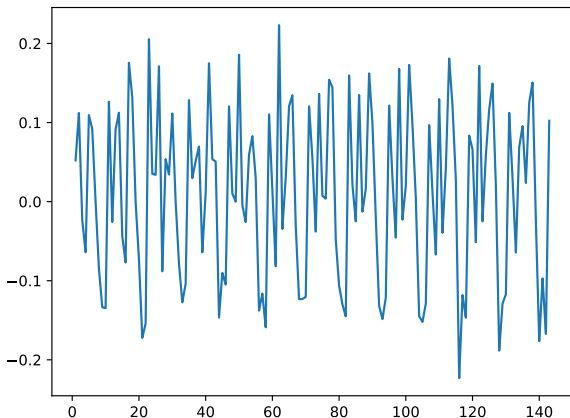
International Airline Passengers, 1949-1960



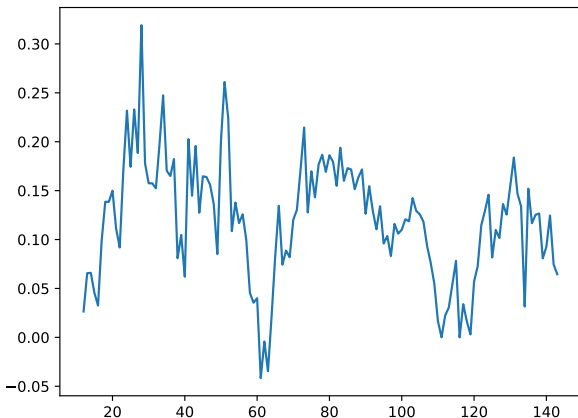
Log(Int. Airline Passengers): 1949-1960



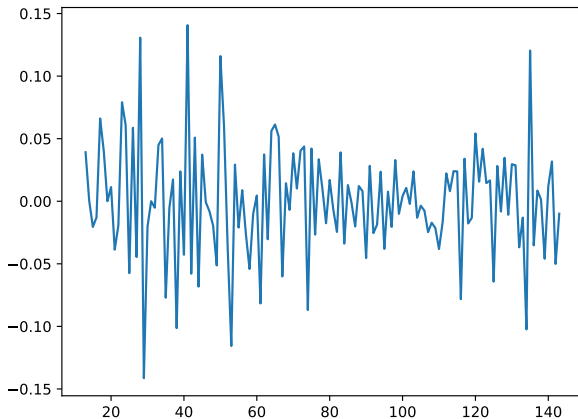
Difference(Log(Airline Passengers))



Seasonal Adj. (12) Log(Airline)



Diff(Seasonal12Log(Airline))



Forecasting with PMDARIMA: 1

```
import numpy as np
import matplotlib.pyplot as plt
import pmdarima as pm
from pmdarima.model_selection import train_test_split
# Load/split data
y = pm.datasets.load_airpassengers()
y = np.log(y)
train, test = train_test_split(y, train_size=100)
# Fit the model
model = pm.auto_arima(train, seasonal=True, m=12)
fit1 = pm.auto_arima(y, m=12, trace=True,
suppress_warnings=True)
Best model: ARIMA(2,1,1)(0,1,0)[12] model.summary()
```

Forecasting with PMDARIMA (100): 2

	coef	std err	z	P> z	[0.025	0.975]
intercept	0.0324	0.015	2.167	0.030	0.003	0.062
ar.L1	0.5509	0.108	5.102	0.000	0.339	0.763
ar.L2	0.2047	0.107	1.915	0.055	-0.005	0.414
ma.S.L12	-0.6353	0.139	-4.558	0.000	-0.909	-0.362
sigma2	0.0013	0.000	6.575	0.000	0.001	0.002

Forecasting with PMDARIMA (120): 3

```
model = pm.auto_arma(train, seasonal=True, m=12)
fit1 = pm.auto_arma(y, m=12, trace=True,
suppress_warnings=True)
```

Best model: ARIMA(2,1,1)(0,1,0)[12]

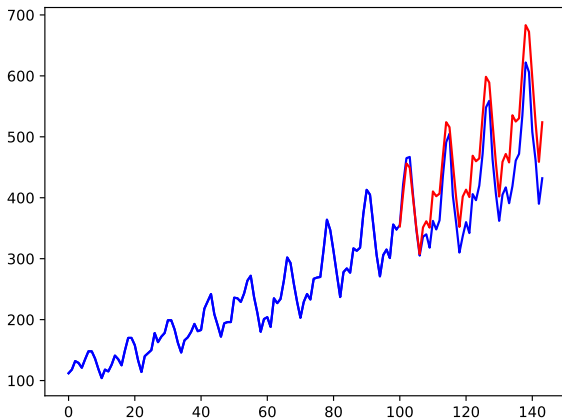
```
model.summary()
```

	coef	std err	z	P> z	[0.025	0.975]
intercept	0.0179	0.010	1.800	0.072	-0.002	0.037
ar.L1	0.6151	0.089	6.948	0.000	0.442	0.789
ar.L2	0.2366	0.092	2.559	0.010	0.055	0.418
ma.S.L12	-0.5564	0.120	-4.626	0.000	-0.792	-0.321
sigma2	0.0013	0.000	7.466	0.000	0.001	0.002

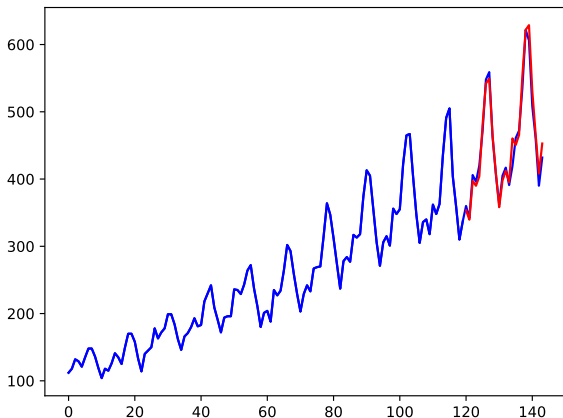
Forecasting with PMDARIMA: 4

```
#make forecasts
forecasts = model.predict(test.shape[0])
y = np.exp(y)
train = np.exp(train)
forecasts = np.exp(forecasts)
# Visualize the forecasts (blue=train, red=forecasts)
x = np.arange(y.shape[0])
plt.plot(x[:100], train, c='blue')
plt.plot(y, c = 'blue')
plt.plot(x[100:], forecasts, c='red')
plt.show()
```

Airline Data, Training = 100, Testing = 40



Airline Data, Training = 120, Testing = 20

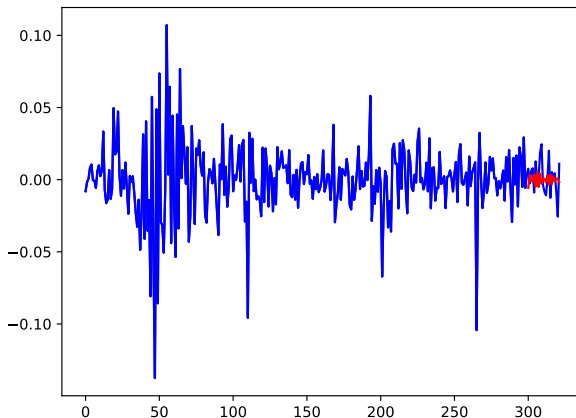


Forecasting IBM 1/1/20-4/14/21 with PMDARIMA

Best model: ARIMA(2,0,2)(0,0,0)[12]

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	-1.7709	0.049	-36.153	0.000	-1.867	-1.675
ar.L2	-0.9062	0.046	-19.660	0.000	-0.997	-0.816
ma.L1	1.6792	0.070	23.899	0.000	1.542	1.817
ma.L2	0.7848	0.071	11.107	0.000	0.646	0.923
sigma2	0.0005	2.55e-05	21.408	0.000	0.000	0.001

IBM Data 1/1/20-4/14/21, Training = 300



ARCH/GARCH Modeling

Background: 1

- ▶ Thus far in the course, we have considered time series models, $\{X_t\}$, which are driven by a weakly stationary noise process, $\{\epsilon_t\}$, which, therefore, have constant variance.
- ▶ This implies that for models like AR, MA, ARMA and ARIMA, the time series also has constant variance.
- ▶ A stylized fact about most financial markets is that volatility is stochastic, and it tends to cluster into periods of low and high volatility.
- ▶ Consequently, standard time series models are not suitable for financial market data.
- ▶ Transformations like Box-Cox won't help as they cannot create volatility clusters.

Background: 2

- ▶ It is important to think about what sort of model we would want to introduce to create volatility clustering.
- ▶ Consider some of the graphs from early in the course - IBM, XOM, S&P 500 etc. The time series plots exhibit extremely different volatility patterns over time, and they tend to cluster.
- ▶ But the volatility clusters tend to be at the same time for all the stocks.
- ▶ This is indicative of a common factor that is impacting all of the stocks simultaneously, rather than the volatility patterns being created within the stocks themselves.

Background: 3

- ▶ GARCH models create volatility patterns “by themselves,” i.e. it is built into their own stochastic model.
- ▶ Stochastic volatility models create an independent or correlated volatility process which then determines the instantaneous volatility of the stock(s).
- ▶ We will consider both in the coming days.

Background: 4

Some of the characteristics about volatility that have been noted in stock price behavior include:

- ▶ Volatility clustering
- ▶ Volatility is continuous, not a jump process
- ▶ Volatility does not diverge to infinity; in fact it is approximately stationary
- ▶ Volatility increases more following big price drops than after big price increases (leverage effect)

Background: 5

- ▶ For a weakly stationary time series model, $\{X_t, t \geq 0\}$
- ▶ If we define $\mu_t = E(X_t|\mathcal{F}_{t-1})$ and $\sigma_t^2 = \text{Var}(X_t|\mathcal{F}_{t-1})$, then for most models we have considered (AR, MA, ARMA, ARIMA), μ_t is non-constant and σ_t^2 is constant.
- ▶ For example, for an AR(1) process,

$$E(X_t|\mathcal{F}_{t-1}) = E(\phi_0 + \phi_1 X_{t-1} + \epsilon_t|\mathcal{F}_{t-1}) = \phi_0 + \phi_1 X_{t-1}$$

$$\text{Var}(X_t|\mathcal{F}_{t-1}) = \text{Var}(\phi_0 + \phi_1 X_{t-1} + \epsilon_t|\mathcal{F}_{t-1}) = \text{Var}(\epsilon_t) = \sigma_\epsilon^2.$$

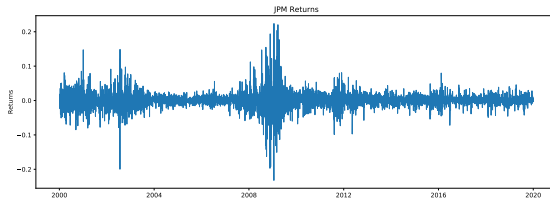
- ▶ The models we next develop are the opposite: μ_t is constant and σ_t^2 is non-constant.

Background: 6

There are two distinct approaches to developing models with these properties:

- ▶ Deterministic Models: $\text{Var}(X_t|\mathcal{F}_{t-1})$ is a deterministic function. ARCH/GARCH models fall into this category
- ▶ Stochastic Models: $\text{Var}(X_t|\mathcal{F}_{t-1})$ is a stochastic process. Stochastic Volatility models (SV) fall into this category.

JPM and GS Returns



ARCH Models: 1

Following and quoting Ruppert p405, we discuss non-constant variance modeling.

- Suppose

$$Y_t = f(X_{1,t}, \dots, X_{p,t}) + \epsilon_t.$$

where $\{\epsilon_t\}$ is independent of the $\{X_{i,t}, 1 \leq i \leq p, t \geq 0\}$ and has constant mean 0 and variance σ_ϵ^2 .

- $E(Y_t | (X_{1,t}, \dots, X_{p,t})) = f(X_{1,t}, \dots, X_{p,t})$ and $Var(Y_t | (X_{1,t}, \dots, X_{p,t})) = \sigma_\epsilon^2$

ARCH Models: 2

- ▶ It is possible to introduce **conditional heteroscedasticity** by letting

$$\text{Var}(Y_t | (X_{1,t}, \dots, X_{p,t})) = \sigma^2(X_{1,t}, \dots, X_{p,t}).$$

- ▶ The model

$$Y_t = f(X_{1,t}, \dots, X_{p,t}) + \epsilon_t \sigma(X_{1,t}, \dots, X_{p,t}),$$

where ϵ_t has, conditional on $(X_{1,t}, \dots, X_{p,t})$, mean 0 and variance 1, and $\sigma(\cdot)$ is non-negative since it is a standard deviation.

- ▶ If $\sigma(\cdot)$ is linear, then the coefficients need to be constrained so that it does not become negative for any values of the X variables.
- ▶ Generally, and in the GARCH case, the function $\sigma(\cdot)$ is non-negative and non-linear. These models are called **variance function models**, and GARCH will use the $\sqrt{\cdot}$ function

ARCH Models: 3

- ▶ The constant volatility models are called **conditionally homoscedastic**, whereas the models with stochastic volatility are called **conditionally heteroscedastic**.
- ▶ The first model, $\{a_t, t \geq 0\}$, is ARCH(1), **Autoregressive Conditionally Heteroscedastic**:

$$E(\epsilon_t | \epsilon_{t-1}, \epsilon_{t-2}, \dots) = 0,$$

$$\text{Var}(\epsilon_t | \epsilon_{t-1}, \epsilon_{t-2}, \dots) = 1,$$

$$a_t = \epsilon_t \sqrt{\omega + \alpha a_{t-1}^2},$$

a variance function model with $f = 0$ and

$$\sigma_t = \sqrt{\omega + \alpha a_{t-1}^2}$$

ARCH Models: 4

- ▶ Ruppert's text takes $\{\epsilon_t, t \geq 0\}$ to be iid Gaussian white noise, but he indicates that more general assumptions are possible.
- ▶ The further assumption $\omega > 0$ and $0 \leq \alpha < 1$ are made to ensure that the argument of the $\sqrt{\cdot}$ is positive.
- ▶ The $\{a_t^2, t \geq 0\}$ process is given by:

$$a_t^2 = \epsilon_t^2(\omega + \alpha a_{t-1}^2).$$

- ▶ This is somewhat similar to the AR(1) model except:
 - ▶ The variable is squared, a_t^2 rather than a_t , and
 - ▶ The noise process, ϵ_t is multiplicative rather than additive, and
 - ▶ The noise term, ϵ_t^2 , is strictly positive and has mean $E(\epsilon_t^2) = 1$.

ARCH Models: 5

We first compute the conditional mean and variance of the a_t process. Let $\mathcal{F}_t$ be the sigma field generated by $\{a_s, s \leq t\}$, then

$$E(a_t | \mathcal{F}_{t-1}) = E(\epsilon_t \sqrt{\omega + \alpha a_{t-1}^2} | \mathcal{F}_{t-1}) = \sqrt{\omega + \alpha a_{t-1}^2} E(\epsilon_t | \mathcal{F}_{t-1}) = 0.$$

$$E(a_t | \mathcal{F}_{t-h}) = E(E(a_t | \mathcal{F}_{t-1}) | \mathcal{F}_{t-h}) = E(0 | \mathcal{F}_{t-h}) = 0.$$

$$\begin{aligned}\sigma_t^2 &= \text{Var}(a_t | \mathcal{F}_{t-1}) \\ &= E(a_t^2 | \mathcal{F}_{t-1}) \\ &= E((\omega + \alpha a_{t-1}^2) \epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= (\omega + \alpha a_{t-1}^2) E(\epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= \omega + \alpha a_{t-1}^2.\end{aligned}$$

ARCH Models: 6

- ▶ Ruppert p408 “If a_{t-1} has an unusually large absolute value, then σ_t is larger than usual and so a_t is also expected to have an unusually large magnitude. This volatility propagates since when a_t has a large magnitude that makes σ_{t+1}^2 large, then a_{t+1} tends to be large in magnitude, and so on,”
- ▶ The same argument applies when a_{t-1} is unusually small.
- ▶ “Because of this behavior, unusual volatility in a_t tends to persist, though not forever. The conditional variance tends to revert to the unconditional variance provided that $\alpha < 1$, so the process is stationary with finite variance.”

ARCH Models: 7

- ▶ Taking expectations we get

$\gamma_a(0) = E(\sigma_t^2) = E(\omega + \alpha a_{t-1}^2) = \omega + \alpha \gamma_a(0)$. Assuming $0 \leq \alpha < 1$ and solving for $\gamma(0)$, we find

$$\gamma_a(0) = \frac{\omega}{1 - \alpha}.$$

- ▶ We can find the covariance function for a_t using similar conditioning methods. For $h > 0$:

$$\begin{aligned}\gamma_a(h) &= E(a_t a_{t-h}) \\ &= E(E(a_t a_{t-h} | \mathcal{F}_{t-h})) \\ &= E(a_{t-h} E(a_t | \mathcal{F}_{t-h})) \\ &= E(a_{t-h} * 0) \\ &= 0.\end{aligned}$$

ARCH Models: 8

- ▶ It follows that the ACF of the ARCH(1) process a_t is $\rho_a(h) = 1$ if $h = 0$ and 0 if $h \neq 0$, a white noise process.
- ▶ The conditional variance expression $\sigma_t^2 = \omega + \alpha a_{t-1}^2$ gives insight into the behavior of ARCH(1) and GARCH models.
- ▶ If a_{t-1}^2 takes on an unusually large value, then $\sigma_t^2 = \omega + \alpha a_{t-1}^2$ will be unusually large, so we expect a_t to be large also (note though $\alpha < 1$). We expect this to create a cluster of large values for the a_t process, although the persistence depends importantly on the size of α .
- ▶ The same argument holds if a_{t-1}^2 takes on an unusually small value. There will tend to be a cluster of small volatility values.
- ▶ In both cases there is mean reversion to the mean value with α governing the speed of mean reversion.

AR(1)/ARCH(1)

We can combine the ARCH noise process with the AR time series model to achieve a process both of whose conditional mean and variance depend on the past.

- ▶ Suppose $\{a_t, t \in \mathcal{Z}\}$ is an ARCH(1) noise process where

$$a_t = \sqrt{\omega + \alpha a_t^2} \epsilon_t,$$

where $\{\epsilon_t, t \in \mathcal{Z}\}$ consists of i.i.d. $N(0,1)$ random variables, $\omega > 0$, $0 \leq \alpha < 1$, $|\beta| < 1$, and define

$$X_t = \mu + \beta(X_{t-1} - \mu) + a_t.$$

- ▶ $\rho_X(h) = \beta^{|h|}$ and $\rho_{a^2} = \alpha^{|h|}$.
- ▶ This model has non-constant conditional mean and variance.

ARCH(p)

- ▶ The order of the ARCH process can easily be increased from 1 to $p > 1$. Let

$$a_t = \sigma_t \epsilon_t,$$

where

$$\sigma_t = \sqrt{\omega + \sum_{i=1}^p \alpha_i a_{t-i}^2}.$$

- ▶ The ARCH(p) process is uncorrelated and has constant conditional and unconditional mean. Its unconditional variance is also constant, but the conditional variance is nonconstant.
- ▶ The ACF of a_t^2 is the same as the ACF of an AR(p) process.

Financial Time Series Analysis

Lecture 9

- ▶ TA Session Friday April 23, 2021, 11:00am - 12:00pm (EDT)
- ▶ TA Session Monday April 26, 2021, 7:30pm - 8:30pm (EDT)

Outline

- ▶ ARCH Review
- ▶ Properties of ARCH(1)
- ▶ Connection with Stylized Facts
- ▶ Generalization to GARCH
- ▶ Data Example

ARCH Review: 1

- ▶ Recall that ARCH/GARCH models are for the volatility component of a time series model, i.e. they are MDS, hence white noise, e.g. AR(1) + ARCH(1).
- ▶ ARCH is an abbreviation for AutoRegressive Conditional Heteroschedasticity. The conditional and unconditional mean of an ARCH model is 0, the unconditional variance is constant, but the conditional variance depends upon the past variability.
- ▶ $\{a_t, -\infty < t < \infty\}$ is ARCH(1) if

$$a_t = \sigma_t \epsilon_t, \text{ and } \sigma_t = \sqrt{\omega + \alpha a_{t-1}^2},$$

where

$\{\epsilon_t, -\infty < t < \infty\}$ is iid with $E(\epsilon_t) = 0$ and $E(\epsilon_t^2) = 1$.

ARCH Review: 2

- ▶ σ_t^2 is the conditional variance at time t , and letting $\mathcal{F}_t = \sigma\{a_s, s \leq t\}$, σ_t is $\mathcal{F}_{t-1}$ measurable, so $E(\sigma_t|\mathcal{F}_{t-1}) = \omega + \alpha a_{t-1}^2$, a deterministic function given $\mathcal{F}_{t-1}$.
- ▶ $E(a_t) = E(\epsilon_t \sigma_t) = E(E(\epsilon_t \sigma_t | \mathcal{F}_{t-1})) = E(\sigma_t E(\epsilon_t | \mathcal{F}_{t-1})) = E(\epsilon_t \cdot 0) = 0$.
- ▶ $Var(a_t) = E(a_t^2) = E(\epsilon_t^2 \sigma_t^2) = E(\epsilon_t^2 (\omega + \alpha a_{t-1}^2)) = 1(\omega + \alpha E(a_{t-1}^2))$.
- ▶ Assuming weak stationarity (therefore constant variance)
- next slide - we solve for $Var(a_t) = \omega / (1 - \alpha)$.

ARCH Properties: 1

$$\begin{aligned}a_t^2 &= \epsilon_t^2 \sigma_t^2 \\&= \epsilon_t^2 (\omega + \alpha a_{t-1}^2) \\&= \omega \epsilon_t^2 + \alpha a_{t-1}^2 \epsilon_t^2 \\&= \omega \epsilon_t^2 + \alpha \epsilon_t^2 \sigma_{t-1}^2 \epsilon_{t-1}^2 \\&= \omega \epsilon_t^2 + \alpha \epsilon_t^2 \epsilon_{t-1}^2 (\omega + \alpha a_{t-2}^2) \\&= \dots \\&= \omega \sum_{j=0}^n \alpha^j \prod_{i=0}^{t-j} \epsilon_{t-i}^2 + \alpha^{n+1} a_{t-n-1}^2 \prod_{j=0}^n \epsilon_{t-j}^2\end{aligned}$$

If $|\alpha| < 1$, the second term converges to 0, thus

$$a_t^2 = \omega \sum_{j=0}^{\infty} \alpha^j \prod_{i=0}^{t-j} \epsilon_{t-i}^2.$$

Clearly $E(a_t^2) = \omega \sum_{j=0}^{\infty} \alpha^j = \omega / (1 - \alpha)$.

ARCH Properties: 2

- One can express $\{a_t^2\}$ as an AR(1) process because

$$\begin{aligned}a_t^2 &= \sigma_t^2 + a_t^2 - \sigma_t^2 \\&= \omega + \alpha a_{t-1}^2 + \sigma_t^2 \epsilon_t^2 - \sigma_t^2 \\&= \omega + \alpha a_{t-1}^2 + \sigma_t^2 (\epsilon_t^2 - 1) \\&= \omega + \alpha a_{t-1}^2 + \nu_t.\end{aligned}$$

Thus $\{a_t^2\}$ is an AR(1) process with noise process $\{\nu_t\}$, where $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$.

- We show that $\{\nu_t\}$ is a white noise process assuming $E(\epsilon_t^4) < \infty$.

ARCH Properties: 3

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \nu_t,$$

where $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$.

- ▶ $E(\nu_t) = E(E(\sigma_t^2(\epsilon_t^2 - 1)|\mathcal{F}_{t-1})) = E(\sigma_t^2 E(\epsilon_t^2 - 1|\mathcal{F}_{t-1})) = E(\sigma_t^2 \cdot 0) = 0$.
- ▶ $\gamma_\nu(h) = E(\nu_{t+h}\nu_t) = E(\sigma_{t+h}^2 \sigma_t^2 (\epsilon_{t+h}^2 - 1)(\epsilon_t^2 - 1)) = E(\epsilon_{t+h}^2 - 1)E((\epsilon_t^2 - 1)\sigma_{t+h}^2 \sigma_t^2) = 0, h \geq 1$. Thus $\{\nu_t\}$ a white noise process.
- ▶ We only need to calculate $Var(\nu_t) = E(\sigma_t^4(\epsilon_t^2 - 1)^2) = E(\sigma_t^4)E(\epsilon_t^4 - 2 \cdot 1 + 1) = 2E(\sigma_t^4)$ assuming ϵ_t has a $N(0,1)$ distribution.

ARCH Properties: 4

Because a_t^2 is an AR(1) process, we know

$$E(a_t^2) = \omega / (1 - \alpha),$$

$$\text{Var}(a_t^2) = \text{Var}(\nu_t) / (1 - \alpha^2) = 2E(\sigma_t^4) / (1 - \alpha^2).$$

To calculate $E(\sigma_t^4)$,

$$\begin{aligned} E(\sigma_t^4) &= E((\omega + \alpha a_{t-1}^2)^2) \\ &= \omega^2 + 2\alpha\omega(\omega / (1 - \alpha)) + \alpha^2(\text{Var}(a_{t-1}^2) + E(a_{t-1}^2)^2) \\ &= \omega^2 + \frac{2\omega^2\alpha}{(1 - \alpha)} + \frac{2E(\sigma_t^4)\alpha^2}{1 - \alpha^2} + \frac{\omega^2\alpha^2}{(1 - \alpha)^2} \end{aligned}$$

Solving for $E(\sigma_t^4)$, we find

ARCH Properties: 5

$$E(\sigma_t^4) \frac{1 - 3\alpha^2}{1 + \alpha} = \frac{\omega^2}{1 - \alpha}.$$

- ▶ This requires $\alpha^2 < 1/3$ or $0 \leq \alpha < .577$, assuming that ϵ_t has a $N(0,1)$ distribution.
- ▶ Solving the above equation for $E(\sigma_t^4)$:

$$E(\sigma_t^4) = \frac{\omega^2(1 + \alpha)}{(1 - 3\alpha^2)(1 - \alpha)}.$$

- ▶ It follows that

$$\begin{aligned} E(a_t^4) &= E(\sigma_t^4 \epsilon_t^4) = 3E(\sigma_t^4) \\ &= 3 \frac{\omega^2(1 + \alpha)}{(1 - \alpha)(1 - 3\alpha^2)} \\ &= 3 \left(\frac{\omega}{(1 - \alpha)} \right)^2 \frac{1 - \alpha^2}{1 - 3\alpha^2} \\ &= 3(E(a_t^2))^2 \frac{1 - \alpha^2}{1 - 3\alpha^2} \end{aligned}$$

ARCH Properties: 6

- ▶ Since the last factor is larger than 1, we conclude that $E(a_t^4)/(E(a_t^2)^2)$ is larger than 3, which is indicative of heavier tails than the $N(0,1)$ distribution.
- ▶ Note also that the AR(1) structure of $\{a_t^2\}$,

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \nu_t,$$

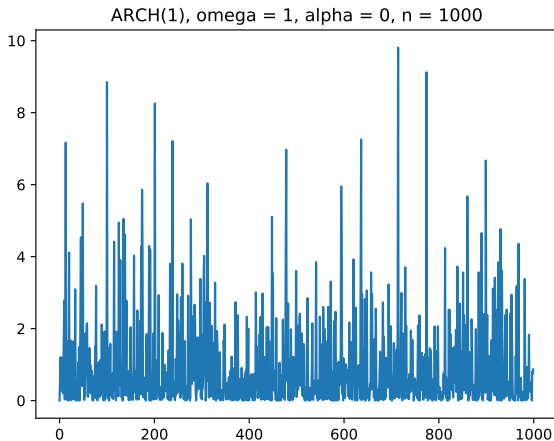
where $\{\nu_t\}$ is white noise allows us to conclude that

$$\rho_{a^2}(h) = \alpha^h \geq 0.$$

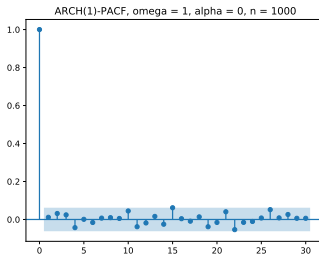
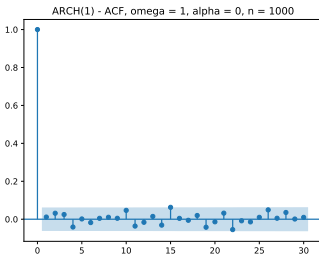
Thus the ACF of $\{a_t^2\}$ is non-negative.

- ▶ That a_t^2 and a_{t+h}^2 are positively correlated (especially for small values of h) is known as the **ARCH Effect**.

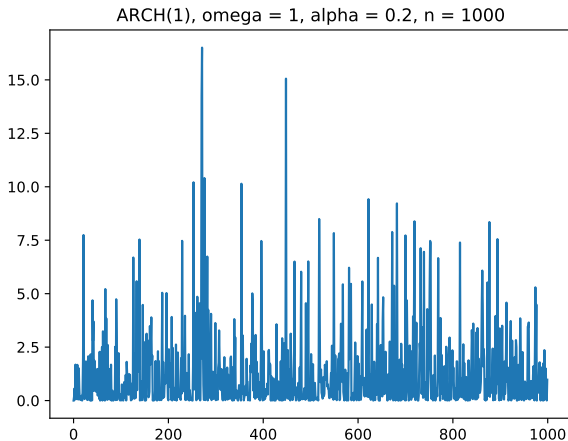
$$a_t^2, \omega = 1, \alpha = 0, n = 1000$$



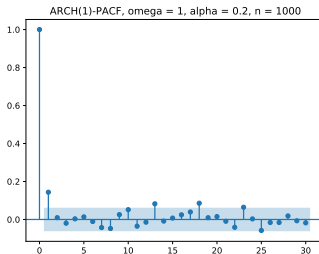
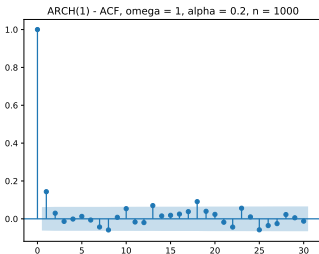
$$a_t^2, \omega = 1, \alpha = 0, n = 1000$$



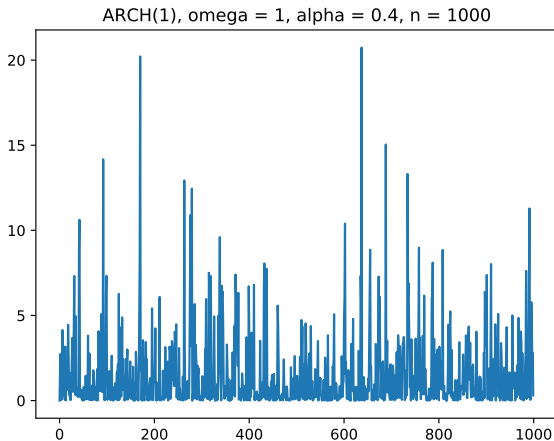
$$a_t^2, \omega = 1, \alpha = .2, n = 1000$$



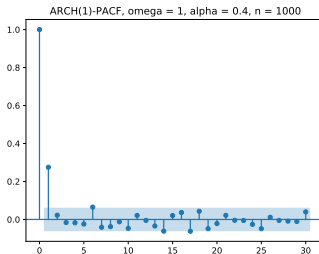
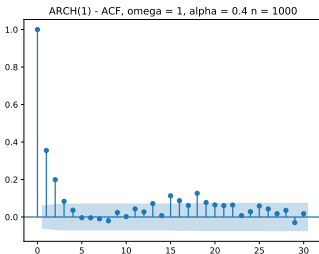
$$a_t^2, \omega = 1, \alpha = .2, n = 1000$$



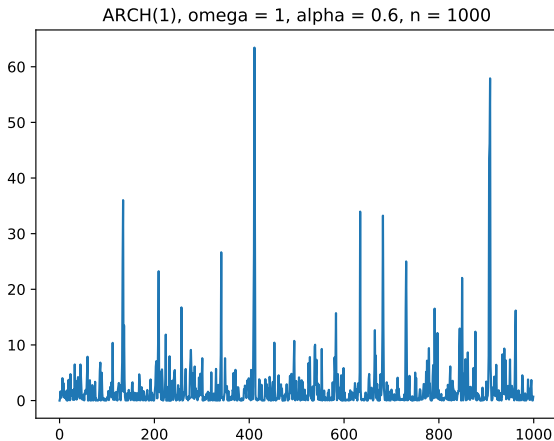
$$a_t^2, \omega = 1, \alpha = .4, n = 1000$$



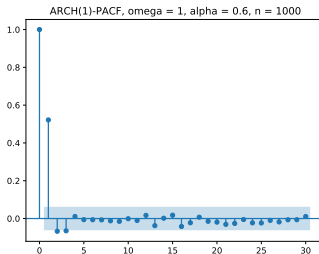
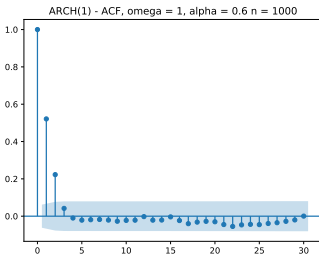
$$a_t^2, \omega = 1, \alpha = .4, n = 1000$$



$$a_t^2, \omega = 1, \alpha = .6, n = 1000$$



$$a_t^2, \omega = 1, \alpha = .6, n = 1000$$



ARCH and Stylized Facts

The ARCH(1) model does support some of the stylized facts:

- ▶ (Absence of Autocorrelation) - ARCH noise models are MDS and are white noise, so $Cov(a_t, a_{t+h}) = 0$, $h \neq 0$; however, $Cov(a_t^2, a_{t+h}^2) > 0$ (ARCH Effect). This implies volatility clustering.
- ▶ Prices are non-stationary, but returns are weakly stationary.
- ▶ Distribution of returns have heavier tails than the normal distribution.
- ▶ ARCH models do not support asymmetry or the leverage effect. $a_t = \epsilon_t \sigma_t = \epsilon_t \sqrt{\omega + \alpha a_{t-1}^2}$. Since the returns depend on the square of a_t , there is no difference between positive and negative values of a_{t-1} . There are variations of the ARCH/GARCH model that overcome this drawback.

Weaknesses of ARCH Model

R. Tsay (2002) notes some weaknesses of the ARCH Model:

- ▶ The model assumes that positive and negative shocks have the same effects on volatility because the model assumes that the next shock depends only on the square of the previous shock(s). The stylized facts indicate that the returns respond differently to positive and negative shocks.
- ▶ The model is restrictive. With $\{\epsilon_t\}$ assumed to be iid $N(0,1)$, $E(a_t^4)$ exists only if $\alpha^2 < 1/3$.
- ▶ The ARCH model does not provide any insight into understanding the source of volatility. It only provides a mechanical way to describe the behavior of the conditional variance. It gives no indication about what causes such behavior to occur. Furthermore, it applies to single stocks and does not account for the effects that are occurring across market segments.

AR(1)/ARCH(1)

We can combine the ARCH noise process with the AR time series model to achieve a process both of whose conditional mean and variance depend on the past.

- ▶ Suppose $\{a_t, t \in \mathcal{Z}\}$ is an ARCH(1) noise process where

$$a_t = \sqrt{\omega + \alpha a_t^2} \epsilon_t,$$

where $\{\epsilon_t, t \in \mathcal{Z}\}$ consists of i.i.d. $N(0,1)$ random variables, $\omega > 0$, $0 \leq \alpha < 1$, $|\beta| < 1$, and define

$$X_t = \mu + \beta(X_{t-1} - \mu) + a_t.$$

- ▶ $\rho_X(h) = \beta^{|h|}$ and $\rho_{a^2} = \alpha^{|h|}$.
- ▶ This model has non-constant conditional mean and variance.

ARCH(p)

- ▶ The order of the ARCH process can easily be increased from 1 to $p > 1$. Let

$$a_t = \sigma_t \epsilon_t,$$

where

$$\sigma_t = \sqrt{\omega + \sum_{i=1}^p \alpha_i a_{t-i}^2}.$$

- ▶ The ARCH(p) process is uncorrelated and has constant conditional and unconditional mean. Its unconditional variance is also constant, but the conditional variance is nonconstant.
- ▶ The ACF of a_t^2 is the same as the ACF of an AR(p) process.

From ARCH to GARCH

- ▶ The original ARCH model was created by R. Engle in 1982. It was generalized to GARCH (generalized autoregressive conditional heteroskedasticity) by Tim Bollerslev in 1986.
- ▶ The GARCH model allows the values of the X process to impact the evolution of the conditional variance, σ_t .
- ▶ The GARCH(p,q) model is given by $\omega \geq 0, \alpha_i \geq 0, \beta_j > 0$:

$$\begin{aligned}a_t &= \sigma_t \epsilon_t, \{\epsilon_t\} \text{ is } N(0, 1) \\ \sigma_t^2 &= \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2\end{aligned}$$

- ▶ There are many generalizations of GARCH(p,q) including EGARCH, GARCH with leverage, IGARCH, Multivariate GARCH, etc.

GARCH(1,1):1

Consider a GARCH(1,1) model with $a_t = \sigma_t \epsilon_t$, and $\{\epsilon_t, t \geq 0\}$ (i.i.d. white noise), a

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2,$$

an ARMA(1,1) process but with a weak white noise error process.

$$\begin{aligned} E(a_t^2 | \mathcal{F}_{t-1}) &= E(\sigma_t^2 \epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= E(\sigma_t^2 | \mathcal{F}_{t-1}) \cdot 1 \\ &= E(\omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2 | \mathcal{F}_{t-1}) \\ &= \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned}$$

GARCH(1,1):2

- ▶ Define $\eta_t = a_t^2 - \sigma_t^2$
- ▶ $E(\eta_t | \mathcal{F}_{t-1}) = E(a_t^2 - \sigma_t^2 | \mathcal{F}_{t-1}) = 0$.
- ▶ For $s < t$, $E(\eta_s \eta_t) = E(\eta_s E(\eta_t | \mathcal{F}_s)) = 0$, by conditioning on $\mathcal{F}_{t-1}$.
- ▶ Consequently the $\{\eta_t, t \geq 0\}$ process is weak white noise.
- ▶ Some algebra gives

$$\sigma_t^2 = \omega + (\alpha + \beta)a_{t-1}^2 - \beta\eta_{t-1}.$$

GARCH(1,1):3

Let $c = \omega / (1 - (\alpha + \beta))$

$$\begin{aligned}a_t^2 - c &= -c + \sigma_t^2 + \eta_t \\&= -c + \omega + (\alpha + \beta)a_{t-1}^2 - \beta\eta_{t-1} + \eta_t \\&= (\alpha + \beta)(a_{t-1}^2 - c) - \beta\eta_{t-1} + \eta_t\end{aligned}$$

- ▶ It follows that $\{a_t^2, t \geq 0\}$ is an ARMA(1,1) process with mean c and AR(1) coefficient $(\alpha + \beta)$.



$$\rho_{a^2}(h) = (\alpha + \beta)^h \frac{\alpha(1 - \alpha\beta - \beta^2)}{1 - 2\alpha\beta - \beta^2}$$

GARCH(1,1):4

- ▶ It follows that $\{a_t^2, t \geq 0\}$ is an ARMA(1,1) process with mean ν and AR(1) coefficient $\phi_1 = \alpha_1 + \beta_1$, and MA(1) coefficient $\theta_1 = -\beta_1$.



$$\rho_{a^2}(h) = (\alpha_1 + \beta_1)^h \frac{\alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)}{1 - 2\alpha_1\beta_1 - \beta_1^2}$$

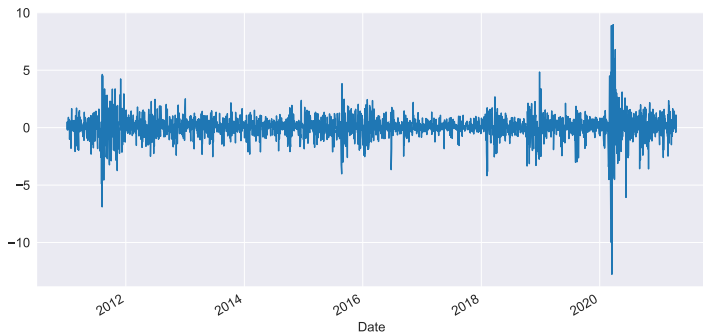
Fitting ARCH to S&P 500 Data: 1

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import datetime as dt
import pandas_datareader as pdr
start = dt.datetime(2011, 1, 1)
end = dt.datetime(2021, 4, 16)
sandp = pdr.get_data_yahoo('GSPC', start, end)
sandp_ac = sandp['Adj Close']
logret = 100*(np.log(sandp_ac/sandp_ac.shift(1)).dropna())
```

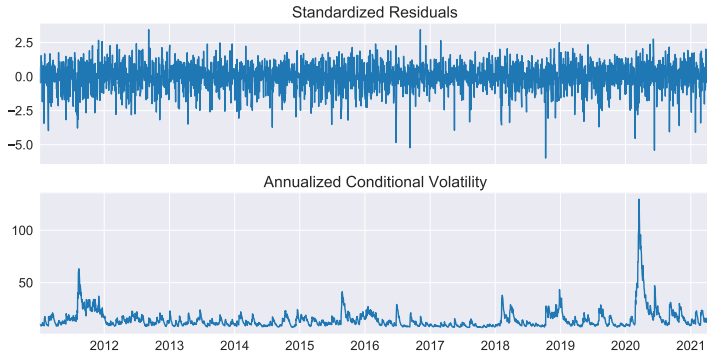

Fitting ARCH to S&P 500 Data: 2

```
figure = logret.plot()  
plt.show()  
from arch import arch_model  
am = arch_model(logret)  
res = am.fit(update_freq=5)  
print(res.summary())  
fig = res.plot(annualize='D')  
plt.show()
```

S&P 500



S&P 500



GARCH(p,q):1

- ▶ We have written the GARCH(1,1) as an ARMA(1,1) model with an MDS noise process. This can be generalized to the GARCH(p,q)

$$a_t = \sigma_t \epsilon_t,$$
$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

where we assume $\{\epsilon_t\}$ are iid with mean 0 and variance 1.

- ▶ If we define $\alpha_i = 0$ if $i > p$ and $\beta_j = 0$ if $j \geq q$, then letting $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \sigma_t^2 + \eta_t$, we find:

$$\begin{aligned} a_t^2 &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j (a_t^2 - \sigma_t^2) \\ &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j \eta_t \end{aligned}$$

GARCH(p,q):2

- ▶ η_t is a MDS, because $E(a_t^2|\mathcal{F}_{t-1}) = \sigma_t^2$, so
 $E(\eta_t|\mathcal{F}_{t-1}) = E(a_t^2|\mathcal{F}_{t-1}) - \sigma_{t-1}^2 = \sigma_t^2 - \sigma_{t-1}^2 = 0$.
- ▶ Consequently, $\{a_t^2\}$ is an ARMA(p,q) with AR coefficients $\alpha_j + \beta_j$ and MA coefficients β_j .
- ▶ The variance of η_t is constant assuming $E(\epsilon_t^4) < \infty$
Consequently, η_t is a white noise process.
- ▶ Because the error process is a white noise process, and all of our previous results hold.

GARCH(p,q):3

- ▶ To ensure that the $\{a_t^2\}$ process is weakly stationary, we need the roots of the polynomial

$$1 - \sum_{j=1}^{\max p, q} (\alpha_j + \beta_j) z^j,$$

to lie outside the unit circle.

- ▶ We usually assume that all of the α_i and β_j are positive, the weak stationary condition becomes

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1.$$

- ▶ If this condition holds, then

$$E(a_t^2) = \omega / (1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j).$$

GARCH Forecasting: 1

- ▶ Suppose we wish to forecast X_{t+k} for some $k \geq 1$ where the $\{X_t\}$ process is $\text{ARMA}(p_m, q_m) + \text{GARCH}(p_v, q_v)$.
- ▶ The $\text{GARCH}(p_v, q_v)$ noise process is white noise, and we developed ARMA forecasting assuming white noise errors. So the point forecasts of X_{t+k} will be the same as if we assumed the errors were iid Gaussian.
- ▶ What is different is that the GARCH noise process will impact the variability of the forecast, specifically the forecast intervals.

GARCH Forecasting: 2

- ▶ We showed that a $\text{GARCH}(p_v, q_v)$ can be represented as an ARMA process with the MDS.
- ▶ Consequently, we can use the ARMA forecasting techniques on the GARCH model.
- ▶ We will first review ARMA forecasting and then apply it to GARCH forecasting.

GARCH Forecasting: 3

- ▶ We showed that the GARCH(1,1) can be written as an ARMA process with MDS noise, hence $WN(0, \sigma_a^2)$. Let us review how to forecast ARMA(p,q) processes:

$$X_t = \phi_0 + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t.$$

- ▶ Recall that for forecasting a future value, say $\hat{X}_{t+k}$, we use the defining model above and forecast future values of the process, $(\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+k})$ sequentially.
- ▶ At each stage, we plug in the current data $(X_t, X_{t-1}, \dots, X_{t-k})$ as called for by the model, and we plug in the estimates of the unknown parameters.

GARCH Forecasting: 4

- ▶ This leaves the ϵ terms in the equation. For current and previous values of ϵ , we plug in $(\hat{\epsilon}_t, \hat{\epsilon}_{t-1}, \dots, \hat{\epsilon}_{t-q})$ going backward in time, and we plug in 0 going forward, i.e. $\hat{\epsilon}_{t+i} = 0$ for $i \geq 1$.
- ▶ For a GARCH(1,1) volatility process with parameters $\alpha_1 > 0$ and $\beta_1 > 0$ with $\alpha_1 + \beta_1 < 1$,

$$a_t^2 = \epsilon_t^2(\omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_t).$$

- ▶ Let the constant unconditional variance be σ^2 . Taking expectations across the equation, we find

$$\sigma^2 = \omega / (1 - \alpha_1 - \beta_1).$$

GARCH Forecasting: 5

- ▶ Suppose that the current time is t and we want to forecast $\hat{\sigma}_{t+k}^2$, the k -step ahead forecast of the volatility given $\mathcal{F}_t$
- ▶ Consider the one-step ahead forecast, $\hat{\sigma}_{t+1}^2$. We have

$$\sigma_{t+1}^2 = \omega + \alpha_1 a_t^2 + \beta_1 \sigma_t^2.$$

The right-side is $\mathcal{F}_t$ measurable, so the above equation provides the one-step-ahead forecast, although estimates must be substituted for the unknown parameters, $\omega, \alpha_1, \beta_1$.

- ▶ For multi-step forecasts, $k \geq 1$, recall that $a_t^2 = \epsilon_t \sigma_t^2$. So the volatility equation can be rewritten as

$$\sigma_{t+1}^2 = \omega + \alpha_1 \epsilon_t \sigma_t^2 + \beta_1 \sigma_t^2$$

GARCH Forecasting: 6

- ▶ or rearranging

$$\sigma_{t+1}^2 = \omega + (\alpha_1 + \beta_1)\sigma_t^2 + \alpha_1\sigma_t^2(\epsilon_t^2 - 1).$$

- ▶ The forecast at time $t + 2$ is $E(\hat{\sigma}_{t+2}^2 | \mathcal{F}_t)$.
- ▶ Changing $t + 1$ to $t + 2$ in the above equation and taking $E(\cdot | \mathcal{F}_t)$ and noting $E(\epsilon_{t+1} - 1 | \mathcal{F}_t) = 0$, we find

$$\hat{\sigma}_{t+2}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+1}^2.$$

This generalizes to

GARCH Forecasting: 7

$$\hat{\sigma}_{t+k}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+k-1}.$$

We let $\lambda = \alpha_1 + \beta_1$ and iterate backward:

$$\begin{aligned}\hat{\sigma}_{t+k}^2 &= \omega + \lambda\hat{\sigma}_{t+k-1}^2 \\ &= \omega(1 + \lambda) + \lambda^2\hat{\sigma}_{t+k-2}^2 \\ &= \dots \\ &= \omega(1 + \lambda + \dots + \lambda^{k-2}) + \lambda^{k-1}\hat{\sigma}_{t+1}^2\end{aligned}$$

As $k \rightarrow \infty$, assuming $\lambda < 1$

$$\hat{\sigma}_{t+k}^2 \rightarrow \sigma^2 = \omega/(1 - \lambda).$$

The **half-life** of the volatility difference is approximately

$$\lambda^T = \frac{1}{2}, \text{ so } T \approx -\frac{\log 2}{\log \lambda} = -\frac{.693}{\log \lambda}$$

Financial Time Series Analysis

Lecture 10

- ▶ TA Session: Friday, April 23, 11:00am-12:00pm
- ▶ TA Session: Monday, April 26, 7:30-8:30pm

Outline

- ▶ Review GARCH(1,1)
- ▶ Representing GARCH processes as ARMA processes
- ▶ Forecasting GARCH Models
- ▶ Example by R. Engle relevant to Risk Management
- ▶ Other GARCH models

From ARCH to GARCH

- ▶ The original ARCH model was created by R. Engle in 1982. It was generalized to GARCH (generalized autoregressive conditional heteroskedasticity) by Tim Bollerslev in 1986.
- ▶ The GARCH model allows the values of the X process to impact the evolution of the conditional variance, σ_t .
- ▶ The GARCH(p,q) model is given by $\omega > 0, \alpha_i > 0, \beta_j > 0$:

$$a_t = \sigma_t \epsilon_t,$$

$$\{\epsilon_t\} \text{ is i.i.d., } E(\epsilon_t) = 0, E(\epsilon_t^2) = 1, E(\epsilon_t^4) < \infty,$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

GARCH(1,1):1

Consider a GARCH(1,1) model with $a_t = \sigma_t \epsilon_t$, and $\{\epsilon_t, t \geq 0\}$ (i.i.d. $E(\epsilon_t) = 0, E(\epsilon_t^2) = 1, E(\epsilon_t^4) < \infty$),

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2,$$

We will show $\{a_t^2\}$ is an ARMA(1,1) process.

$$\begin{aligned} E(a_t^2 | \mathcal{F}_{t-1}) &= E(\sigma_t^2 \epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= \sigma_t^2 E(\epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= \sigma_t^2 E(\epsilon_t^2) \\ &= \sigma_t^2 \cdot 1 \\ &= \sigma_t^2 \end{aligned}$$

GARCH(1,1):2

- ▶ Define $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \eta_t + \sigma_t^2$
- ▶ $E(\eta_t | \mathcal{F}_{t-1}) = E(a_t^2 - \sigma_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2 - \sigma_t^2 = 0$.
- ▶ Consequently the $\{\eta_t, t \geq 0\}$ is a MDS and weak white noise, provided it has constant variance (homoscedastic).
- ▶ Some algebra gives

$$\begin{aligned}\sigma_t^2 &= \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 (a_{t-1}^2 - \sigma_{t-1}^2) \\ &= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 \eta_{t-1}\end{aligned}$$

GARCH(1,1):3

► Thus

$$a_t^2 = \sigma_t^2 + \eta_t = \omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_{t-1} + \eta_t.$$

- Let $\nu = \omega/(1 - (\alpha_1 + \beta_1))$, so $\omega = \nu(1 - (\alpha_1 + \beta_1))$.
- Taking expectations across the equation and assuming stationarity, we find $E(a_t^2) = \nu$ and

$$a_t^2 = \nu(1 - (\alpha_1 + \beta_1)) + (\alpha_1 + \beta_1)(a_{t-1}^2) - \beta_1\eta_{t-1} + \eta_t,$$

or

$$a_t^2 - \nu = (\alpha_1 + \beta_1)(a_{t-1}^2 - \nu) - \beta_1\eta_{t-1} + \eta_t.$$

GARCH(1,1):4

- ▶ It follows that $\{a_t^2, t \geq 0\}$ is an ARMA(1,1) process with mean ν and AR(1) coefficient $\phi_1 = \alpha_1 + \beta_1$, and MA(1) coefficient $\theta_1 = -\beta_1$.
- ▶ Recall that for an ARMA(1,1), given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

$$\rho_X(h) = \phi_1^{h-1} \frac{(\phi_1 + \theta_1)(1 + \phi_1 \theta_1)}{1 + 2\phi_1 \theta_1 + \theta_1^2},$$

so

▶

$$\rho_{a^2}(h) = (\alpha_1 + \beta_1)^{h-1} \frac{\alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)}{1 - 2\alpha_1\beta_1 - \beta_1^2}$$

GARCH(p,q):1

- ▶ We have written the GARCH(1,1) as an ARMA(1,1) model with an MDS noise process. This can be generalized to the GARCH(p,q)

$$a_t = \sigma_t \epsilon_t,$$
$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i a_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

where we assume $\{\epsilon_t\}$ are iid with mean 0 and variance 1.

- ▶ If we define $\alpha_i = 0$ if $i > p$ and $\beta_j = 0$ if $j > q$, then letting $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \sigma_t^2 + \eta_t$, we find:

$$\begin{aligned} a_t^2 &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j (a_{t-j}^2 - \sigma_{t-j}^2) \\ &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j \eta_{t-j} \end{aligned}$$

GARCH(p,q):2

- ▶ η_t is a MDS, because $E(a_t^2|\mathcal{F}_{t-1}) = \sigma_t^2$, so $E(\eta_t|\mathcal{F}_{t-1}) = E(a_t^2|\mathcal{F}_{t-1}) - \sigma_{t-1}^2 = \sigma_{t-1}^2 - \sigma_{t-1}^2 = 0$.
- ▶ Consequently, $\{a_t^2\}$ is an ARMA(p,q) with AR coefficients $\alpha_i + \beta_i$ and MA coefficients $-\beta_i$.
- ▶ The variance of η_t is constant assuming $E(\epsilon_t^4) < \infty$ This can be done following the methods given in slides 7-9 in Lecture 9 given there for the ARCH(1) process. Consequently, η_t is an MDS and a white noise process.
- ▶ Because the error process is a white noise process, $\{a_t^2\}$ is an ARMA(p,q) process.

GARCH(p,q):3

- ▶ To ensure that the $\{a_t^2\}$ process is weakly stationary, we need the roots of the polynomial

$$1 - \sum_{j=1}^{\max(p,q)} (\alpha_j + \beta_j)z^j,$$

to lie outside the unit circle.

- ▶ We usually assume that all of the α_j and β_j are positive, the weak stationary condition becomes

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1.$$

- ▶ If this condition holds, then

$$E(a_t^2) = \omega / (1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j).$$

GARCH Forecasting: 1

- ▶ Suppose we wish to forecast X_{t+k} for some $k \geq 1$ where the $\{X_t\}$ process is $\text{ARMA}(p_m, q_m) + \text{GARCH}(p_v, q_v)$.
- ▶ The $\text{GARCH}(p_v, q_v)$ noise process is white noise, and we developed ARMA forecasting assuming white noise errors. So the point forecasts of X_{t+k} will be the same as if we assumed the errors were iid Gaussian.
- ▶ What is different is that the GARCH noise process will impact the variability of the forecast, specifically the forecast intervals.
- ▶ This is particularly important is **risk management**, as the concern is the quantile (e.g. 95% or 99%) of the loss distribution. The variance will impact the forecast of that quantile.

GARCH Forecasting: 2

- ▶ We showed that a $\text{GARCH}(p_v, q_v)$ can be represented as an ARMA process with an MDS/white noise error process.
- ▶ Consequently, we can use the ARMA forecasting techniques on the GARCH model.
- ▶ We will first review ARMA forecasting and then apply it to GARCH forecasting.

GARCH Forecasting: 3

- ▶ We showed that the GARCH(1,1) can be written as an ARMA process with MDS noise, hence $WN(0, \sigma_a^2)$. Let us review how to forecast ARMA(p,q) processes:

$$X_t = \phi_0 + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t.$$

- ▶ Recall that for forecasting a future value, say $\hat{X}_{t+k}$, we use the defining model above and forecast future values of the process, $(\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+k})$ sequentially.
- ▶ At each stage, we plug in the current data $(X_t, X_{t-1}, \dots, X_{t-k})$ as called for by the model, and we plug in the estimates of the unknown parameters.

GARCH Forecasting: 4

- ▶ This leaves the ϵ_t terms in the equation. For current and previous values of ϵ_t , we plug in $(\hat{\epsilon}_t, \hat{\epsilon}_{t-1}, \dots, \hat{\epsilon}_{t-q})$ going backward in time, and we plug in 0 going forward, i.e. $\hat{\epsilon}_{t+i} = 0$ for $i \geq 1$.
- ▶ For a GARCH(1,1) volatility process with parameters $\alpha_1 > 0$ and $\beta_1 > 0$ with $\alpha_1 + \beta_1 < 1$,

$$a_t^2 = \epsilon_t^2(\omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_t).$$

- ▶ Let the constant unconditional variance be σ_a^2 . Taking expectations across the equation, we find

$$\sigma_a^2 = \omega / (1 - \alpha_1 - \beta_1).$$

GARCH Forecasting: 5

- ▶ Suppose that the current time is t and we want to forecast $\hat{\sigma}_{t+k}^2$, the k -step ahead forecast of the volatility given $\mathcal{F}_t$
- ▶ Consider the one-step ahead forecast, $\hat{\sigma}_{t+1}^2$. We have

$$\sigma_{t+1}^2 = \omega + \alpha_1 a_t^2 + \beta_1 \sigma_t^2.$$

The right-side is $\mathcal{F}_t$ measurable, so the above equation provides the one-step-ahead forecast, although estimates must be substituted for the unknown parameters, $\omega, \alpha_1, \beta_1$.

- ▶ For multi-step forecasts, $k \geq 1$, recall that $a_t^2 = \epsilon_t^2 \sigma_t^2$. So the volatility equation can be rewritten as

$$\sigma_{t+1}^2 = \omega + \alpha_1 \epsilon_t^2 \sigma_t^2 + \beta_1 \sigma_t^2$$

GARCH Forecasting: 6

- ▶ Or rearranging

$$\sigma_{t+1}^2 = \omega + (\alpha_1 + \beta_1)\sigma_t^2 + \alpha_1\sigma_t^2(\epsilon_t^2 - 1).$$

- ▶ The forecast at time $t + 2$ is $E(\hat{\sigma}_{t+2}^2 | \mathcal{F}_t)$.
- ▶ Changing $t + 1$ to $t + 2$ in the above equation and taking $E(\cdot | \mathcal{F}_t)$ and noting $E(\epsilon_{t+1}^2 - 1 | \mathcal{F}_t) = 0$, we find

$$\hat{\sigma}_{t+2}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+1}^2.$$

This generalizes to

GARCH Forecasting: 7

$$\hat{\sigma}_{t+k}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+k-1}.$$

We let $\lambda = \alpha_1 + \beta_1$ and iterate backward:

$$\begin{aligned}\hat{\sigma}_{t+k}^2 &= \omega + \lambda\hat{\sigma}_{t+k-1}^2 \\ &= \omega(1 + \lambda) + \lambda^2\hat{\sigma}_{t+k-2}^2 \\ &= \dots \\ &= \omega(1 + \lambda + \dots + \lambda^{k-2}) + \lambda^{k-1}\hat{\sigma}_{t+1}^2\end{aligned}$$

As $k \rightarrow \infty$, assuming $\lambda < 1$

$$\hat{\sigma}_{t+k}^2 \rightarrow \sigma^2 = \omega/(1 - \lambda).$$

The **half-life** of the volatility difference is approximately

$$\lambda^T = \frac{1}{2}, \text{ so } T \approx -\frac{\log 2}{\log \lambda} = -\frac{.693}{\log \lambda}$$

Engle Test for ARCH Effects

IBM 1/1/2011 - 4/16/21

Fit by GARCH(1,1)

```
am = arch_model(logret)
```

```
res = am.fit(update_freq=5)
```

```
std_resid = res.resid/res.conditional_volatility
```

```
statsmodels.stats.diagnostic.het_arch(std_resid)
```

(7.865826004902203, 0.9999359555318335, Engle test

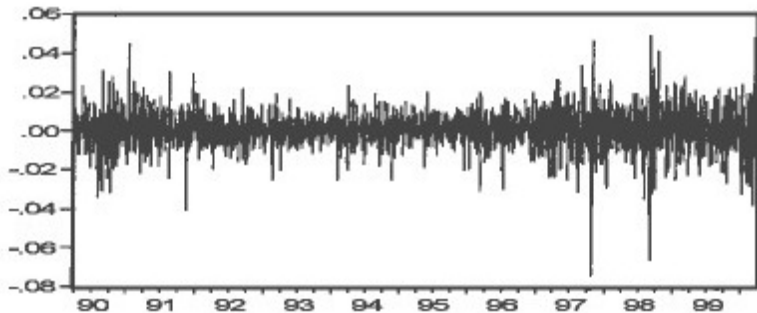
0.27859604468156574, 0.9999390381653693) F test

GARCH Forecasting Example

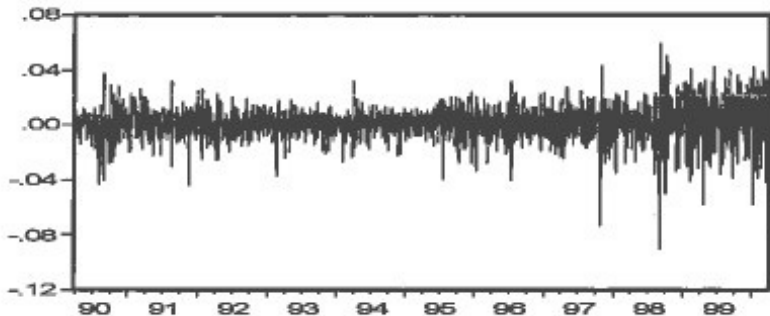
R. Engle presented an example of the use of GARCH models in Risk Management, 'GARCH 101: The Use of ARCH/GARCH models in applied econometrics,' *Journal of Economic Perspectives*, **15(4)**, 2001, 157-168.

- ▶ He used a GARCH(1,1) model to estimate $\text{VaR}_{.99}$ for a \$1M portfolio on March 23, 2000.
- ▶ The portfolio consists of 50% NASDAQ, 30% Dow Jones and 20% long bonds (10 year constant maturity treasury bond). The proportion was kept constant through rebalancing.
- ▶ This date was chosen to be just before a large market slide and a time of high volatility.
- ▶ Consider the 10 year prior history:

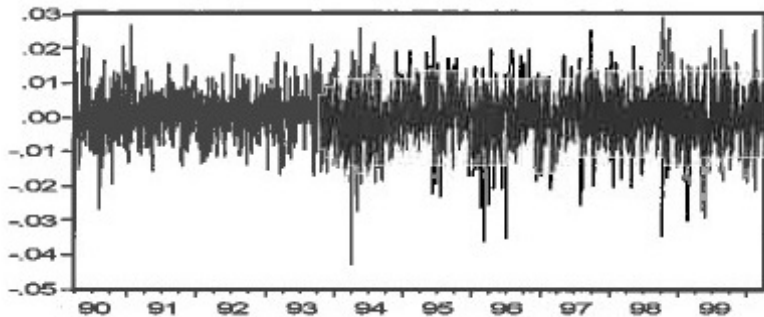
Dow Jones



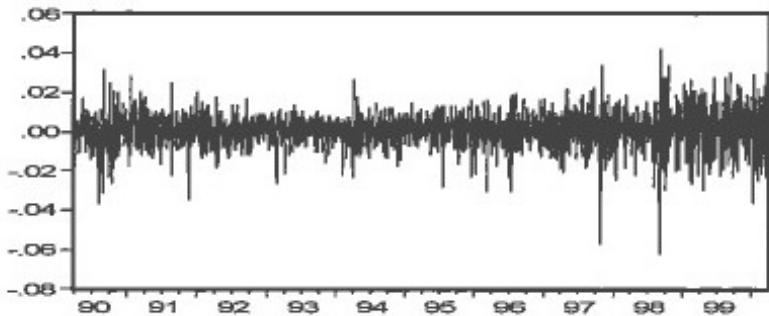
NASDAQ



Rates



Portfolio



Relevance to Risk Management: 1

3/23/1990 to 3/23/2000

	Dow Jones	NASDAQ	Rate	Portfolio
Mean	0.0005	0.0009	0.0001	0.0007
Std Dev	0.0090	0.0115	0.0073	0.0083
Skewness	-0.3593	-0.5310	-0.2031	-0.4738
Kurtosis	8.3288	7.4936	4.9579	7.0026

Relevance to Risk Management: 2

- ▶ If we consider a portfolio whose current value is $V(t)$. Suppose the portfolio is held constant over a time window of length Δ . The loss over this period is
$$L = V(t) - V(t + \Delta)$$
- ▶ L is a random variable whose value depends upon Δ and the change in risk factors over the interval (stock prices, rates, commodity prices, etc.).
- ▶ A risk measure is some “feature” of this random variable, like its standard deviation.
- ▶ VaR = value at risk is a particular quantile of the loss distribution.
- ▶ Shortfall risk is $E(L|L > Var_\alpha)$

Relevance to Risk Management: 3

- ▶ The VaR value can be computed by a number of methods:
 - ▶ Historical data - look at the last n values of the portfolio. Use the α quantile of the empirical distribution to estimate the VaR value.
 - ▶ Model the loss distribution and compute the quantile from the model.
 - ▶ Forecast, say using GARCH, to generate a point prediction for the portfolio and a standard deviation for the time of the forecast.. Use these two quantities to generate a VaR (quantile) estimate.

Relevance to Risk Management: 4

“I now turn to the question of how the econometrician can possibly estimate an equation like the GARCH(1,1) equation:

$$X_t = m_t + \epsilon_t \sigma_t,$$

$$\sigma_{t+1}^2 = \omega + \alpha(X_t - m_t)^2 + \beta\sigma_t^2.$$

when the only variable on which there are data is $\{X_t\}$, the returns or log-returns. The simple answer is to use maximum likelihood by substituting σ_t^2 in the normal likelihood and then maximizing with respect to the parameters. Many software packages will facilitate this automatically.”

Relevance to Risk Management: 5

The values of the three parameters in the GARCH(1,1) model are

- ▶ $\omega = 1.4E - 06(.0018)$
- ▶ $\alpha_1 = .0772(.0000)$
- ▶ $\beta_1 = .9046(.0000)$

“Consider a trader who knows that the long-run average daily standard deviation of the S&P 500 is 1 percent, that the forecast made yesterday was 2 percent and the unexpected return observed today is 3 percent. Obviously, this is a high volatility period, and today is especially volatile, which suggests that the forecast for tomorrow could be even higher. However, the fact that the long-term average is only 1 percent might lead the forecaster to lower the forecast.”

ACF of m_t

Table 2
Autocorrelations of Squared Portfolio Returns

	<i>AC</i>	<i>Q-Stat</i>	<i>Prob</i>
1	0.210	115.07	0.000
2	0.183	202.64	0.000
3	0.116	237.59	0.000
4	0.082	255.13	0.000
5	0.122	294.11	0.000
6	0.163	363.85	0.000
7	0.090	384.95	0.000
8	0.099	410.77	0.000
9	0.081	427.88	0.000
10	0.081	445.03	0.000
11	0.069	457.68	0.000
12	0.080	474.29	0.000
13	0.076	489.42	0.000
14	0.074	503.99	0.000
15	0.083	521.98	0.000

Sample: March 23, 1990 to March 23, 2000.

ACF of a_t^2 model

Table 4
Autocorrelations of Squared Standardized Residuals

	AC	Q-Stat	Prob
1	0.005	0.0589	0.808
2	0.039	4.0240	0.134
3	-0.011	4.3367	0.227
4	-0.017	5.0981	0.277
5	0.002	5.1046	0.403
6	0.009	5.3228	0.503
7	-0.015	5.8836	0.553
8	-0.013	6.3272	0.611
9	-0.024	7.8169	0.553
10	-0.006	7.9043	0.638
11	-0.023	9.3163	0.593
12	-0.013	9.7897	0.634
13	-0.003	9.8110	0.709
14	0.009	10.038	0.759
15	-0.012	10.444	0.791

Relevance to Risk Management: 6

“The best strategy depends upon the dependence between days. If these 3 numbers are each squared and weighted equally, then the new forecast would be

$$\sqrt{(1 + 4 + 9)/3} = 2.16.$$

However, rather than weighting these equally, it is generally found for daily data that weights such as those in the empirical example (.02, .9, .08) are much more accurate. Hence the forecast is

$$\sqrt{.02 * 1 + .9 * 4 + .08 * 9} = 2.08.$$

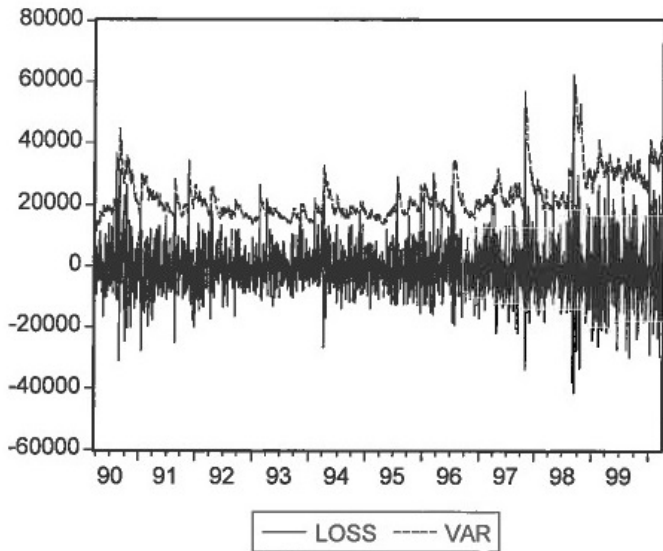
Relevance to Risk Management: 7

- ▶ The $\text{VaR}_{.99}$ values for the portfolio calculated from historical data:
 - ▶ \$22,477 using the 10-year sample
 - ▶ \$24,653 using the most recent year
 - ▶ \$35,159 using 1/1/2000 to 3/23/2000
- ▶ Fitting a GARCH(1,1) model: parameters estimates and p-value:
 - ▶ $\omega = 1.4E - 06(.0018)$
 - ▶ $\alpha_1 = .0772(.0000)$
 - ▶ $\beta_1 = .9046(.0000)$

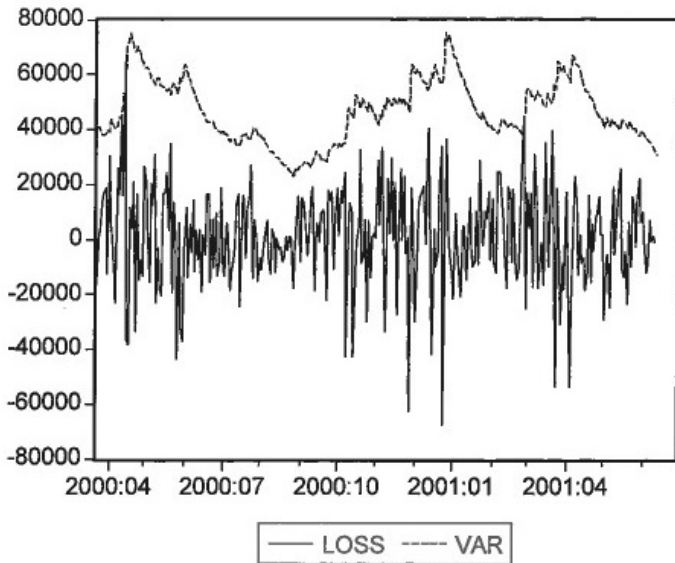
Relevance to Risk Management: 8

- ▶ Good model fit and the estimated coefficients sum to less than 1, but are close to 1, so we have slow mean reversion.
- ▶ The forecast next day standard deviation is 0.0146, which is much larger than .0083, the 10 year average standard deviation. So this day is in a period of high volatility.
- ▶ The residuals are relatively heavy tailed, so multiply by 2.844 (instead of 2.327 for a standard normal distribution).
- ▶ $\text{VaR}_{.99}$ value is \$39,996.

Portfolio



Portfolio



Other Volatility Models: 1

- ▶ There are many variations on the GARCH model to address shortcomings and extend its applicability. We mention just a few of the more common ones.
- ▶ Consider **GARCH-M (GARCH in the mean)**. Normally, we have a mean model and a separate volatility model.
- ▶ The GARCH-M model includes a volatility term in the mean equation. This can arise because it might be appropriate to include a risk premium associated with volatility in the returns part of the model

Other Volatility Models: 2

- ▶ This model can be expressed as

$$X_t = \mu + c\sigma_t^2 + a_t,$$

$$a_t = \epsilon_t \sigma_t,$$

$$\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

- ▶ Here μ and c are constant. If $c > 0$ then the return process is positively related to the past volatility. Other specifications of the volatility term have been used including $c\sigma_t$ and $c \log(\sigma_t^2)$.

Other Volatility Models: 3

- ▶ One notable shortcoming of the GARCH model is the presence of volatility terms being squared ($a_{t-1}^2, \sigma_{t-1}^2$)
- ▶ This means that both positive and negative changes are treated the same in updating the volatility equation.
- ▶ However, one of the stylized facts is that the relationship should be asymmetric, with larger increases in volatility coming from negative changes than come from positive changes. the **EGARCH (Exponential GARCH)** model addresses this shortcoming.
- ▶ This was introduced in 1991 by Nelson. He considered weighting the innovations:

Other Volatility Models: 4

- ▶ Nelson began with the $\{\epsilon_t, t \geq 0\}$ innovations and weighted them $\{g(\epsilon_t), t \geq 0\}$ where

$$g(\epsilon_t) = \theta\epsilon_t + \gamma[|\epsilon_t| - E(|\epsilon_t|)],$$

where θ and γ are constants. Both components are iid with mean 0. The asymmetry can be seen by writing out the two cases:

$$g(\epsilon_t) = \begin{cases} (\theta + \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t \geq 0, \\ (\theta - \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t < 0, \end{cases}$$

Financial Time Series Analysis

Lecture 11

- ▶ TA Session: Friday, April 30, 11:00am-12:00pm
- ▶ TA Session: Monday, May 3, 7:30-8:30pm

Outline

- ▶ Final Exam, Monday, May 10, 10:00am (EDT)
- ▶ Other GARCH Models
- ▶ Multiple Time Series
 - ▶ Basic Details
 - ▶ VAR Processes (Vector Autoregressive Processes)
- ▶ Cointegration

Nature of the Final Exam: 1

- ▶ Final exam is comprehensive, from stylized facts, stationarity and MDS to state space models (lecture 13).
- ▶ Exam length will be 1.5 or 2 hours, partly multiple choice, partly calculations (like homework calculation problems).
- ▶ Administered under ProctorTrack and Canvas quiz with an open zoom link.
- ▶ Closed book but one 8.5x11 (two-sided) cheat sheet) is permitted. No other computational or communication devices except to download and upload the exam.

Nature of the Final Exam: 2

- ▶ The test **is not** intended to be a test of memory, especially the exact use of time series commands like: `statsmodels.tsa.stattools.adfuller`
- ▶ Rather the test is intended to see whether you have an understanding of the time series concepts, models and methods and their appropriate use.
- ▶ I will post a previous exam next week to help you study for the exam.
- ▶ The basis for the final will be the lecture slides and the homework problems. The Course Notes also provide a study guide.

Other Volatility Models: 1

- ▶ There are many variations on the GARCH model to address shortcomings and extend its applicability. We mention just a few of the more common ones.
- ▶ Consider **GARCH-M (GARCH in the mean)**. Normally, we have a mean model and a separate volatility model.
- ▶ The GARCH-M model includes a volatility term in the mean equation. This can arise because it might be appropriate to include a risk premium associated with volatility in the returns part of the model (see homework problem 5.1).

Other Volatility Models: 2

- ▶ This model can be expressed as

$$X_t = \mu + c\sigma_t^2 + a_t,$$

$$a_t = \epsilon_t \sigma_t,$$

$$\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

- ▶ Here μ and c are constant. If $c > 0$ then the return process is positively related to the past volatility. Other specifications of the volatility term have been used including $c\sigma_t$ and $c \log(\sigma_t^2)$.

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- ▶ This means that both positive and negative changes are treated the same in updating the volatility equation.
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- ▶ This was introduced in 1991 by Nelson. He considered weighting the innovations:

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- ▶ Nelson began with the $\{\epsilon_t, t \geq 0\}$ innovations and weighted them $\{g(\epsilon_t), t \geq 0\}$ where

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where θ and γ are constants. Both components are iid with mean 0. The asymmetry can be seen by writing out the two cases:

$$g(\epsilon_t) = \begin{cases} (\theta + \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t \geq 0, \\ (\theta - \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t < 0, \end{cases}$$

For the leverage effect we would expect to have $\theta < 0$ and $\gamma > 0$.

Other Volatility Models: 5

The name, E-GARCH, comes from the modeling of σ_t :

$$a_t = \sigma_t \epsilon_t, \{\epsilon_t\}, i.i.d., E(\epsilon_t) = 0,$$

$$\log(\sigma_t^2) = \omega + \sum_{i=1}^p \beta_i \log(\sigma_{t-i}^2) + \sum_{j=1}^q g_j(\epsilon_{t-j}),$$

$$g_j(\epsilon_t) = \alpha_j \epsilon_t + \gamma_j (|\epsilon_t| - E(|\epsilon_t|)).$$

- ▶ The model can retain the good properties of the GARCH model and in addition add the leverage effect.
- ▶ The Python package ARCH includes a number of variations on the GARCH model including E-GARCH, FIGARCH, and APARCH among others.

Multivariate Time Series: 1

- ▶ Many of the ideas we have studied for univariate time series carry over to the multivariate case. This is important because we would like to conduct an analysis that not only deals with the individual aspects of some asset but its relationship with other assets.
- ▶ Unfortunately, as we go to the multivariate case, the number of parameters can skyrocket - parameters for the individual components and parameters that deal with dependency relationships.
- ▶ We begin with a m -dimensional time series $\mathbf{X}_t = (X_{1t}, \dots, X_{mt})$.

Multivariate Time Series: 2

- ▶ As each component is a time series, each will have a mean function μ_{it} and an autocovariance function, $\Gamma_{ii}(t+h, t) = \text{Cov}(X_{i(t+h)}, X_{it}) = E(X_{i(t+h)}X_{it}) - \mu_{it}^2$.
- ▶ The covariance function (now a matrix) is given by

$$\mathbf{\Gamma}(t+h, t) = E((\mathbf{X}_{t+h} - \boldsymbol{\mu}_{t+h})(\mathbf{X}_t - \boldsymbol{\mu}_t)^T).$$

- ▶ We can define a weakly stationary multivariate time series by requiring that the first two moments, the mean vector, $\boldsymbol{\mu}_t$ and the autocovariance matrix, $\mathbf{\Gamma}(t+h, t)$, are independent of t .
- ▶ The diagonal elements of the $\mathbf{\Gamma}(h)$, $\gamma_{ii}(h)$, are the autocovariance functions of the individual component time series.

Multivariate Time Series: 3

- ▶ We can also create the autocorrelation matrix, $\mathbf{R}(h) = \rho_{ij}(h)$, and a typical entry can be computed by

$$\rho_{ij}(h) = \frac{\gamma_{ij}(h)}{(\gamma_{ii}(0)\gamma_{jj}(0))^{1/2}}.$$

- ▶ From these definitions, we can define a vector white noise process.

Definition

$\{\epsilon_t, t \geq 0\}$ is $\text{WN}(\mathbf{0}, \mathbf{\Sigma})$ if and only if

- ▶ The ϵ_t process is weakly stationary,
- ▶ $\mu = \mathbf{0}$,
- ▶

$$\Gamma(h) = \begin{cases} \mathbf{\Sigma} & \text{if } h = 0, \\ \mathbf{0} & \text{if } h \neq 0. \end{cases}$$

Multivariate Time Series: 4

Note that $\mathbf{R} = (R_{ij})$.

- ▶ $R_{ii}(h)$ is the autocorrelation function of X_{it} .
- ▶ $R_{ij}(h) = \rho(X_{i(t+h)}, X_{jt})$ is called the **cross-correlation function (CCF)** of $\{X_{it}\}$ and $\{X_{jt}\}$
- ▶ Note that $R_{ij}(h) = \rho(X_{i(t+h)}, X_{jt}) \neq \rho(X_{it}, X_{j(t+h)})$.
- ▶ Rather $R_{ij}(-h) = \rho(X_{it}, X_{j(t+h)}) = \rho(X_{j(t+h)}, X_{it}) = R_{ji}(h)$
- ▶ In matrix form, $\mathbf{R}(h) = \mathbf{R}(-h)^T$.
- ▶ The sample estimates for the first and second moments are:

Multivariate Time Series: 5

The usual estimate of the common mean of $\mathbf{X}$, $E(\mathbf{X}) = \boldsymbol{\mu} = (\mu_1, \dots, \mu_m)$ is the sample mean,

$$\bar{\mathbf{X}}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i.$$

- ▶ Under the assumption that the multivariate time series is weakly stationary, then the sample mean, $\bar{\mathbf{X}}_n$, converges to $\boldsymbol{\mu}$ and is asymptotically normal.
- ▶ That is, $\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ converges weakly to a $N_d(\mathbf{0}, \boldsymbol{\Sigma}^*)$ where $\boldsymbol{\Sigma}^*$ depends upon the properties of the underlying white noise process.

Multivariate Time Series: 6

- ▶ We next turn to the estimation:

$$\boldsymbol{\Gamma}(h) = E((\mathbf{X}_{t+h} - \boldsymbol{\mu})(\mathbf{X}_t - \boldsymbol{\mu})^T),$$

- ▶ The obvious sample estimate is

$$\hat{\boldsymbol{\Gamma}}(h) = \frac{1}{n} \sum_{i=1}^{n-h} (\mathbf{X}_{t+h} - \bar{\mathbf{X}}_n)(\mathbf{X}_t - \bar{\mathbf{X}}_n)^T, \quad 0 \leq h \leq n-1.$$

- ▶ Letting $\hat{\boldsymbol{\Gamma}}(h) = (\hat{\gamma}_{ij})$ and $\hat{\mathbf{R}}(h) = (\hat{\rho}_{ij})$, we have



$$\hat{\rho}_{ij}(h) = \frac{\hat{\gamma}_{ij}(h)}{(\hat{\gamma}_{ii}(0)\hat{\gamma}_{jj}(0))^{1/2}}.$$

Vector Autoregressive Processes

- ▶ We want to deal with the simultaneous behavior of financial time series in pairs trading and portfolio construction.
- ▶ The generalization of univariate AR processes to multivariate AR processes is straightforward. These are called **Vector Autoregressive Processes** or VAR(p).
- ▶ Let $\mathbf{X}_t = (X_{1t}, \dots, X_{mt})$ be an m-dimensional vector time series satisfying

$$\mathbf{X}_t = \mathbf{a}_0 + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\epsilon}_t,$$

where $\mathbf{a}_0$ is an m-dimensional vector, $\mathbf{A}_1, \dots, \mathbf{A}_p$ are $m \times m$ matrices and $\{\boldsymbol{\epsilon}_t\}$ are i.i.d. m-dimensional random vectors with mean $\mathbf{0}$ and some covariance matrix $\boldsymbol{\Gamma}$.

- ▶ For stationarity, we would want each of the individual series to be stationary as well as the vector process.

Stationarity of VAR(1) Processes

- ▶ Suppose we consider the VAR(1) process

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \epsilon_t.$$

- ▶ One can calculate $\Sigma_{\mathbf{X}_t}$ by iteration. In the simple case with $\mathbf{a} = 0$ and $\Gamma = \sigma^2 \mathbf{I}$, then as $t \rightarrow \infty$

$$\Sigma_{\mathbf{X}_t} \rightarrow \sigma^2 (\mathbf{I} - \mathbf{A}\mathbf{A}^T)^{-1}.$$

- ▶ This is the multivariate analog of the univariate result in which

$$\text{Var}(X_t) \rightarrow \frac{\sigma^2}{1 - \beta^2}.$$

- ▶ Need the roots of the equation $\det(\mathbf{I} - \mathbf{A}x) = 0$ to have modulus larger than 1, in analogy with the AR(1) process.

Stationarity of VAR(p) processes

- ▶ For the VAR(p) processes, the stationarity condition is similar to that for AR(p). Consider the roots of the polynomial equation

$$\det\left(I - \sum_{i=1}^p \mathbf{A}_i \lambda^i\right) = 0.$$

- ▶ The VAR(p) process is stationary if and only if the roots of the above equation have modulus strictly larger than 1.

Cointegration: 1

In the introduction to their 1987 paper, Engle and Granger wrote

An individual economic variable, viewed as a time series, can wander extensively and yet some pairs of series may be expected to move so that they do not drift too far apart. Typically economic theory will propose forces which tend to keep such series together. Examples might be short and long term interest rates, capital appropriations and expenditures, household income and expenditures, and prices of the same commodity in different markets or close substitutes in the same market.

Cointegration: 2

- ▶ In financial markets, the variables in question are stock prices. No single stock price can be expected to be stationary but will vary, potentially widely.
- ▶ The crucial idea is to find a subset of these processes that “move together” in the sense that some linear combination is a stationary process.
- ▶ If two stock price processes are “cointegrated,” then over the long run these two processes are “tied together” and make a good candidate for a trading strategy.
- ▶ This strategy, known as “pairs trading” would track two stock price paths (denominated in standard units). We expect to see mean reversion.

Cointegration: 3

- ▶ If one of the stocks' price rises above the other, when the difference is sufficiently large, short the “overpriced” stock and long the “underpriced” stock. Close the trade when they revert to their mean.
- ▶ This strategy is considered to be zero cost and market neutral.
- ▶ Zero cost because your portfolio is long and short the same amount of money (although there may be extra costs in shorting stocks and other technical difficulties).
- ▶ The strategy is considered to be market neutral. As long as the cointegration relationship holds, it does not matter if markets move up or down, as this pair is moving together, and the profit comes from collecting the spread at the time the trade is initiated.

Gatev et al: Pairs Trading

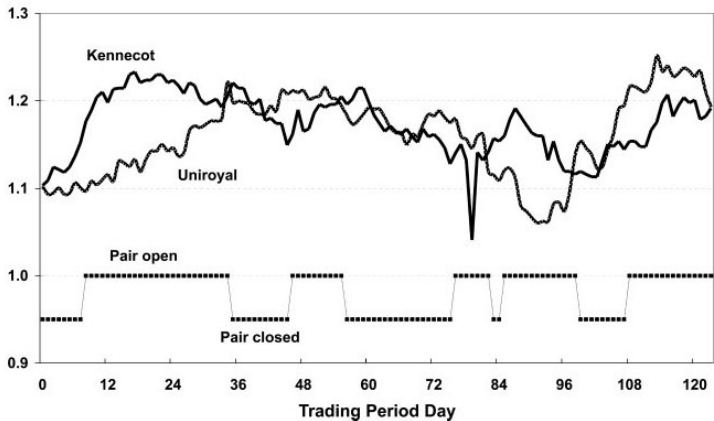


Figure 1
Daily normalized prices: Kennecott and Uniroyal (pair 5)
Trading period August 1963–January 1964.

Cointegration: 4

- ▶ Definition: The components of $\mathbf{X}$ are said to be **cointegrated of order d, b** (denoted $CI(d, b)$) if:
 - a) All components of $\mathbf{X}_t$ are $\mathcal{I}(d)$,
 - b) There exists a vector $\alpha \neq \mathbf{0}$ for which if $Z_t = \alpha \mathbf{X}_t$ is $\mathcal{I}(d - b)$ with $b > 0$.
- ▶ The vector α is called a **cointegrating vector**.

Cointegration: 5

- ▶ Recall that 2 or more time series are said to be **cointegrated** if some linear combination is stationary.
- ▶ Key Questions:
 - ▶ Is cointegration the same as correlation?
 - ▶ How to determine if two or more series are cointegrated?
 - ▶ How to find a “cointegrating vector?”
 - ▶ How to find cointegrated series in a large universe of stocks?

Cointegration: 6

- ▶ Consider three time series, X_t, Y_t, W_t and three independent sequences of i.i.d. random variables, $\{\epsilon_{X,t}\}$, $\{\epsilon_{Y,t}\}$, and $\{\epsilon_t\}$.
- ▶ Define

$$W_t = W_{t-1} + \epsilon_t,$$

$$X_t = W_t + \epsilon_{X,t},$$

$$Y_t = W_t + \epsilon_{Y,t},$$

- ▶ The X_t and Y_t are simply noisy versions of W_t , and W_t is a random walk.
- ▶ Note that $\nabla X_t = W_t + \epsilon_{X,t} - W_{t-1} - \epsilon_{X,t-1} = \epsilon_{X,t} - \epsilon_{X,t-1} + \epsilon_t$ which is stationary (a linear combination of stationary processes).
- ▶ It follows that X_t is $\mathcal{I}(1)$.

Cointegration: 7

- ▶ Similarly, $\nabla Y_t = W_t + \epsilon_{Y,t} - W_{t-1} - \epsilon_{Y,t-1} = \epsilon_{Y,t} - \epsilon_{Y,t-1} + \epsilon_t$ which is stationary.
- ▶ It follows that Y_t is $\mathcal{I}(1)$.
- ▶ X_t and Y_t are cointegrated since both are $\mathcal{I}(1)$ and $X_t - Y_t = \epsilon_{X,t} - \epsilon_{Y,t}$, which is a stationary noise process with cointegrating vector $(1, -1)$.
- ▶ Even though these two processes are cointegrated, they may have very low correlation when $\text{Var}(\epsilon_{X,t}) = \sigma_X^2$ and $\text{Var}(\epsilon_{Y,t}) = \sigma_Y^2$ are large compared with $\text{Var}(\epsilon_t) = \sigma^2$.

$$\text{Var}(X_t) = \text{Var}(W_t + \epsilon_X) = t\sigma^2 + \sigma_X^2,$$

$$\text{Var}(Y_t) = \text{Var}(W_t + \epsilon_Y) = t\sigma^2 + \sigma_Y^2.$$

Cointegration: 8

$$\begin{aligned}\text{Cov}(X_t, Y_t) &= \text{Cov}(W_t + \epsilon_{X,t}, W_t + \epsilon_{Y,t}) \\ &= \text{Cov}(W_t, W_t) \\ &= t\sigma^2\end{aligned}$$

The correlation between X_t and Y_t is given by

$$t\sigma^2 / \sqrt{(t\sigma^2 + \sigma_X^2)(t\sigma^2 + \sigma_Y^2)},$$

which gives

$$1 / \sqrt{(1 + \frac{\sigma_X^2}{t\sigma^2})(1 + \frac{\sigma_Y^2}{t\sigma^2})},$$

Cointegration: 9

- ▶ If one or both of σ_X^2 and σ_Y^2 is large compared with $t\sigma^2$, then the denominator can be large compared with the numerator (1).
- ▶ Thus, even though the X_t and Y_t are cointegrated, their correlation can be small.
- ▶ Nevertheless, as t increases, this ratio is decrease and the correlation between X_t and Y_t will increase to 1. We can, however, modify this example

$$W_t = W_0 + \epsilon_t,$$

$$X_t = W_t + \epsilon_{X,t},$$

$$Y_t = W_t + \epsilon_{Y,t},$$

Here, rather than performing a random walk, W_t randomly moves about a fixed point W_0 .

Cointegration: 10

- ▶ With this modification, the quantity $t\sigma^2$ becomes σ^2 , and if either or both of σ_X and σ_Y is large compared with σ^2 , the correlation between X_t and Y_t will be near 0.
- ▶ When two time series, X_1 and X_2 are cointegrated, there will be a unique cointegrating vector of the form $(1, \alpha)$.
- ▶ The uniqueness holds since if there were two such vectors, $(1, \alpha_1)$ and $(1, \alpha_2)$ then $(X_1 + \alpha_1 X_2) - (X_1 + \alpha_2 X_2) = (\alpha_1 - \alpha_2)X_2$ would be stationary, in contradiction to the assumption that X_2 is not stationary.
- ▶ This uniqueness fails when $m > 2$. There may be as many as $m - 1$ linearly indep. cointegrating vectors,
- ▶ Those cointegrating vectors will span the cointegrating space, but the choice of vectors is not unique,

Cointegration: 11

- ▶ Cointegration does not imply correlation. The reverse is also true. Series that are highly correlated are not necessarily cointegrated. Consider three time series, $\{X_t\}$, $\{Y_t\}$ and $\{Z_t\}$ with

$$X_t = X_{t-1} + \epsilon_{X,t},$$

$$Y_t = Y_{t-1} + \epsilon_{Y,t},$$

$$Z_t = X_t + Y_t,$$

where $\{\epsilon_{X,t}\}$ and $\{\epsilon_{Y,t}\}$ are iid mean 0 processes independent of each other.

- ▶ $\{X_t\}$ and $\{Y_t\}$ are independent random walks whose innovations have variance σ_X^2 and σ_Y^2 while $\{Z_t\}$ the sum of these two random walks.

Cointegration: 12

- ▶ X_t and Z_t are not cointegrated. Although each is $\mathcal{I}(1)$ (as is Y_t), any linear combination of the two will involve both X_t and Y_t , neither of which is stationary.
- ▶ $\text{Cov}(X_t, Z_t) = \text{Var}(X_t) = t\sigma_X^2$
- ▶ $\text{Var}(X_t) = t\sigma_X^2$ and $\text{Var}(Z_t) = t\sigma_X^2 + t\sigma_Y^2$
- ▶ $\rho_{X_t, Z_t} = 1/\sqrt{1 + (\sigma_Y^2/\sigma_X^2)}$,
- ▶ If σ_Y/σ_X is small, then the correlation will be large. Hence this is an example of two series that are not cointegrated but can have very high correlation.

Cointegration: 13

- ▶ Both $\{X_t\}$ and $\{Y_t\}$ are $\mathcal{I}(1)$ by direct differencing.
- ▶ Every linear combination of $\{X_t\}$ and $\{Y_t\}$ is also $\mathcal{I}(1)$, hence they are not cointegrated.
- ▶ The squared correlation coefficient is $1/(1 + \frac{\sigma_Y^2}{\sigma_X^2})$. Thus if σ_X is small compared with σ_Y , then the correlation will be near 1.
- ▶ Thus this pair can be highly correlated but is not cointegrated.

Cointegration: 14

- ▶ There are two major approaches to determining the presence of cointegration of two or more time series:

1. Granger-Engle
2. Johansen

- ▶ **Granger-Engle Method**

Represent one time series as a linear combination of the others:

$$X_{jt} = \sum_{i \neq j} \beta_i X_{it} + \epsilon_t.$$

- ▶ Fit using least squares and test whether the residuals are stationary using Dickey-Fuller
- ▶ If the residuals are stationary, then the cointegrating vector is $(\beta_1, \dots, \beta_{j-1}, \mathbf{1}, \beta_{j+1}, \dots, \beta_m)$. There can be several cointegrating vectors.

Cointegration: Johansen 1

- ▶ We begin with a VAR(1) process and later generalize to an AR(p) process.
- ▶ We assume that the process is $\mathcal{I}(1)$. If it is $\mathcal{I}(0)$, then the original series are all stationary, and there is no substance to cointegration.
- ▶ If the process is $\mathcal{I}(k)$, $k \geq 2$, then we difference $k - 1$ times to bring it back to $\mathcal{I}(1)$. Let

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

- ▶ Subtract $\mathbf{X}_{t-1}$ from both sides to obtain

$$\nabla \mathbf{X}_t = \mathbf{a} + (\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

Cointegration: Johansen 2

- ▶ Now if each component of $\mathbf{X}$ is $\mathcal{I}(1)$, then $\nabla \mathbf{X}$ is stationary. Thus the right side of the equation must also be stationary.
- ▶ But $(\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1}$ is stationary if and only if the rows of $\mathbf{A} - \mathbf{I}$ are cointegrating vectors.
- ▶ One possibility is that $(\mathbf{A} - \mathbf{I})$ has rank 0, in which case we do not have any cointegrating vectors.
- ▶ But if $(\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1}$ is stationary and the rank of $\mathbf{A} - \mathbf{I}$ is $R \geq 1$ then there are R cointegrating vectors (linearly independent combinations of the components of stationary).

Cointegration: Johansen 3

- ▶ If $R = 0$, then there are no cointegrating vectors.
- ▶ If $1 \leq R \leq n - 1$, then there are R linearly independent cointegrating vectors.
- ▶ The Johansen method associates cointegration with determining the rank of $\mathbf{A} - \lambda \mathbf{I}$ or equivalently the number of eigenvalues of $\mathbf{A} - \lambda \mathbf{I}$ that are non-zero.
- ▶ These vectors are not uniquely determined, but they span the space of cointegrating vectors.

Cointegration: Johansen 4

The analysis for VAR(1) processes extends in a straightforward way to VAR(p) processes

$$\mathbf{X}_t = \mathbf{a} + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\epsilon}_t.$$

- ▶ Again, we assume that $\mathbf{X}_t$ is $\mathcal{I}(1)$. If not, then the vector needs to be differenced enough times to bring it to $\mathcal{I}(1)$.
- ▶ Subtract $\mathbf{X}_{t-1}$ from both sides to find:

Cointegration: Johansen 5

$$\begin{aligned}\nabla \mathbf{X}_t &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\mathbf{X}_{t-1} + \sum_{i=2}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t \\ &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\nabla \mathbf{X}_{t-1} + (\mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I})\mathbf{X}_{t-2} \\ &\quad + \sum_{i=3}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t\end{aligned}$$

- ▶ This continues sequentially.
- ▶ Let

$$\mathbf{B}_i = (\mathbf{A}_1 + \cdots + \mathbf{A}_i - \mathbf{I}),$$

then subtracting and adding $(\sum_{j=1}^k \mathbf{A}_j - \mathbf{I})$ on the right side yields:

Cointegration: Johansen 6

$$\nabla \mathbf{X}_t = \mathbf{a} + \sum_{j=1}^{p-1} \mathbf{B}_j \nabla \mathbf{X}_{t-j} + \mathbf{B}_p \mathbf{X}_{t-p} + \epsilon_t.$$

- ▶ Since we assumed $\mathbf{X}_t$ is $\mathcal{I}(1)$, all terms involving $\nabla \mathbf{X}_{t-j}$ are stationary, hence $\mathbf{B}_p \mathbf{X}_p$ must be stationary.
- ▶ The term $\mathbf{B}_p \mathbf{X}_{t-p}$ must be consistent with stationarity, so the arguments given for the $p = 1$ case applied to $(\mathbf{A} - \mathbf{I})$ now must be applied to $\mathbf{B}_p$.
- ▶ It's rank determines the number of cointegrating vectors, and the identity of those vectors.

Financial Time Series Analysis

Lecture 12

- ▶ TA Session: Friday, April 30, 11:00am-12:00pm
- ▶ TA Session: Monday, May 3, 7:30-8:30pm

Outline

- ▶ Final Exam, Monday, May 10, 10:00am (EDT)
- ▶ Cointegration
 - ▶ Granger-Engle
 - ▶ Johansen
 - ▶ Phillips-Ouliaris
- ▶ State Space Models

Cointegration

- ▶ There are two major approaches to determining the presence of cointegration of two or more time series:

1. Granger-Engle
2. Johansen

- ▶ **Granger-Engle Method**

Represent one time series as a linear combination of the others:

$$X_{jt} = \sum_{i \neq j} \beta_i X_{it} + \epsilon_t.$$

- ▶ Fit using least squares and test whether the residuals are stationary using Dickey-Fuller
- ▶ If the residuals are stationary, then the cointegrating vector is $(\beta_1, \dots, \beta_{j-1}, -1, \beta_{j+1}, \dots, \beta_m)$. There can be several cointegrating vectors.

Cointegration: Johansen 1

- ▶ We begin with a VAR(1) process and later generalize to an AR(p) process.
- ▶ We assume that the process is $\mathcal{I}(1)$. If it is $\mathcal{I}(0)$, then the original series are all stationary, and there is no substance to cointegration.
- ▶ If the process is $\mathcal{I}(k)$, $k \geq 2$, then we difference $k - 1$ times to bring it back to $\mathcal{I}(1)$. Let

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

- ▶ Subtract $\mathbf{X}_{t-1}$ from both sides to obtain

$$\nabla \mathbf{X}_t = \mathbf{a} + (\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

Cointegration: Johansen 2

- ▶ Note $\mathbf{A} - \mathbf{I}$ cannot be full rank. If it were then

$$(\mathbf{A} - \mathbf{I})^{-1}(\nabla \mathbf{X}_t - \mathbf{a} - \epsilon_t) = \mathbf{X}_{t-1},$$

which would be stationary, contradicting $\mathbf{X}_t$ is $\mathcal{I}(1)$.

- ▶ Now if each component of $\mathbf{X}_t$ is $\mathcal{I}(1)$, then $\nabla \mathbf{X}$ is stationary. Thus the right side of the equation must also be stationary.

Cointegration: Johansen 3

- ▶ But $(\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1}$ is stationary if and only if the rows of $\mathbf{A} - \mathbf{I}$ are cointegrating vectors.
- ▶ One possibility is that $(\mathbf{A} - \mathbf{I})$ has rank 0, in which case we do not have any cointegrating vectors.
- ▶ But if $(\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1}$ is stationary and the rank of $\mathbf{A} - \mathbf{I}$ is $R \geq 1$ then there are R cointegrating vectors (linearly independent combinations of the components of stationary processes).

Cointegration: Johansen 4

- ▶ If $R = 0$, then there are no cointegrating vectors.
- ▶ If $1 \leq R \leq n - 1$, then there are R linearly independent cointegrating vectors.
- ▶ The Johansen method associates cointegration with determining the rank of $\mathbf{A} - \lambda \mathbf{I}$ or equivalently the number of eigenvalues of $\mathbf{A} - \lambda \mathbf{I}$ that are non-zero.
- ▶ These vectors are not uniquely determined, but they span the space of cointegrating vectors.

Cointegration: Johansen 5

- ▶ Thus determining whether two or more processes are cointegrated using the Johansen approach assuming a VAR(1) process comes down to determining the rank of the $\mathbf{A} - \mathbf{I}$. Cointegrating vectors can be found as a basis for this matrix.
- ▶ In multivariate statistical analysis, there are two traditional ways to test for the rank of a matrix: $H_0 R = 0$ vs $R > 0$. These involve converting a test of the rank into a test of the eigenvalues:
 - ▶ The maximum eigenvalue: $H_0 : \lambda_{max} = 0$ vs $\lambda_{max} > 0$.
 - ▶ The trace of the matrix: $Tr(\mathbf{A} - \mathbf{I}) = \sum_{i=1}^n = 0$ vs $Tr(\mathbf{A} - \mathbf{I}) > 0$.
- ▶ Note, there is a series of tests to find all the cointegrating vectors and the rank. For example, in the maximum eigenvalue test, if it is rejected, then there is a second test, whether the second largest eigenvalue is 0, and so on.

Cointegration: Johansen 6

The analysis for VAR(1) processes extends in a straightforward way to VAR(p) processes

$$\mathbf{X}_t = \mathbf{a} + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\epsilon}_t.$$

- ▶ Again, we assume that $\mathbf{X}_t$ is $\mathcal{I}(1)$. If not, then the vector needs to be differenced enough times to bring it to $\mathcal{I}(1)$.
- ▶ Subtract $\mathbf{X}_{t-1}$ from both sides to find: $\mathbf{v}$

Cointegration: Johansen 7

$$\begin{aligned}\nabla \mathbf{X}_t &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\mathbf{X}_{t-1} + \sum_{i=2}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t \\ &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\nabla \mathbf{X}_{t-1} + (\mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I})\mathbf{X}_{t-2} \\ &\quad + \sum_{i=3}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t\end{aligned}$$

- ▶ This continues sequentially.
- ▶ Let

$$\mathbf{B}_i = (\mathbf{A}_1 + \cdots + \mathbf{A}_i - \mathbf{I}),$$

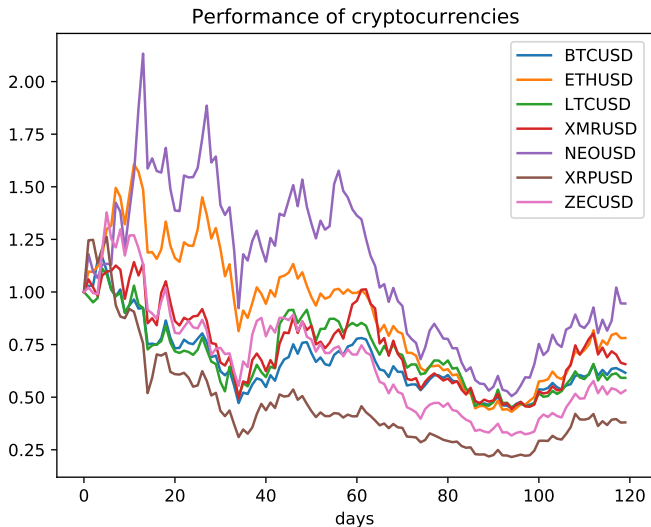
then subtracting and adding $(\sum_{j=1}^k \mathbf{A}_j - \mathbf{I})$ on the right side yields:

Cointegration: Johansen 8

$$\nabla \mathbf{X}_t = \mathbf{a} + \sum_{j=1}^{p-1} \mathbf{B}_j \nabla \mathbf{X}_{t-j} + \mathbf{B}_p \mathbf{X}_{t-p} + \epsilon_t.$$

- ▶ Since we assumed $\mathbf{X}_t$ is $\mathcal{I}(1)$, all terms involving $\nabla \mathbf{X}_{t-j}$ are stationary, hence $\mathbf{B}_p \mathbf{X}_p$ must be stationary.
- ▶ The term $\mathbf{B}_p \mathbf{X}_{t-p}$ must be consistent with stationarity, so the arguments given for the $p = 1$ case applied to $(\mathbf{A} - \mathbf{I})$ now must be applied to $\mathbf{B}_p$.
- ▶ It's rank determines the number of cointegrating vectors, and the identity of those vectors.

Cryptocurrency



Example: 1

```
import quandl
import pandas as pd
import matplotlib.pyplot as plt
import requests
import statsmodels.tsa.stattools as ts
from statsmodels.tsa.vector_ar.vecm import coint_johansen
def get_bitfinex_asset(asset, ts_ms_start, ts_ms_end):
    url = 'https://api.bitfinex.com/v2/candles/trade:1D:t' + asset +
'/hist'
    params = 'start': ts_ms_start, 'end': ts_ms_end, 'sort': 1
    r = requests.get(url, params = params)
    data = r.json()
    return pd.DataFrame(data)[2]
```

Example: 2

```
start_date = 1514768400000 # 1 January 2018, 00:00:00
end_date = 1527811199000 # 31 May 2018, 23:59:59
assets = ['BTCUSD', 'ETHUSD', 'LTCUSD', 'XMRUSD', 'NEOUSD',
          'XRPUSD', 'ZECUSD']
#Bitcoin, Ethereum, Litecoin, Monero, NEO, Ripple XRP, Zcash
crypto_prices = pd.DataFrame()
for a in assets:
    print('Downloading ' + a)
    crypto_prices[a] = get_bitfinex_asset(asset = a, ts_ms_start=
    start_date, ts_ms_end = end_date)
crypto_prices.head()
```

Example: 3

```
# Normalize prices by first value
norm_prices = crypto_prices.divide(crypto_prices.iloc[0])
plt.figure(figsize = (15, 10))
plt.plot(norm_prices)
plt.xlabel('days')
plt.title('Performance of cryptocurrencies')
plt.legend(assets)
plt.show()
for a1 in crypto_prices.columns:
    for a2 in crypto_prices.columns:
        if a1 != a2:
            test_result = ts.coint(crypto_prices[a1], crypto_prices[a2])
            print(a1 + ' and ' + a2 + ': p-value = ' +
str(test_result[1]))
```

Example: 4

BTCUSD and ETHUSD: p-value = 0.0657697980426905

BTCUSD and LTCUSD: p-value = 0.07347140678451415

BTCUSD and XMRUSD: p-value = 0.02157088942418218

BTCUSD and NEOUSD: p-value = 0.10239483419041967

BTCUSD and XRPUSD: p-value = 0.00900122457399112

BTCUSD and ZECUSD: p-value = 0.16378128244807383

ETHUSD and LTCUSD: p-value = 0.6090758251850134

ETHUSD and XMRUSD: p-value = 0.17284643088427865

ETHUSD and NEOUSD: p-value = 0.1287696772206111

ETHUSD and XRPUSD: p-value = 0.8353724771161406

ETHUSD and ZECUSD: p-value = 0.008516903095807193

Example: 5

LTCUSD and XMRUSD: p-value = 0.17400005498761367

LTCUSD and NEOUSD: p-value = 0.1249504285029151

LTCUSD and XRPUSD: p-value = 0.34798612459501893

LTCUSD and ZECUSD: p-value = 0.2208460047208785

XMRUSD and NEOUSD: p-value = 0.15900072498514933

XMRUSD and XRPUSD: p-value = 0.10535218181116529

XMRUSD and ZECUSD: p-value = 0.08687381831452784

NEOUSD and XRPUSD: p-value = 0.7028115231545062

NEOUSD and ZECUSD: p-value = 0.19217634013579027

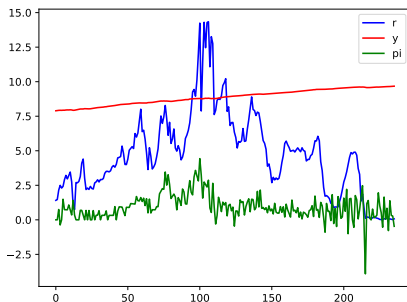
XRPUSD and ZECUSD: p-value = 0.6935781798945646

Example: 6

The data set contains three time series, r (the 91-day T-bill rate), y (the log of real GDP), and π (the inflation rate). The goal is to perform an analysis to determine if any of these series are cointegrated.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from arch.unitroot import engle_granger
from arch.unitroot.cointegration import phillips_ouliaris
from statsmodels.tsa.vector_ar.vecm import coint_johansen
data = pd.read.csv('TbGdpPi.csv')
```

Example: 7



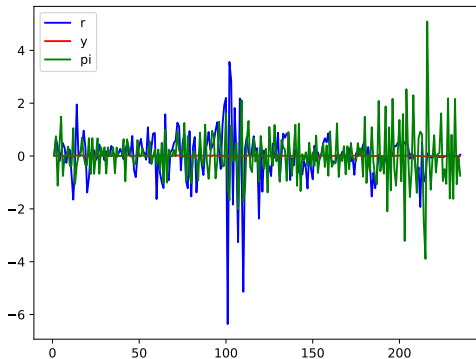
```
plt.plot(data['r'], 'b-', label = 'r')  
plt.plot(data['y'], 'r-', label = 'y')  
plt.plot(data['pi'], 'g-', label = 'pi')  
plt.legend(loc = 'best')  
plt.show()
```

Example: 8

- ▶ `sm.tsa.stattools.adfuller(data['y'])`
(-1.45, 0.56, 2, 233, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
- ▶ `sm.tsa.stattools.adfuller(data['r'])`
(-1.58, 0.49, 12, 223, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
- ▶ `sm.tsa.stattools.adfuller(data['pi'])`
(-1.78, 0.39, 15, 220, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
- ▶ All are nonstationary and H_0 (unit root) is clearly not rejected.
- ▶ `diffdata = data.diff()[1:]`

Example: 9

```
plt.plot(data_diff['r'], 'b-', label = 'r')  
plt.plot(data_diff['y'], 'r-', label = 'y')  
plt.plot(data_diff['pi'], 'g-', label = 'pi')  
plt.legend(loc = 'best')  
plt.show()
```



Example: 10

```
sm.tsa.stattools.adfuller(diffdata['y'])  
(-7.87, 4.89e-12, 1, 233, '1%': -3.46, '5%': -2.87, '10%':  
-2.57,)  
sm.tsa.stattools.adfuller(diffdata['r'])  
(-5.35, 4.25e-06, 11, 223, '1%': -3.46, '5%': -2.87, '10%':  
-2.57)  
sm.tsa.stattools.adfuller(diffdata['pi'])  
(-5.98, 1.87e-07, 14, 220, '1%': -3.46, '5%': -2.87, '10%':  
-2.57)
```

All three H_0 are rejected, so all are $\mathcal{I}(1)$

```
eg_test = engle_granger(data['r'], data['pi'], trend = 'n')  
eg_test
```

Example: 11

Engle-Granger Cointegration Test (r, pi)

Statistic: -2.377

P-value: 0.119

Null: No Cointegration, Alternative: Cointegration

ADF Lag length: 3

Trend: c

Estimated Root: 0.856

```
eg_testry = engle_granger(data['r'],data['y'])
```

```
eg_testry
```

Engle-Granger Cointegration Test (r,y)

Statistic: -2.29

P-value: 0.378

Null: No Cointegration, Alternative: Cointegration

ADF Lag length: 3

Trend: c

Example: 12

```
po_zt_testrpi = phillips_ouliaris(data['r'],data['pi'], trend = 'n',  
test_type = 'Zt')
```

```
po_zt_testrpi.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
```

Test Statistic	-11.755
P-value	0.000
Kernel	Bartlett
Bandwidth	12.323

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

Example: 13

```
po_zt_testry = phillips_ouliaris(data['r'],data['y'],trend =  
'n',test_type = 'Zt')
```

```
po_zt_testry.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
```

Test Statistic	-2.051
P-value	40.217
Kernel	Bartlett
Bandwidth	4.313

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

Example: 14

```
po_zt_testpiy = phillips_ouliaris(data['pi'],data['y'], trend =  
'n',test_type = 'zt')
```

```
po_zt_testpiy.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
```

Test Statistic	-11.880
P-value	0.000
Kernel	Bartlett
Bandwidth	11.959

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

Example: 15

```
johansen_rst=coint_johansen(data,det_order=0, k_ar_diff=0)
johansen_rst.lr1
#tracestatistic
([131.27584764, 10.78357386, 4.18281327])
johansen_rst.lr2 # max eigenvalue statistic
([120.49227378, 6.60076059, 4.18281327])
johansen_rst.Results()
johansen_rst.cvt, critical values (.90,.95,.99) trace
((27.0669, 29.7961, 35.4628),
(13.4294, 15.4943, 19.9349),
( 2.7055, 3.8415, 6.6349))
johansen_rst.cvm critical values (.90,.95,.99) max eigenvalue
((18.8928, 21.1314, 25.865 ),
(12.2971, 14.2639, 18.52 ),
( 2.7055, 3.8415, 6.6349))
```

Example: 16

```
trace_rst = 'Reject': list(johansen_rst.lr1 >  
johansen_rst.cvt[:,1])  
trace_rst = pd.DataFrame(trace_rst)  
trace_rst.index = ['r', 'y', 'pi']  
trace_rst
```


Example: 17

```
max_eigv_rst  
= 'Reject': list(johansen_rst.lr2 > johansen_rst.cvm[:,1])  
max_eigv_rst = pd.DataFrame(max_eigv_rst)  
max_eigv_rst.index = ['r', 'y', 'pi']  
max_eigv_rst
```

State Space Modeling: 1

- ▶ State space modeling Has a long history and is connected to important technology problems (e.g. Hidden Markov Models, which have been central in speech and image recognition)
- ▶ The idea is that there is a stochastic process which described the true state of a system (the state model), $\{\mathbf{X}_t, t \geq 0\}$.
- ▶ There is a second time series that represents the observation of this state process (the observation process), $\{\mathbf{Y}_t, t \geq 0\}$. These can be thought of as a noisy version of the true state process.
- ▶ The models can be univariate or multivariate, and are parameterized like the following:

$$\mathbf{X}_{t+1} = \mathbf{F}_t \mathbf{X}_t + \mathbf{V}_t,$$

$$\mathbf{Y}_t = \mathbf{G}_t \mathbf{X}_t + \mathbf{W}_t,$$

State Space Models: 2

- ▶ The two noise processes, $\mathbf{V}_t$ and $\mathbf{W}_t$ are assumed to be $WN(\mathbf{0}, \boldsymbol{\Sigma}_{X,t})$ and $WN(\mathbf{0}, \boldsymbol{\Sigma}_{Y,t})$ respectively.
- ▶ The two noise processes are uncorrelated, i.e. $E(\mathbf{V}_s \mathbf{W}_t) = \mathbf{0}, \forall s, t$.
- ▶ Here is a simple example - writing an AR(1) model in state-space form (setting $E(\mathbf{X}_t) = 0$):

$$X_t = \phi_1 X_{t-1} + \epsilon_{X,t},$$

$$X_{t-1} = X_{t-1},$$

$$Y_t = X_t + \epsilon_{Y,t}.$$

State Space Models: 3

- ▶ The state space modeling form is very flexible and allows for the expression of a wide range of models. There are also three distinct types of problems that arise in this context, each solved by **Kalman Recursions: prediction/forecasting, filtering, and smoothing.**
- ▶ Suppose there is an unknown, latent state vector process $\mathbf{X}_t$. We have noisy observations of this state vector, $\mathbf{Y}_t$ and some initial state Y_0 . We want to use the observations, $\mathbf{Y}_t$, to make some inference about $\mathbf{X}_t$ of the following type:

State Space Models: 4

► Prediction or Forecasting

Estimate $\mathbf{X}_{t+1}$ or $\mathbf{X}_{t+k}$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. This can be denoted $\hat{\mathbf{X}}_{(t+k)|t}$

► Filtering

Estimate $\mathbf{X}_t$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. We denote this $\hat{\mathbf{X}}_{t|t}$

► Smoothing

Estimate $\mathbf{X}_t$ given $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T$, for $1 \leq t < T$. This is denoted $\hat{\mathbf{Y}}_{t|T}$ indicating the estimate of $\mathbf{X}_t$ conditional on $\mathcal{F}_T$.

State Space Models: 4

- ▶ In all three cases there are **Kalman recursions** to solve these problems based on minimizing mean square error. The recursions allow one to solve these problems “on-line,” that is in the case of forecasting and filtering to use the current results and the new data to obtain new estimates/forecasts without needing the entire data set that preceeded it.

Kalman Prediction: 1

- ▶ In Kalman prediction, the goal is to predict/forecast $\mathbf{X}_{t+1}$ given $(\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t)$, a future value of $\mathbf{X}$ given the information from $\mathbf{Y}$ up to time t .
- ▶ We let the mean squared error of the prediction be denoted by

$$\Omega_t = E((\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})(\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})^T).$$

- ▶ The Kalman prediction recursion begins with $\hat{\mathbf{X}}_1$ and proceeds one-step at a time to $\hat{\mathbf{X}}_{t+1}$, as follows:

Kalman Prediction: 2

$$\hat{\mathbf{X}}_1 = E(\mathbf{X}_1 | \mathbf{Y}_0),$$

$$\hat{\mathbf{X}}_{t+1} = \mathbf{F}_t \hat{\mathbf{X}}_t + \mathbf{\Theta} \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

where

$$\mathbf{\Delta}_t = \mathbf{G}_t \mathbf{\Omega}_t \mathbf{G}_t^T + \mathbf{R}_t,$$

$$\mathbf{\Theta}_t = \mathbf{F}_t \mathbf{\Omega}_t \mathbf{G}_t^T.$$

A Simple Example: 1

- ▶ Consider a simple linear regression model

$$Y_t = bZ_t + \epsilon_t,$$

where ϵ_t are i.i.d. $N(0, \sigma^2)$ and the predictor Z_t is known.

- ▶ We first express this as a state-space model with the goal being to estimate the unknown parameter b :

$$X_t = b,$$

$$Y_t = G_t X_t + W_t,$$

where $G_t = Z_t$, $R_t = \sigma^2$, $F = 1$, $Q = 0$

- ▶ Estimate X_t given Y_i , $i \leq t$.

A Simple Example: 2

- ▶ We know the answer from simple least squares

$$\hat{b}_t = \sum_{i=1}^t Y_i Z_i / \sum_{i=1}^t Z_i^2.$$

- ▶ How can this come from the Kalman equations, and how can $\hat{b}_t$ be computed recursively?

$$\begin{aligned}\hat{b}_t &= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} \frac{\sum_{i=1}^{t-1} Y_i Z_i + Y_t Z_t}{\sum_{i=1}^{t-1} Z_i^2} \\ &= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} (\hat{b}_{t-1} + \frac{Y_t Z_t}{\sum_{i=1}^{t-1} Z_i^2}) \\ &= \text{next}\end{aligned}$$

A Simple Example: 3

$$\begin{aligned}\hat{b}_t &= \hat{b}_{t-1} - \frac{Z_t^2}{\sum_{i=1}^t Z_i^2} \hat{b}_{t-1} + \frac{Y_t Z_t}{\sum_{i=1}^t Z_i^2} \\ &= \hat{b}_{t-1} + \frac{Z_t}{\sum_{i=1}^t Z_i^2} (Y_t - Z_t \hat{b}_{t-1})\end{aligned}$$

► The quantity $K_t = Z_t / \sum_{i=1}^t Z_i^2$, the “Kalman Gain.”



$$\text{Var}(\hat{b}_t) = (1 - K_t Z_t) \text{Var}(\hat{b}_{t-1}).$$

Kalman Filtering

- ▶ For Kalman filtering, we want to predict/estimate the current state, $\mathbf{X}_t$ given $(\mathbf{Y}_0, \dots, \mathbf{Y}_t)$. This is denoted by $\mathbf{X}_{t|t}$.
- ▶ The estimate is a combination of the one-step-ahead prediction from time $t - 1$, $\hat{\mathbf{X}}_t$ and the difference between the observed $\mathbf{Y}_t$ and its predicted value using the observation equation, $\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t$

$$\mathbf{X}_{t|t} = \hat{\mathbf{X}}_t + \mathbf{\Omega}_t \mathbf{G}_t^T \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

where the relevant terms were defined earlier.

Financial Time Series Analysis

Lecture 13

- ▶ TA Session: Friday, May 7, 11:00am-12:00pm
- ▶ Final Exam: Monday, May 10, 10:00am
- ▶ Please take the course evaluation (for all classes)

Outline

- ▶ Details and Coverage of Final Exam
- ▶ Cointegration Example
- ▶ State Space Models
 - ▶ “Filtering in Finance”
 - ▶ Stochastic volatility model

Final Exam: 1

- ▶ The final exam will be comparable to the posted Time Series Exam, 90 minutes in length.
- ▶ It will be comprehensive, covering the entire course.
- ▶ It will consist largely of short answers (multiple choice, single and multiple answers) and a smaller number of problems requiring pen and ink.
- ▶ The test is closed book, notes and other sources as well except for a one-page (two-sided) cheat sheet. The use of computers/tablets, cell phones, calculators, etc. is forbidden, except for purposes of responding to, downloading and uploading the exam.
- ▶ No communication with others except the instructor, who will be available by zoom link.

Final Exam: 2

- ▶ Cell phones may be used to contact John Lehoczky during the exam for questions and technology problems as well as uploading any pen and ink solutions.
- ▶ Each student may have access to a 1-page (8.5x11, two-sided) “cheat sheet.” that can be in arbitrarily small font. It can be handwritten or computer generated.
- ▶ The final exam is not intended to be a test of memory. Rather it is intended to focus on an understanding of and the use of time series concepts.
- ▶ The final will be based on material covered in lectures, the course notes, and the homework. While no questions from homework will be replicated, homework and lecture should serve as a guide to doing well on the exam.

Final Exam Coverage: 1

- ▶ Rama Cont's stylized facts about financial markets
- ▶ Stationarity
 - ▶ Strong stationarity
 - ▶ Weak stationarity
 - ▶ Martingale Difference Sequences
- ▶ Autocovariance and autocorrelation functions
 - ▶ ACF, PACF -their definition and plots for a few simple time series like AR and MA
 - ▶ Bartlett's test (confidence limits for ACF plots for simple time series)
 - ▶ Ljung-Box test

Final Exam Coverage: 2

- ▶ Basic time series models (AR, MA, ARMA, ARIMA)
 - ▶ Know or be able to work out simple facts (mean, variance and covariance functions) for simple models.
- ▶ Transforming to achieve stationarity
 - ▶ Trend and Seasonal adjustments
 - ▶ Box-Cox transformation
 - ▶ Backshift (B) and ∇ operators and notation.
 - ▶ Differencing
 - ▶ Dickey Fuller test
- ▶ Parameter estimation using Yule Walker equations
- ▶ Forecasting AR, MA, ARMA models

Final Exam Coverage: 3

- ▶ ARCH/GARCH models
 - ▶ Understanding construction of ARCH/GARCH models
 - ▶ Tests to identify ARCH effects
 - ▶ Ljung-Box test
 - ▶ Engle's test
 - ▶ GARCH(p, q) and $GARCH^2(p, q)$ as ARMA
 - ▶ Forecasting GARCH

Final Exam Coverage: 4

- ▶ Multiple Time Series (VAR(p) process)
- ▶ Cointegration
 - ▶ Understanding the meaning of cointegration
 - ▶ Cointegration vs correlation
 - ▶ Tests for cointegration (Engle-Grange, Phillips-Ouliaris, Johansen)
- ▶ State Space Models
 - ▶ Their formulation
 - ▶ Understanding the “Kalman problems” (filtering, prediction and smoothing)

Cointegration

- ▶ There are two major approaches to determining the presence of cointegration of two or more time series:

1. Granger-Engle
2. Johansen

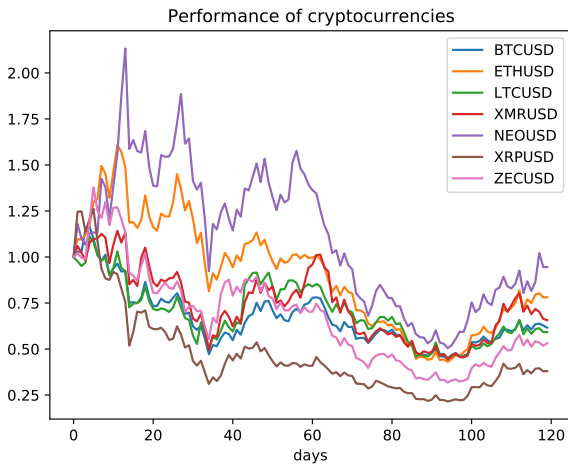
- ▶ **Granger-Engle Method**

Represent one time series as a linear combination of the others:

$$X_{jt} = \sum_{i \neq j} \beta_i X_{it} + \epsilon_t.$$

- ▶ Fit using least squares and test whether the residuals are stationary using Dickey-Fuller
- ▶ If the residuals are stationary, then the cointegrating vector is $(\beta_1, \dots, \beta_{j-1}, -1, \beta_{j+1}, \dots, \beta_m)$. There can be several cointegrating vectors.

Cryptocurrency



Example: 1

```
import quandl
import pandas as pd
import matplotlib.pyplot as plt
import requests
import statsmodels.tsa.stattools as ts
from statsmodels.tsa.vector_ar.vecm import coint_johansen
def get_bitfinex_asset(asset, ts_ms_start, ts_ms_end):
    url = 'https://api.bitfinex.com/v2/candles/trade:1D:t' + asset +
'/hist'
    params = 'start': ts_ms_start, 'end': ts_ms_end, 'sort': 1
    r = requests.get(url, params = params)
    data = r.json()
    return pd.DataFrame(data)[2]
```

Example: 2

```
start_date = 1514768400000 # 1 January 2018, 00:00:00
end_date = 1527811199000 # 31 May 2018, 23:59:59
assets = ['BTCUSD', 'ETHUSD', 'LTCUSD', 'XMRUSD', 'NEOUSD',
          'XRPUSD', 'ZECUSD']
#Bitcoin, Ethereum, Litecoin, Monero, NEO, Ripple XRP, Zcash
crypto_prices = pd.DataFrame()
for a in assets:
    print('Downloading ' + a)
    crypto_prices[a] = get_bitfinex_asset(asset = a, ts_ms_start=
    start_date, ts_ms_end = end_date)
crypto_prices.head()
```


Example: 3

```
# Normalize prices by first value
norm_prices = crypto_prices.divide(crypto_prices.iloc[0])
plt.figure(figsize = (15, 10))
plt.plot(norm_prices)
plt.xlabel('days')
plt.title('Performance of cryptocurrencies')
plt.legend(assets)
plt.show()
for a1 in crypto_prices.columns:
    for a2 in crypto_prices.columns:
        if a1 != a2:
            test_result = ts.coint(crypto_prices[a1], crypto_prices[a2])
            print(a1 + ' and ' + a2 + ': p-value = ' +
str(test_result[1]))
```

Example: 4

BTCUSD and ETHUSD: p-value = 0.0657697980426905

BTCUSD and LTCUSD: p-value = 0.07347140678451415

BTCUSD and XMRUSD: p-value = 0.02157088942418218

BTCUSD and NEOUSD: p-value = 0.10239483419041967

BTCUSD and XRPUSD: p-value = 0.00900122457399112

BTCUSD and ZECUSD: p-value = 0.16378128244807383

ETHUSD and LTCUSD: p-value = 0.6090758251850134

ETHUSD and XMRUSD: p-value = 0.17284643088427865

ETHUSD and NEOUSD: p-value = 0.1287696772206111

ETHUSD and XRPUSD: p-value = 0.8353724771161406

ETHUSD and ZECUSD: p-value = 0.008516903095807193

Example: 5

LTCUSD and XMRUSD: p-value = 0.17400005498761367

LTCUSD and NEOUSD: p-value = 0.1249504285029151

LTCUSD and XRPUSD: p-value = 0.34798612459501893

LTCUSD and ZECUSD: p-value = 0.2208460047208785

XMRUSD and NEOUSD: p-value = 0.15900072498514933

XMRUSD and XRPUSD: p-value = 0.10535218181116529

XMRUSD and ZECUSD: p-value = 0.08687381831452784

NEOUSD and XRPUSD: p-value = 0.7028115231545062

NEOUSD and ZECUSD: p-value = 0.19217634013579027

XRPUSD and ZECUSD: p-value = 0.6935781798945646

Example: 6

```
cj = coint_johansen(crypto_prices, det_order = 0, k_ar_diff = 0)
```

```
cj.lr1
```

```
array([177.07125622, 112.17759288, 63.09167506,  
40.91449977, 24.41337134, 10.94471608, 4.37697074])
```

```
cj.cvt
```

```
array([[120.3673, 125.6185, 135.9825],  
      [ 91.109 , 95.7542, 104.9637],  
      [ 65.8202, 69.8189, 77.8202],  
      [ 44.4929, 47.8545, 54.6815],  
      [ 27.0669, 29.7961, 35.4628],  
      [ 13.4294, 15.4943, 19.9349],  
      [ 2.7055, 3.8415, 6.6349]])
```

Example: 7

cj.lm2

```
array([64.89366334, 49.08591782, 22.17717529,  
16.50112843, 13.46865526, 6.56774534, 4.37697074])
```

cj.cvm

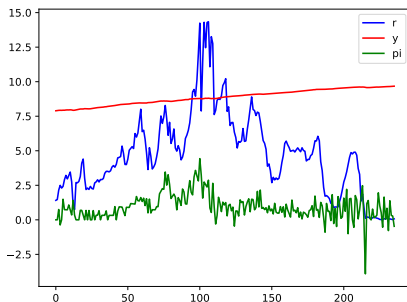
```
array([[43.2947, 46.2299, 52.3069],  
       [37.2786, 40.0763, 45.8662],  
       [31.2379, 33.8777, 39.3693],  
       [25.1236, 27.5858, 32.7172],  
       [18.8928, 21.1314, 25.865 ],  
       [12.2971, 14.2639, 18.52 ],  
       [ 2.7055, 3.8415, 6.6349]])
```

Example: 8

The data set contains three time series, r (the 91-day T-bill rate), y (the log of real GDP), and π (the inflation rate). The goal is to perform an analysis to determine if any of these series are cointegrated.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from arch.unitroot import engle_granger
from arch.unitroot.cointegration import phillips_ouliaris
from statsmodels.tsa.vector_ar.vecm import coint_johansen
data = pd.read.csv('TbGdpPi.csv')
```

Example: 9



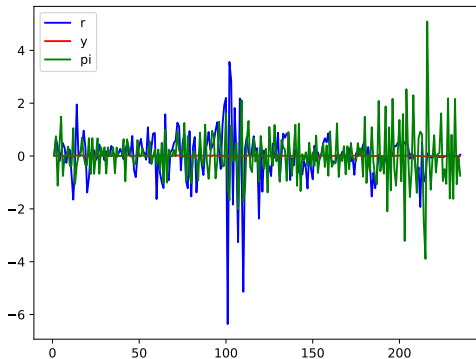
```
plt.plot(data['r'], 'b-', label = 'r')  
plt.plot(data['y'], 'r-', label = 'y')  
plt.plot(data['pi'], 'g-', label = 'pi')  
plt.legend(loc = 'best')  
plt.show()
```

Example: 10

- ▶ `sm.tsa.stattools.adfuller(data['y'])`
(-1.45, 0.56, 2, 233, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
- ▶ `sm.tsa.stattools.adfuller(data['r'])`
(-1.58, 0.49, 12, 223, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
- ▶ `sm.tsa.stattools.adfuller(data['pi'])`
(-1.78, 0.39, 15, 220, '1%': -3.46, '5%': -2.87, '10%': -2.57,)
- ▶ All are nonstationary and H_0 (unit root) is clearly not rejected.
- ▶ `diffdata = data.diff()[1:]`

Example: 11

```
plt.plot(data_diff['r'], 'b-', label = 'r')  
plt.plot(data_diff['y'], 'r-', label = 'y')  
plt.plot(data_diff['pi'], 'g-', label = 'pi')  
plt.legend(loc = 'best')  
plt.show()
```



Example: 12

```
sm.tsa.stattools.adfuller(diffdata['y'])  
(-7.87, 4.89e-12, 1, 233, '1%': -3.46, '5%': -2.87, '10%':  
-2.57,)  
sm.tsa.stattools.adfuller(diffdata['r'])  
(-5.35, 4.25e-06, 11, 223, '1%': -3.46, '5%': -2.87, '10%':  
-2.57)  
sm.tsa.stattools.adfuller(diffdata['pi'])  
(-5.98, 1.87e-07, 14, 220, '1%': -3.46, '5%': -2.87, '10%':  
-2.57)
```

All three H_0 are rejected, so all are $\mathcal{I}(1)$

```
eg_test = engle_granger(data['r'], data['pi'], trend = 'n')  
eg_test
```

Example: 13

Engle-Granger Cointegration Test (r, pi)

Statistic: -2.377

P-value: 0.119

Null: No Cointegration, Alternative: Cointegration

ADF Lag length: 3

Trend: c

Estimated Root: 0.856

```
eg_testry = engle_granger(data['r'],data['y'])
```

```
eg_testry
```

Engle-Granger Cointegration Test (r,y)

Statistic: -2.29

P-value: 0.378

Null: No Cointegration, Alternative: Cointegration

ADF Lag length: 3

Trend: c

Example: 14

```
po_zt_testrpi = phillips_ouliaris(data['r'],data['pi'], trend = 'n',  
test_type = 'Zt')
```

```
po_zt_testrpi.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
```

Test Statistic	-11.755
P-value	0.000
Kernel	Bartlett
Bandwidth	12.323

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

Example: 15

```
po_zt_testry = phillips_ouliaris(data['r'],data['y'],trend =  
'n',test_type = 'Zt')
```

```
po_zt_testry.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
```

Test Statistic	-2.051
P-value	40.217
Kernel	Bartlett
Bandwidth	4.313

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

Example: 16

```
po_zt_testpiy = phillips_ouliaris(data['pi'],data['y'], trend =  
'n',test_type = 'zt')
```

```
po_zt_testpiy.summary()
```

Phillips-Ouliaris zt Cointegration Test

```
=====
```

Test Statistic	-11.880
P-value	0.000
Kernel	Bartlett
Bandwidth	11.959

Trend: Constant

Critical Values: -2.48 (10%), -2.79 (5%), -3.39 (1%)

Null Hypothesis: No Cointegration

Alternative Hypothesis: Cointegration

Distribution Order: 2

Example: 17

```
johansen_rst=coint_johansen(data,det_order=0, k_ar_diff=0)
johansen_rst.lr1
#tracestatistic
([131.27584764, 10.78357386, 4.18281327])
johansen_rst.lr2 # max eigenvalue statistic
([120.49227378, 6.60076059, 4.18281327])
johansen_rst.Results()
johansen_rst.cvt, critical values (.90,.95,.99) trace
((27.0669, 29.7961, 35.4628),
(13.4294, 15.4943, 19.9349),
( 2.7055, 3.8415, 6.6349))
johansen_rst.cvm critical values (.90,.95,.99) max eigenvalue
((18.8928, 21.1314, 25.865 ),
(12.2971, 14.2639, 18.52 ),
( 2.7055, 3.8415, 6.6349))
```

Example: 18

```
trace_rst = 'Reject': list(johansen_rst.lr1 >  
johansen_rst.cvt[:,1])  
trace_rst = pd.DataFrame(trace_rst)  
trace_rst.index = ['r', 'y', 'pi']  
trace_rst
```


Example: 19

```
max_eigv_rst  
= 'Reject': list(johansen_rst.lr2 > johansen_rst.cvm[:,1])  
max_eigv_rst = pd.DataFrame(max_eigv_rst)  
max_eigv_rst.index = ['r', 'y', 'pi']  
max_eigv_rst
```

State Space Modeling: 1

- ▶ State space modeling Has a long history and is connected to important technology problems (e.g. Hidden Markov Models, which have been central in speech and image recognition)
- ▶ The idea is that there is a stochastic process which described the true state of a system (the state model), $\{\mathbf{X}_t, t \geq 0\}$.
- ▶ There is a second time series that represents the observation of this state process (the observation process), $\{\mathbf{Y}_t, t \geq 0\}$. These can be thought of as a noisy version of the true state process.
- ▶ The models can be univariate or multivariate, and are parameterized like the following:

$$\mathbf{X}_{t+1} = \mathbf{F}_t \mathbf{X}_t + \mathbf{V}_t,$$

$$\mathbf{Y}_t = \mathbf{G}_t \mathbf{X}_t + \mathbf{W}_t,$$

State Space Models: 2

- ▶ The two noise processes, $\mathbf{V}_t$ and $\mathbf{W}_t$ are assumed to be $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{X,t})$ and $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{Y,t})$ respectively.
- ▶ The two noise processes are uncorrelated, i.e. $E(\mathbf{V}_s \mathbf{W}_t) = \mathbf{0}, \forall s, t.$
- ▶ The standard time series models (AR, MA, ARMA) have state space representations.

State Space Models: 3

For example, consider an AR(2) model with mean 0.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t.$$

Define $\mathbf{X}_t = (X_t, X_{t-1})^T$, $\epsilon_t = (\epsilon_t, 0)^T$,

$$\mathbf{X}_t = \mathbf{F}\mathbf{X}_{t-1} + \epsilon_t$$

$$Y_t = (1, 0)\mathbf{X}_t.$$

with

$$\mathbf{F} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix}.$$

State Space Models: 4

Consider a MA(2) model with mean 0

$$X_t = \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \epsilon_t.$$

Define $\mathbf{X}_t = (\epsilon_t, \epsilon_{t-1}, \epsilon_{t-2})^T$, $\epsilon_t = (\epsilon_t, 0, 0)^T$,

$$\mathbf{X}_t = \mathbf{F}\mathbf{X}_{t-1} + \epsilon_t$$

$$Y_t = (1, \theta_1, \theta_2)\mathbf{X}_t,$$

with

$$\mathbf{F} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

State Space Models: 5

- ▶ The state space modeling form is very flexible and allows for the expression of a wide range of models. There are also three distinct types of problems that arise in this context, each solved by **Kalman Recursions: prediction/forecasting, filtering, and smoothing**.
- ▶ Suppose there is an unknown, latent state vector process $\mathbf{X}_t$. We have noisy observations of this state vector, $\mathbf{Y}_t$ and some initial state Y_0 . We want to use the observations, $\mathbf{Y}_t$, to make some inference about $\mathbf{X}_t$ of the following type:

State Space Models: 6

► Prediction or Forecasting

Estimate $\mathbf{X}_{t+1}$ or $\mathbf{X}_{t+k}$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. This can be denoted $\hat{\mathbf{X}}_{(t+k)|t}$

► Filtering

Estimate $\mathbf{X}_t$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. We denote this $\hat{\mathbf{X}}_{t|t}$

► Smoothing

Estimate $\mathbf{X}_t$ given $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T$, for $1 \leq t < T$. This is denoted $\hat{\mathbf{Y}}_{t|T}$ indicating the estimate of $\mathbf{X}_t$ conditional on $\mathcal{F}_T$.

State Space Models: 7

- ▶ In all three cases there are **Kalman recursions** to solve these problems based on minimizing mean square error. The recursions allow one to solve these problems “on-line,” that is in the case of forecasting and filtering to use the current results and the new data to obtain new estimates/forecasts without needing the entire data set that preceeded it.

Kalman Prediction: 1

- ▶ In Kalman prediction, the goal is to predict/forecast $\mathbf{X}_{t+1}$ given $(\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t)$, a future value of $\mathbf{X}$ given the information from $\mathbf{Y}$ up to time t .
- ▶ We let the mean squared error of the prediction be denoted by

$$\Omega_t = E((\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})(\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})^T).$$

- ▶ The Kalman prediction recursion begins with $\hat{\mathbf{X}}_1$ and proceeds one-step at a time to $\hat{\mathbf{X}}_{t+1}$, as follows:

Kalman Prediction: 2

$$\hat{\mathbf{X}}_1 = E(\mathbf{X}_1 | \mathbf{Y}_0),$$

$$\hat{\mathbf{X}}_{t+1} = \mathbf{F}_t \hat{\mathbf{X}}_t + \mathbf{\Theta} \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

where

$$\mathbf{\Delta}_t = \mathbf{G}_t \mathbf{\Omega}_t \mathbf{G}_t^T + \mathbf{R}_t,$$

$$\mathbf{\Theta}_t = \mathbf{F}_t \mathbf{\Omega}_t \mathbf{G}_t^T.$$

A Simple Example: 1

- ▶ Consider a simple linear regression model

$$Y_t = bZ_t + \epsilon_t,$$

where ϵ_t are i.i.d. $N(0, \sigma^2)$ and the predictor Z_t is known.

- ▶ We first express this as a state-space model with the goal being to estimate the unknown parameter b :

$$X_t = b,$$

$$Y_t = G_t X_t + W_t,$$

where $G_t = Z_t$, $R_t = \sigma^2$, $F = 1$, $Q = 0$

- ▶ Estimate X_t given Y_i , $i \leq t$.

A Simple Example: 2

- ▶ We know the answer from simple least squares

$$\hat{b}_t = \sum_{i=1}^t Y_i Z_i / \sum_{i=1}^t Z_i^2.$$

- ▶ How can this come from the Kalman equations, and how can $\hat{b}_t$ be computed recursively?

$$\begin{aligned}\hat{b}_t &= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} \frac{\sum_{i=1}^{t-1} Y_i Z_i + Y_t Z_t}{\sum_{i=1}^{t-1} Z_i^2} \\ &= \frac{\sum_{i=1}^{t-1} Z_i^2}{\sum_{i=1}^t Z_i^2} (\hat{b}_{t-1} + \frac{Y_t Z_t}{\sum_{i=1}^{t-1} Z_i^2}) \\ &= \text{next}\end{aligned}$$

A Simple Example: 3

$$\begin{aligned}\hat{b}_t &= \hat{b}_{t-1} - \frac{Z_t^2}{\sum_{i=1}^t Z_i^2} \hat{b}_{t-1} + \frac{Y_t Z_t}{\sum_{i=1}^t Z_i^2} \\ &= \hat{b}_{t-1} + \frac{Z_t}{\sum_{i=1}^t Z_i^2} (Y_t - Z_t \hat{b}_{t-1})\end{aligned}$$

► The quantity $K_t = Z_t / \sum_{i=1}^t Z_i^2$, the “Kalman Gain.”



$$\text{Var}(\hat{b}_t) = (1 - K_t Z_t) \text{Var}(\hat{b}_{t-1}).$$

Kalman Filtering

- ▶ For Kalman filtering, we want to predict/estimate the current state, $\mathbf{X}_t$ given $(\mathbf{Y}_0, \dots, \mathbf{Y}_t)$. This is denoted by $\mathbf{X}_{t|t}$.
- ▶ The estimate is a combination of the one-step-ahead prediction from time $t - 1$, $\hat{\mathbf{X}}_t$ and the difference between the observed $\mathbf{Y}_t$ and its predicted value using the observation equation, $\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t$

$$\mathbf{X}_{t|t} = \hat{\mathbf{X}}_t + \mathbf{\Omega}_t \mathbf{G}_t^T \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

where the relevant terms were defined earlier.

Stochastic Volatility: 1

- ▶ Recall we have develop ARCH/GARCH modeling to create models that better conform to the stylized facts of financial markets: heteroscedasticity, volatility clustering, heavy tail behavior.
- ▶ The GARCH model created this behavior “within itself,” not based on external events/shocks. Yet we have seen that many stocks (particularly those impacted by the same factors, such as the market, industry sectors, etc) exhibit similar behavior with respect to volatility clustering.
- ▶ An alternative model would allow for exogenous processes to impact the volatility behavior of a financial asset.
- ▶ Such models are called stochastic volatility models, and a state space formulation of these models allows us to employ “Kalman methodology” to filter and predict.

Stochastic Volatility: 2

- ▶ In state space models, there is an unobserved or latent variable(s) modeled and an observation/measurement process. We often want to make inference about the latent process.
- ▶ A very good example of this situation is volatility. Volatility is unobserved, yet crucial to the valuation of assets (e.g. an option price).
- ▶ We want to model volatility as the unobserved state and then have an observation process that can be used to estimate (filter) and predict the latent process.
- ▶ An example using the Heston model is presented in the paper “Filtering in Finance.”

Stochastic Volatility: 3

- ▶ The Heston model involves the state process, v_t , and the observation process, S_t , defined by the SDE:

$$dv(t) = (\omega - \theta v_t)dt + \xi \sqrt{v_t} dW_t^v,$$

$$dS_t = rS_t dt + \sqrt{v(t)} S_t dW_t^S.$$

- ▶ The volatility process is a CIR process, whereas the stock price process is GBM with stochastic volatility where:
 - ▶ r is the risk free rate,
 - ▶ θ is the speed of mean reversion,
 - ▶ ω/θ is the long-run equilibrium variance
 - ▶ ξ is the volatility of the variance process.
 - ▶ ρ is the correlation between W_t^v and W_t^S
- ▶ Conditional on v_t , the price process is a GBM

Stochastic Volatility: 4

- ▶ The Heston model is defined in continuous time, and we need to convert it to discrete time, and put it into state space form.
- ▶ The parameter set $(\omega, \theta, \xi, \rho)$ must be calibrated with market data to have the model in risk-neutral form.
- ▶ We can rewrite the stock price equation

$$\log S_{t+\Delta} = \log S_t + (r - .5v_t)\Delta + \sqrt{v_t}\Delta Z_t^S.$$

- ▶ We next use an Euler discretization for the two SDEs with time step Δ :

Stochastic Volatility: 5

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}\Delta}Z_{(i+1)\Delta}^S,$$

$$v_{(i+1)\Delta} = v_{i\Delta} + (\omega - \theta v_{i\Delta})\Delta + \xi \sqrt{v_{i\Delta}\Delta}Z_{(i+1)\Delta}^v,$$

where $\{(Z_i^S, Z_i^v), 1 \leq i \leq N\}$, $\Delta = T/N$, are iid standard bivariate normal random vectors with correlation ρ .

- It is possible to eliminate the correlation parameter ρ by introducing the independent pair $(\tilde{Z}_i^v, Z_i^S)$ where

$$Z_i^v = \rho Z_i^S + \sqrt{1 - \rho^2} \tilde{Z}_i^v$$

and substituting $\tilde{Z}_{i\Delta}^v = \frac{1}{\sqrt{1-\rho^2}}(Z_{i\Delta}^v - \rho \tilde{Z}_{i\Delta}^S)$ into the volatility equation.

Stochastic Volatility: 6

- Substituting $\sqrt{v_k}\Delta Z_k^S = \log(S_{(i+1)\Delta}/S_{i\Delta}) - (r_{i\Delta} - .5v_{i\Delta})\Delta$ into the volatility equation, we arrive at the state space version of the Euler discretized Heston model with independent noise terms:

$$\begin{aligned}v_{(i+1)\Delta} &= v_{i\Delta} + [\omega - \rho\xi r_{i\Delta} - (\theta - .5\rho\xi)v_{i\Delta}]\Delta \\ &+ \rho\xi \log(S_{(i+1)\Delta}/S_{i\Delta}) + \xi\sqrt{1 - \rho^2}\sqrt{v_{i\Delta}}\Delta\tilde{Z}_k^v.\end{aligned}$$

for the state equation and

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}}\Delta Z_k^S,$$

for the measurement equation.

Stochastic Volatility: 7

- ▶ Once the stochastic volatility model is in state space form, we can utilize the Kalman procedures for statistical inference.
- ▶ In the particular case, the volatility is of great interest, and inferring the current volatility using Kalman filtering has significant implications (e.g. in valuation of options, comparing current volatility estimates with implied volatility)
- ▶ There are a number of stochastic volatility models, e.g. the Hull-White model we used in the simulation course.
- ▶ Estimation of these models is known to be more complicated than GARCH models.

Financial Time Series Analysis

Lecture 14

- ▶ **The practice ProctorTrack quiz is ready to be taken. 23 students are not yet authorized for ProctorTrack and MUST take it in order to take the final exam on Monday**
- ▶ TA Session: Friday, May 7, 11:00am-12:00pm
- ▶ Final Exam: Monday, May 10, 10:00am
- ▶ Please take the course evaluation (for all classes).
Deadline is Thursday 11:59pm.

Outline

- ▶ State Space Models
 - ▶ Bayesian approach
 - ▶ “Filtering in Finance” Heston stochastic volatility model
 - ▶ Stochastic volatility model

State Space Models: 2

- ▶ The two noise processes, $\mathbf{V}_t$ and $\mathbf{W}_t$ are assumed to be $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{X,t})$ and $\text{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{Y,t})$ respectively.
- ▶ The two noise processes are uncorrelated, i.e. $E(\mathbf{V}_s \mathbf{W}_t) = \mathbf{0}, \forall s, t$.
- ▶ The standard time series models (AR, MA, ARMA) have state space representations.

State Space Models: 5

► Prediction or Forecasting

Estimate $\mathbf{X}_{t+1}$ or $\mathbf{X}_{t+k}$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. This can be denoted $\hat{\mathbf{X}}_{(t+k)|t}$

► Filtering

Estimate $\mathbf{X}_t$ using $\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t$. We denote this $\hat{\mathbf{X}}_{t|t}$

► Smoothing

Estimate $\mathbf{X}_t$ given $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T$, for $1 \leq t < T$. This is denoted $\hat{\mathbf{Y}}_{t|T}$ indicating the estimate of $\mathbf{X}_t$ conditional on $\mathcal{F}_T$.

Bayesian Background to Kalman Methods: 1

- ▶ Recall the Bayesian approach to statistics: unknown parameters have probability distributions (prior distribution).
- ▶ When one has data that was generated by a mechanism involving the unknown parameters, this generates a likelihood function, $f(\text{data}|\text{parameters})$.
- ▶ The (prior) distribution of the parameters coupled with the likelihood function can be combined to generate a new (posterior) distribution of the parameters using Bayes theorem.

$$P(\text{parameters}|\text{data}) = \frac{P(\text{parameters}) * P(\text{data}|\text{parameters})}{P(\text{data})}.$$

Bayesian Background to Kalman Methods: 2

- ▶ If minimum mean square error is the criterion, then given only the prior distribution, the minimum mean square error predictor of the parameters is the mean of the prior distribution. The associated minimum mean square error is the variance of the prior distribution, $\text{Var}(\text{parameters})$.
- ▶ Given the data, the minimum mean square error predictor is the mean of the posterior distribution, and the associated mean square error is the variance of the posterior distribution, $\text{Var}(\text{parameters}|\text{data})$.
- ▶ These ideas are best understood with a simple example using the normal distribution.

Bayesian Background to Kalman Methods: 3

- ▶ Suppose that there is an unknown parameter, θ . We have a prior distribution for θ given by $N(\mu, V_0^2)$. We generate data, $X_1, X_2, \dots$ that are i.i.d. $N(\theta, \sigma^2)$. Suppose the data arrives one at a time.
- ▶ After observing X_1 we can update the prior to a posterior,

$$\theta|X_1 = x_1 \text{ has a } N(\mu_1, V_1^2),$$

where

$$\mu_1 = \frac{\sigma^2}{\sigma^2 + V_0^2} \mu + \frac{V_0^2}{\sigma^2 + V_0^2} X_1$$

$$V_1^2 = \frac{\sigma^2}{\sigma^2 + V_0^2} V_0^2.$$

Bayesian Background to Kalman Methods: 4

This easily extend to $X_1, \dots, X_n$ i.i.d. $N(\theta, \sigma^2)$. We can reduce to the sufficient statistic $\bar{X}_n$, which has $N(\theta, \sigma^2/n)$. Thus the posterior distribution is $N(\mu_n, V_n^2)$ and

$$\mu_n = \frac{\sigma^2}{\sigma^2 + nV_0^2} \mu + \frac{nV_0^2}{\sigma^2 + nV_0^2} \bar{X}_n,$$

$$V_n^2 = \frac{\sigma^2}{\sigma^2 + nV_0^2} V_0^2.$$

Equivalently, the updating can be done sequentially (on-line)

$$\mu_n = \frac{\sigma^2}{\sigma^2 + V_0^2} \mu_{n-1} + \frac{V_0^2}{\sigma^2 + V_0^2} X_{n-1}$$

$$V_n^2 = \frac{\sigma^2}{\sigma^2 + V_{n-1}^2} V_{n-1}^2.$$

Bayesian Background to Kalman Methods: 5

What is the relevance to the state space models and the Kalman methodology? We simplify by considering the univariate case, although everything carries over to the multivariate case.

- ▶ In the state space model we have a dynamic version of the problem. That is, there is an unknown state X_t that is changing with time according to the process

$$X_{t+1} = FX_t + V_t,$$

where V_t has $N(0, Q_t)$ distribution.

- ▶ At each step, there is an observation, Y_t given by

$$Y_t = GX_t + W_t,$$

where W_t has a $N(0, R_t)$ distribution.

Bayesian Background to Kalman Methods: 6

- ▶ We begin by assuming that there is an initial normal prior distribution for X_1 .
- ▶ At stage $t-1$, we have observed $Y_1, \dots, Y_{t-1}$ and using those data, we have a posterior distribution for X_{t-1} .
- ▶ Unlike the previous example, this must be done a step at a time because the X_t process is dynamic, not just an unknown constant.
- ▶ Nevertheless, each step can be carried out using the Bayesian machinery.
- ▶ Recall Glasserman pages on joint, marginal and conditional distributions of multivariate normals.

Bayesian Background to Kalman Methods: 7

- ▶ To do Kalman filtering, we want to estimate X_t conditional on $\mathcal{F}_t = \sigma(Y_1, \dots, Y_t)$.
- ▶ We have the posterior distribution for X_{t-1} given $\mathcal{F}_{t-1}$, let's say it is $N(\mu_{t-1}, \sigma_{t-1}^2)$.
- ▶ Using the evolution of the state equation, the conditional distribution of X_t given X_{t-1} is given by $N(FX_{t-1}, Q_t)$
- ▶ Also the conditional distribution of $Y_t|X_t$ is given by the observation equation and is $N(GX_t, R_t)$.
- ▶ Now we can consider the triplet (X_{t-1}, X_t, Y_t) given $\mathcal{F}_{t-1}$.

Bayesian Background to Kalman Methods: 8

- ▶ This triplet has a trivariate normal distribution, the marginal of X_{t-1} , which is the posterior given $\mathcal{F}_{t-1}$ times the conditional of $X_t|X_{t-1}$ times the conditional of $Y_t|X_t$.
- ▶ For Kalman filtering, we want the conditional distribution of X_t given Y_t (note: $\mathcal{F}_{t-1}$ is incorporated in the posterior of X_{t-1}).
- ▶ We can reduce this to the bivariate distribution of (X_t, Y_t) , which we have, and we seek to generate the conditional distribution of $X_t|Y_t$.
- ▶ This requires Bayes formula or the Glasserman formulas.

Bayesian Background to Kalman Methods: 9

- ▶ Recall if X_{t-1} has a $N(\mu_{t-1}, \sigma_{t-1}^2)$ distribution and $X_t|X_{t-1} = FX_t + V_t$ which has a $N(FX_t, Q_t)$ distribution, then
- ▶ the distribution of marginal distribution of X_t is $N(F\mu_{t-1}, F^2\sigma_{t-1}^2 + Q_t)$
- ▶ So we begin with this as the prior distribution for X_t and seek to find the posterior of X_t given Y_t , the Kalman filter.

The Kalman Recursions

- ▶ In all three cases there are **Kalman recursions** to solve these problems based on minimizing mean square error. The recursions allow one to solve these problems “on-line,” that is in the case of forecasting and filtering to use the current results and the new data to obtain new estimates/forecasts without needing the entire data set that preceeded it.

Kalman Prediction: 1

- ▶ In Kalman prediction, the goal is to predict/forecast $\mathbf{X}_{t+1}$ given $(\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_t)$, a future value of $\mathbf{X}$ given the information from $\mathbf{Y}$ up to time t .
- ▶ We let the mean squared error of the prediction be denoted by

$$\Omega_{t+k} = E((\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})(\mathbf{X}_{t+k} - \hat{\mathbf{X}}_{t+k})^T).$$

- ▶ The Kalman prediction recursion begins with $\hat{\mathbf{X}}_1$ and proceeds one-step at a time to $\hat{\mathbf{X}}_{t+1}$, as follows:

Kalman Prediction: 2

$$\hat{\mathbf{X}}_1 = E(\mathbf{X}_1 | \mathbf{Y}_0),$$

$$\hat{\mathbf{X}}_{t+1} = \mathbf{F}_t \hat{\mathbf{X}}_t + \mathbf{\Theta} \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

$$\mathbf{\Omega}_{t+1} = \mathbf{F}_t \mathbf{\Omega}_t \mathbf{F}_t^T + \mathbf{Q}_t - \mathbf{\Theta}_t \mathbf{\Delta}_t^{-1} \mathbf{\Theta}_t^T,$$

where

$$\mathbf{\Delta}_t = \mathbf{G}_t \mathbf{\Omega}_t \mathbf{G}_t^T + \mathbf{R}_t,$$

$$\mathbf{\Theta}_t = \mathbf{F}_t \mathbf{\Omega}_t \mathbf{G}_t^T.$$

Kalman Filtering

- ▶ For Kalman filtering, we want to predict/estimate the current state, $\mathbf{X}_t$ given $(\mathbf{Y}_0, \dots, \mathbf{Y}_t)$. This is denoted by $\mathbf{X}_{t|t}$.
- ▶ The estimate is a combination of the one-step-ahead prediction from time $t - 1$, $\hat{\mathbf{X}}_t$ and the difference between the observed $\mathbf{Y}_t$ and its predicted value using the observation equation, $\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t$

$$\mathbf{X}_{t|t} = \hat{\mathbf{X}}_t + \mathbf{\Omega}_t \mathbf{G}_t^T \mathbf{\Delta}_t^{-1} (\mathbf{Y}_t - \mathbf{G}_t \hat{\mathbf{X}}_t),$$

where the relevant terms were defined earlier.

Stochastic Volatility: 1

- ▶ Recall we have develop ARCH/GARCH modeling to create models that better conform to the stylized facts of financial markets: heteroscedasticity, volatility clustering, heavy tail behavior.
- ▶ The GARCH model created this behavior “within itself,” not based on external events/shocks. Yet we have seen that many stocks (particularly those impacted by the same factors, such as the market, industry sectors, etc) exhibit similar behavior with respect to volatility clustering.
- ▶ An alternative model would allow for exogenous processes to impact the volatility behavior of a financial asset.
- ▶ Such models are called stochastic volatility models, and a state space formulation of these models allows us to employ “Kalman methodology” to filter and predict.

Stochastic Volatility: 2

- ▶ In state space models, there is an unobserved or latent variable(s) modeled and an observation/measurement process. We often want to make inference about the latent process.
- ▶ A very good example of this situation is volatility. Volatility is unobserved, yet crucial to the valuation of assets (e.g. an option price).
- ▶ We want to model volatility as the unobserved state and then have an observation process that can be used to estimate (filter) and predict the latent process.
- ▶ An example using the Heston model is presented in the paper “Filtering in Finance.”

Stochastic Volatility: 3

- ▶ The Heston model involves the state process, v_t , and the observation process, S_t , defined by the SDE:

$$dv(t) = (\omega - \theta v_t)dt + \xi \sqrt{v_t} dW_t^v,$$

$$dS_t = rS_t dt + \sqrt{v(t)} S_t dW_t^S.$$

- ▶ The volatility process is a CIR process, whereas the stock price process is GBM with stochastic volatility where:
 - ▶ r is the risk free rate,
 - ▶ θ is the speed of mean reversion,
 - ▶ ω/θ is the long-run equilibrium variance
 - ▶ ξ is the volatility of the variance process.
 - ▶ ρ is the correlation between W_t^v and W_t^S
- ▶ Conditional on v_t , the price process is a GBM

Stochastic Volatility: 4

- ▶ The Heston model is defined in continuous time, and we need to convert it to discrete time, and put it into state space form.
- ▶ The parameter set $(\omega, \theta, \xi, \rho)$ must be calibrated with market data to have the model in risk-neutral form.
- ▶ We can rewrite the stock price equation

$$\log S_{t+\Delta} = \log S_t + (r - .5v_t)\Delta + \sqrt{v_t}\Delta Z_t^S.$$

- ▶ We next use an Euler discretization for the two SDEs with time step Δ :

Stochastic Volatility: 5

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}\Delta}Z_{(i+1)\Delta}^S,$$

$$v_{(i+1)\Delta} = v_{i\Delta} + (\omega - \theta v_{i\Delta})\Delta + \xi \sqrt{v_{i\Delta}\Delta}Z_{(i+1)\Delta}^v,$$

where $\{(Z_i^S, Z_i^v), 1 \leq i \leq N\}$, $\Delta = T/N$, are iid standard bivariate normal random vectors with correlation ρ .

- It is possible to eliminate the correlation parameter ρ by introducing the independent pair $(\tilde{Z}_i^v, Z_i^S)$ where

$$Z_i^v = \rho Z_i^S + \sqrt{1 - \rho^2} \tilde{Z}_i^v$$

and substituting $\tilde{Z}_{i\Delta}^v = \frac{1}{\sqrt{1-\rho^2}}(Z_{i\Delta}^v - \rho \tilde{Z}_{i\Delta}^S)$ into the volatility equation.

Stochastic Volatility: 6

- ▶ Substituting $\sqrt{v_k}\Delta Z_i^S = \log(S_{(i+1)\Delta}/S_{i\Delta}) - (r_{i\Delta} - .5v_{i\Delta})\Delta$ into the volatility equation, we arrive at the state space version of the Euler discretized Heston model with independent noise terms:

$$\begin{aligned}v_{(i+1)\Delta} &= v_{i\Delta} + [\omega - \rho\xi r_{i\Delta} - (\theta - .5\rho\xi)v_{i\Delta}]\Delta \\ &+ \rho\xi \log(S_{(i+1)\Delta}/S_{i\Delta}) + \xi\sqrt{1 - \rho^2}\sqrt{v_{i\Delta}}\Delta\tilde{Z}_k^v.\end{aligned}$$

for the state equation and

$$\log S_{(i+1)\Delta} = \log S_{i\Delta} + (r_{i\Delta} - .5v_{i\Delta})\Delta + \sqrt{v_{i\Delta}}\Delta Z_k^S,$$

for the measurement equation.

Stochastic Volatility: 7

- ▶ Once the stochastic volatility model is in state space form, we can utilize the Kalman procedures for statistical inference.
- ▶ In the particular case, the volatility is of great interest, and inferring the current volatility using Kalman filtering has significant implications (e.g. in valuation of options, comparing current volatility estimates with implied volatility)
- ▶ There are a number of stochastic volatility models, e.g. the Hull-White model we used in the simulation course.
- ▶ Estimation of these models is known to be more complicated than GARCH models.